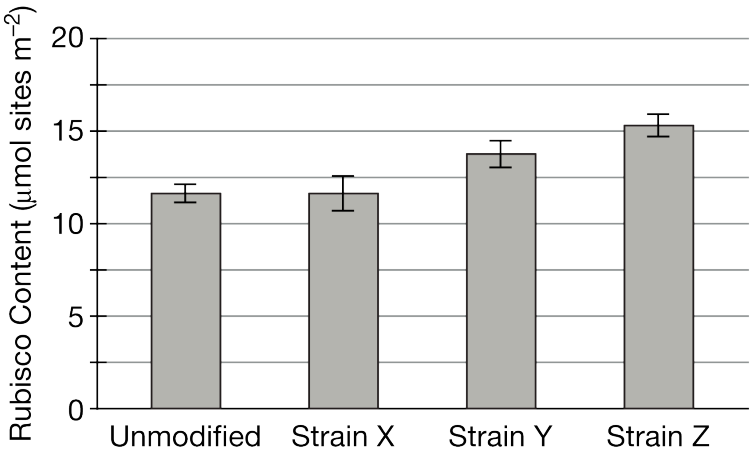
FRQ

1. Read each question carefully. Write your response in the space provided for each part of each question. Answers must be written out in paragraph form. Outlines, bulleted lists, or diagrams alone are not acceptable and will not be scored.

Increasing the efficiency of photosynthesis is one way to increase crop yield and help to feed human populations. One enzyme vital to the Calvin-Benson cycle of photosynthesis is Rubisco. Rubisco catalyzes the fixation of atmospheric CO_2 into organic molecules so that the carbon atoms can be used to produce carbohydrates. Rubisco is composed of two different types of polypeptide subunits: large subunits and small subunits. It is hypothesized that another protein called Rubisco assembly factor (RAF) is needed to help the polypeptide subunits fold together to form a functional enzyme.

Researchers engineered three genetically modified strains of maize (corn). Strain X was modified to produce additional Rubisco polypeptides of both types in the cells. Strain Y was modified to produce additional RAF protein. Strain Z was modified to produce both additional Rubisco polypeptides and additional RAF protein. The Rubisco content of each of the maize strains studied is shown in Figure 1.

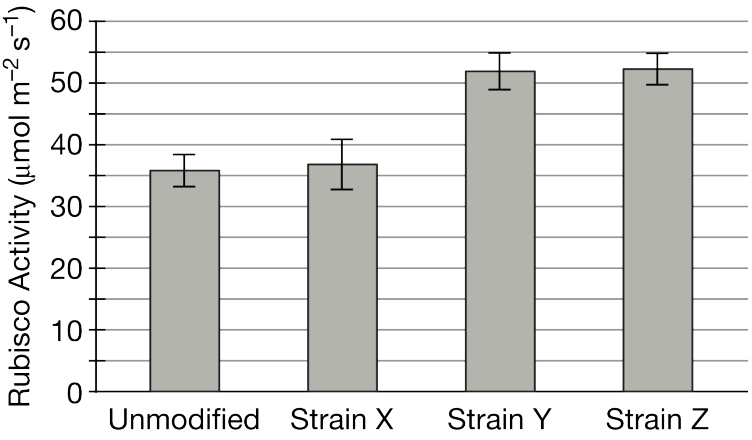


Additional Rubisco Polypeptides	−	+	−	+
Additional Rubisco Assembly Factor (RAF)	−	−	+	+

Figure 1. Rubisco content in genetically modified and unmodified strains of maize. (−) indicates unmodified levels of Rubisco polypeptides and/or RAF protein. (+) indicates additional Rubisco polypeptides and/or RAF protein.

The researchers then grew plants of each strain at a light intensity of $500\ \mu mol\ of\ photons \cdot m^{-2} \cdot s^{-1}$ and a temperature of $25^{\circ}C$. After 40 days, the amount of Rubisco activity (as determined by the rate of carbon fixation) in each modified strain was determined (Figure 2).

FRQ



Additional Rubisco Polypeptides	−	+	−	+
Additional Rubisco Assembly Factor (RAF)	−	−	+	+

Figure 2. Rubisco activity (in $\mu mol\ of\ photons \cdot m^{-2} \cdot s^{-1}$) in genetically modified and unmodified strains of maize. $(-)$ indicates unmodified levels of Rubisco polypeptides and/or RAF protein. $(+)$ indicates additional Rubisco polypeptides and/or RAF protein.

Researchers also evaluated several characteristics of the plants of each strain: the height, the fresh mass of the plants, and the mass of plants dehydrated in an oven (dry mass). The data are presented in Table 1.

TABLE 1. AVERAGE PLANT HEIGHT, FRESH MASS, AND DRY MASS OF FOUR STRAINS OF MAIZE

Strain	Rubisco Production	RAF Production	Average Plant Height (cm) $\pm 2\mathrm{SE}_{\bar{x}}$	Average Fresh Mass (g) $\pm 2\mathrm{SE}_{\bar{x}}$	Average Dry Mass (g) $\pm 2\mathrm{SE}_{\bar{x}}$
Unmodified	$-$	$-$	73.0 ± 4.0	230.0 ± 25.0	29.0 ± 2.0
Strain X	$+$	$-$	83.0 ± 1.0	292.0 ± 13.0	32.0 ± 2.0
Strain Y	$-$	$+$	81.0 ± 3.0	281.0 ± 37.0	34.0 ± 2.0
Strain Z	$+$	$+$	86.0 ± 6.0	294.0 ± 32.0	37.0 ± 1.0

(a) The small subunit of Rubisco is made up of many monomers. **Describe** the general structure of one of these

FRQ

monomers, including the characteristic that would allow it to interact with a negatively charged monomer in a second Rubisco polypeptide.

(b) **Identify** the dependent variable in the experiment whose data are shown in Figure 1. Based on the data in Table 1, **identify** the variable measured that best represents the net amount of carbon fixation that occurs in the experimental plants. Based on the data in Table 1, **describe** the effect of additional Rubisco polypeptide production and normal RAF production on the dry mass of plants, indicating the strains you compare. **Calculate** the percent mass of the fresh plants in strain Z that was due to water.

(c) Researchers engineer a strain of maize with a mutation that results in a decrease in the expression of RAF. **Predict** the most likely effect of this RAF mutation on $\mathrm{CO_2}$ consumption in plants grown under the same conditions as those of the experimental strains. **Justify** your prediction. In addition to binding $\mathrm{CO_2}$, Rubisco can bind $\mathrm{O_2}$ when $\mathrm{CO_2}$ is not readily available. Once Rubisco binds to $\mathrm{O_2}$, an energetically wasteful cycle called photorespiration must be completed before Rubisco can bind $\mathrm{CO_2}$ again. **Predict** the initial effect of a higher than normal $\mathrm{O_2 : CO_2}$ ratio in the atmosphere on the rate of carbon fixation in most plants.

(d) A mutation was found to be naturally occurring in switchgrass, a grass species related to maize. This mutation leads to overexpression of RAF. Using the experimental results shown in Figure 2, **explain** how the mutant switchgrass could have increased fitness compared with nonmutant switchgrass. The small subunits of Rubisco are encoded by a nuclear gene, while the large subunits of Rubisco are encoded by a chloroplast gene. A particular maize plant with a mutation in the allele encoding the large subunit of Rubisco is crossed with a plant that contains the normal allele. **Explain** how examination of the offspring will enable a researcher to determine whether the mutant plant contributed the ovule or the pollen to the cross.

Part A

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1 ✓

The response includes a description that the monomers (amino acids) have an amino group, a carboxyl group, and a variable (R) group with a positive/partially positive charge.

Part B

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1	2	3	4 ✓

The response includes all of the following criteria.

- [ ] An identification that the dependent variable is Rubisco content/level/amount
- [ ] An identification that the variable measured that best represents the net amount of carbon fixation that occurs in the experimental plants is average dry mass

FRQ

- [ ] A description that there is no (statistical) difference between the dry mass of plants with additional Rubisco production (strain X) and control plants (Unmodified)
- [ ] The percent mass of the fresh plants in strain Z that was due to water is calculated to be 87.4%.

Part C

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1	2	3 ✓

The response includes all of the following criteria.

- [ ] A prediction that $\mathrm{CO_2}$ consumption will be reduced
- [ ] A justification that Rubisco activity will be reduced resulting in less carbon fixation/$\mathrm{CO_2}$ entering the Calvin-Benson cycle
- [ ] A prediction that the rate of carbon fixation will decrease

Part D

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1	2 ✓

The response includes both of the following criteria.

- [ ] An explanation that because additional RAF would allow increased Rubisco activity in the mutant strain, the mutant switchgrass could fix more carbon/produce more carbohydrates/increase biomass more quickly
- [ ] An acceptable explanation. Acceptable explanations include the following.
  · If all of the offspring have the mutation, the mutant plant contributed the ovules, because chloroplast genes are only passed on by the ovule.
  · If none of the offspring have the mutation, the mutant plant contributed the pollen, because the pollen does not pass down chloroplast genes.

FRQ

2. Read each question carefully. Write your response in the space provided for each part of each question. Answers must be written out in paragraph form. Outlines, bulleted lists, or diagrams alone are not acceptable and will not be scored.

Hemoglobin is an iron-containing protein in red blood cells that transports oxygen from the lungs to cells throughout the body. Hemoglobin is a protein complex that is composed of four polypeptide subunits. Typically, two of the polypeptide subunits are encoded by the *HBA1* gene and are called alpha-globin subunits, while the other two polypeptide subunits are encoded by the *HBB* gene and are called beta-globin subunits. Each globin subunit contains a single iron atom capable of binding oxygen.

Prior to birth and during early infancy, expression of the *HBB* gene is low and, instead, cells produce a third polypeptide called gamma-globin. During this time, the hemoglobin complex consists of two alpha-globin subunits and two gamma-globin subunits and is referred to as fetal hemoglobin. Fetal hemoglobin is able to bind oxygen more tightly than does the typical adult hemoglobin, which enables the fetus (unborn offspring) to remove oxygen from the hemoglobin in the mother's blood. A model of the expression levels of all three types of globin polypeptides during the weeks leading up to and following birth is shown in Figure 1.

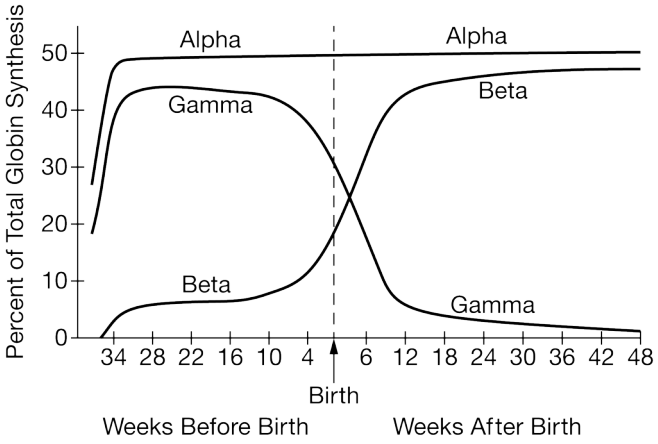
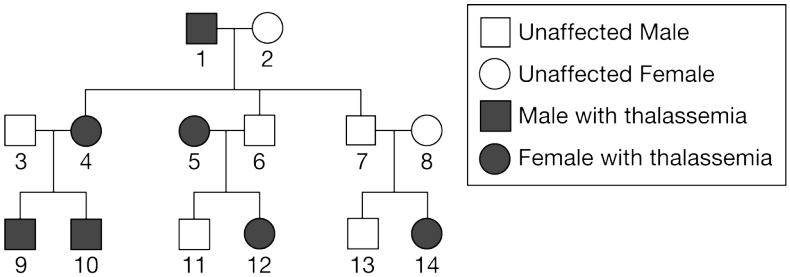


Figure 1. Model of globin expression in fetal and infant tissues

Mutations in the genes that encode globin subunits may affect the function of hemoglobin. Beta-thalassemia is an inherited disorder that results from a variety of mutations to the *HBB* gene that encodes the beta-globin subunit. The incidence of beta-thalassemia in one particular family is shown in Figure 2.



FRQ

Figure 2. Incidence of beta-thalassemia in three generations of a family

The inability to produce functional beta-globin results in an imbalance of alpha- and beta-globin subunits. The accumulation of free alpha-globin subunits causes red blood cells to lyse (burst) or undergo apoptosis (Figure 3). The extra alpha-globin subunits are unable to correctly fold into a hemoglobin protein complex and instead form structures called alpha-hemichromes. The alpha-hemichromes are unable to bind oxygen molecules and become insoluble in the cytoplasm of the red blood cells. The alpha-hemichromes are then deposited in solid form on the surface of the red blood cells, causing the red blood cells to lyse.

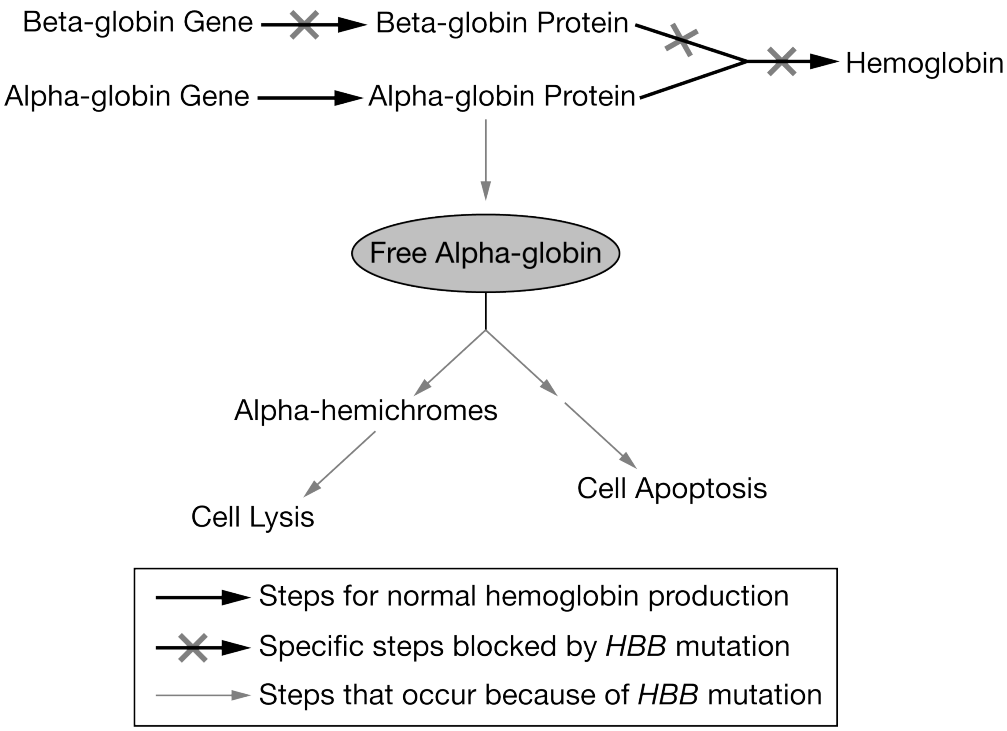


Figure 3. Model of globin synthesis and hemoglobin assembly and the effects of beta-thalassemia. Black arrows with gray Xs indicate steps in the normal hemoglobin assembly that are blocked by the HBB mutation. Gray arrows indicate steps that occur because of the HBB mutation.

(a) Individual 13 in Figure 2 undergoes genetic testing and it is determined that there is a 0% chance that he will pass along the mutant allele. **Describe** the HBB alleles inherited by individual 13, including the parent from which each allele came.

(b) The binding of one oxygen molecule to one subunit of the hemoglobin protein complex facilitates the binding of oxygen to the remaining three subunits. **Explain** how this could happen. Both adult and fetal hemoglobins bind to CO more tightly than to O_2. **Explain** why the hemoglobin of a one-week-old infant is able to bind CO more tightly than the hemoglobin of an adult is able to bind CO.

(c) Researchers investigating the effects of mutant beta-globin found that many individuals with beta-thalassemia have a higher concentration of lactic acid in the blood than do individuals without the disease. **Predict** a possible cause for the increase in lactic acid. **Justify** your prediction.

(d) When abnormal human hemoglobin genes are introduced into mice, the mice develop symptoms similar to

FRQ

those of humans with the abnormal hemoglobin genes. Provide reasoning to **support** the claim that mutant human genes can be expressed in mice to model human genetic conditions.

Part A

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1 ✓

The response includes a description that individual 13 inherited a normal allele from both his parents.

Part B

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1	2 ✓

The response includes both of the following criteria.

- ☐ An explanation that the binding of the first oxygen molecule could alter the shape/charge of other subunits, increasing the compatibility between the binding sites and the oxygen molecule
- ☐ An explanation that a one-week-old infant still has gamma globin in addition to alpha globin, and its hemoglobin will bind more tightly to oxygen than adult hemoglobin does, so it is expected that fetal hemoglobin will also bind CO more tightly

Part C

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1	2 ✓

The response includes both of the following criteria.

- ☐ A prediction that individuals with beta-thalassemia (have less oxygen available and) perform more fermentation than do individuals without the disease
- ☐ A justification that if individuals with beta-thalassemia cannot properly deliver oxygen they will likely rely more heavily on glycolysis to produce ATP than a healthy individual does. Fermentation, which produces lactic acid, will need to occur to keep glycolysis going in the absence of oxygen.

FRQ

Part D

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1 ✓

The response includes supportive reasoning that because the genetic code is shared by mice and humans, human genes can be expressed in mouse cells.

FRQ

3. Read each question carefully. Write your response in the space provided for each part of each question. Answers must be written out in paragraph form. Outlines, bulleted lists, or diagrams alone are not acceptable and will not be scored.

Brook trout (*Salvelinus fontinalis*) is a species of fish that lives in freshwater streams. Brook trout are adapted to tolerate wide ranges in several environmental factors, including temperature and water flow. Brook trout are well adapted to water temperatures from near freezing ($3^{\circ}C$) to about $18^{\circ}C$.

A researcher investigated whether the phenotypes of the brook trout are affected by the rate of water flow. The researcher collected a random sample of $5{,}000$ brook trout eggs and incubated the eggs until the embryos matured. Four weeks after hatching, the juvenile fish were randomly divided into two groups of equal numbers. One group was placed in tanks that simulated a freshwater stream with slowly moving water. The other group was placed in tanks that simulated a freshwater stream with rapidly moving water. After 16 weeks, the researcher collected the fish from the tanks and measured several phenotypic traits (Figure 1). Some of the researcher's measurements are shown in Table 1.

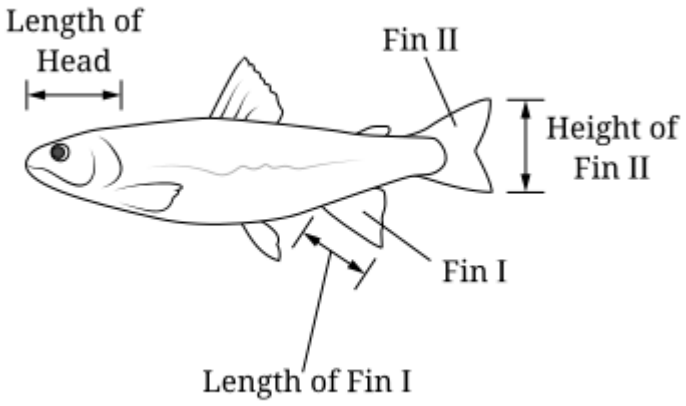


Figure 1. Phenotypic traits of the brook trout measured under different water flow rates

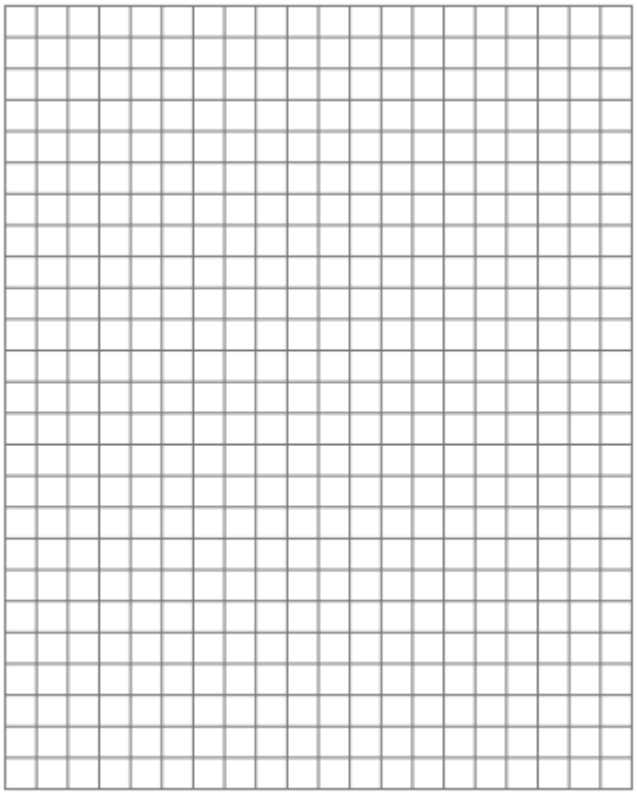
TABLE 1. MEASUREMENTS OF PHENOTYPIC TRAITS OF THE JUVENILE FISH

FRQ

Phenotypic Trait	Juvenile Fish in Slowly Moving Water $(\text{mm}; \text{mean} \pm 2\text{SE}_{\bar{x}})$	Juvenile Fish in Rapidly Moving Water $(\text{mm}; \text{mean} \pm 2\text{SE}_{\bar{x}})$
Length of head	20.95 ± 0.20	20.96 ± 0.20
Length of fin I	15.83 ± 0.70	14.98 ± 0.78
Height of fin II	18.08 ± 0.64	19.62 ± 0.34

(a) Brook trout that live in bodies of freshwater with surface temperatures that reach freezing (0°C) can survive in the water under sheets of ice. **Describe** a property of water that allows the ice to float.

(b) Using the template in the space provided for your response, **construct** an appropriately labeled graph that represents the data shown in Table 1. Based on the data, **determine** whether there is a statistical difference in the length of fin I between the groups of fish in slowly moving water and rapidly moving water.



(c) Based on the data, **describe** the effect of the rate of water flow on head length and the height of fin II.

(d) In a follow-up experiment, the researcher collects fertilized eggs produced by fish raised in a tank with rapidly moving water and places the eggs into a tank with slowly moving water. The researcher measures the phenotypic

FRQ

traits when the offspring reach the same age as the fish measured in the original experiment. Assuming no natural selection occurred over the course of the experiments, **predict** the phenotypes of the offspring in comparison with the phenotypes of their parents. **Explain** how the results of the original experiment provide evidence of phenotypic plasticity.

Part A

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1 ✓

The response does not include an acceptable description. Acceptable descriptions include the following.

· Frozen water is less dense than liquid water.

· More space is present between the water molecules at $0^{\circ}C$ than at higher temperatures.

Part B

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1	2	3	4 ✓

The response includes all of the following criteria.

- ☐ Appropriate labeling
- ☐ Data are plotted in a bar graph or modified bar graph
- ☐ Accurately plotted error bars
- ☐ A determination that there is no (statistical) difference in the length of fin I between the groups of fish in slowly moving water and rapidly moving water

Part C

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1 ✓

The response includes a description that head length was not different for the fish in slowly or rapidly moving water and

FRQ

that the height of fin II was greater for the fish in rapidly moving water.

Part D

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1	2 ✓

The response includes both of the following criteria.

- An acceptable prediction. Acceptable predictions include the following.
  · The fish will appear more like the fish in the tank with slowly moving water (and less like the parent fish).
  · The head and fin I of the offspring fish will be similar to those of their parents, but fin II will be shorter (than that of the parents).
- An acceptable explanation that the fish in each treatment group were genetically similar, but they showed measurable phenotypic differences in different environments, suggesting that phenotypes are influenced by water flow/environmental factors

FRQ

4. Read each question carefully. Write your response in the space provided for each part of each question. Answers must be written out in paragraph form. Outlines, bulleted lists, or diagrams alone are not acceptable and will not be scored.

In order to investigate the possible toxic effects of chemicals on organisms living in rivers, scientists collected water samples from locations downstream of four factories that released wastewater into a river.

The scientists established six treatment groups of bulbs of the common onion, *Allium cepa*, to which they added clean freshwater, an aqueous solution of 300 mM mutagen X, or a water sample from one of the four river locations (A through D). Mutagen X is a compound that causes chromosomal mutations. After incubating the bulbs for 24 hours, the scientists removed root tips emerging from the bulbs in each treatment group and examined individual root tip cells under a microscope.

The scientists examined 6,000 root tip cells from each slide and recorded both the total number of cells that were in some stage of mitosis and the total number of mitotic cells in which either the chromosomes or the mitotic arrangement of the chromosomes appeared to be abnormal (Table 1). They calculated the mitotic index, the percent of all cells examined that were in mitosis. The mitotic index of treatment groups 2 through 6 was then determined relative to the mitotic index of cells in freshwater, which was set at 100 (Figure 1).

TABLE 1. MITOTIC APPEARANCE OF ROOT TIP CELLS AFTER TREATMENT

Treatment Group	Solution Added to Bulbs	Number of Cells in Mitosis	Number of Cells with Chromosomal or Mitotic Abnormalities
1	Freshwater	301	6
2	Mutagen X	144	117
3	Water from location A	150	41
4	Water from location B	210	35
5	Water from location C	186	29
6	Water from location D	270	5

FRQ

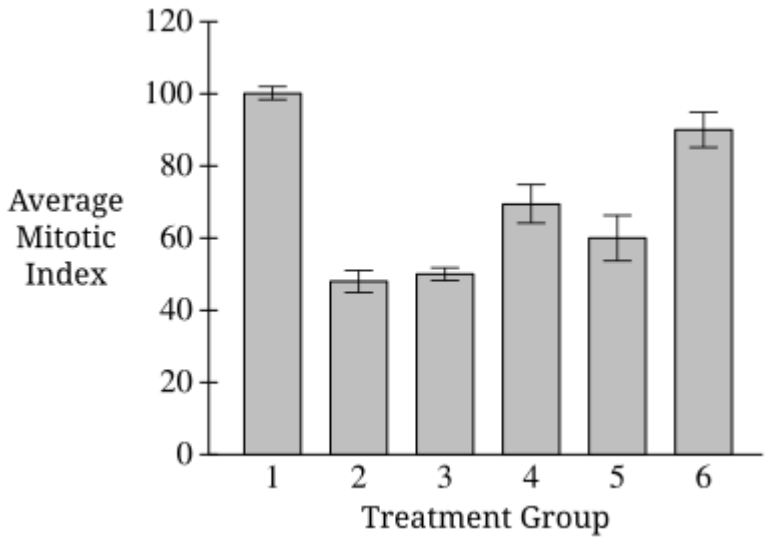


Figure 1. Average mitotic index of root tip cells in each treatment group after 24 hours of incubation. Error bars represent $\pm 2\text{SE}_{\bar{x}}$.

(a) **Describe** the structural features of the two ends of a single nucleotide chain. **Explain** how the addition of a chemical that disrupts hydrogen bonds will alter the structure of DNA isolated from onion root tip cells.

(b) Based on Table 1, **identify** an independent variable in the scientists' investigation. **Justify** the use of mutagen X in one of the treatment groups. **Justify** the scientists' examining the same number of cells in each treatment group.

(c) Based on the data in Table 1 and Figure 1, **describe** the general relationship between the mitotic index and the percent of cells with chromosomal or mitotic abnormalities.

(d) The scientists then analyzed the DNA from root tip cells of *A. cepa* bulbs that were maintained in water from location A for one week and then transferred to freshwater for one week. Assuming mutations do not result in cell death, use the data in Table 1 and Figure 1 to **predict** how the amount of cells with mutations in these bulbs will compare with the amount of cells expected to have mutations in bulbs maintained in freshwater and in bulbs maintained in water from location A for the full two-week period. Based on the data, provide reasoning to **justify** your prediction. **Explain** why the mitotic index of cells in a root tip might be altered by a mutation that affects the mitotic spindle.

Part A

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1	2 ✓

The response includes both of the following criteria.

FRQ

- [ ] A description that a phosphate group/-PO_4 is attached to a $5'$ carbon of the sugar/one end, and a hydroxyl group/-OH is attached to a $3'$ carbon of the sugar/the other end
- [ ] An explanation that the hydrogen bonds (that hold the complementary base pairs of the two nucleotide chains together) will be broken, causing the two chains/backbones to separate

Part B

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1	2	3 ✓

The response includes all of the following criteria.

- [ ] An identification that the independent variable is the incubation/treatment solution
- [ ] A justification that the use of mutagen X was a (positive) control for the effects of a chemical that is known to cause mutations
- [ ] A justification that the scientists' examining the same number of cells was necessary to (accurately) compare the fraction of mitotic cells in each sample

Part C

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1 ✓

The response includes an acceptable description. Acceptable descriptions include the following.

- As the mitotic index decreases, the percent/number of cells (with chromosomal or mitotic abnormalities) increases.
- As the mitotic index increases, the percent/number of cells (with chromosomal or mitotic abnormalities) decreases.
- There is an inverse/negative relationship.

Part D

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1	2	3 ✓

FRQ

The response includes all of the following criteria.

- A prediction that the amount (of cells with mutations in the bulbs that were moved from location A water to freshwater) should be intermediate between the number of cells with mutations in the bulbs in freshwater and in the bulbs in water from location A
- A justification that water from location A induces more mutations than does freshwater, so the longer the bulbs are in location A water, the more mutations they should have
- An explanation that (if the mutation is in a gene encoding a spindle protein, then) chromosomes will not align/separate correctly (and the cells may not complete cell division)

FRQ

5. Read each question carefully. Write your response in the space provided for each part of each question. Answers must be written out in paragraph form. Outlines, bulleted lists, or diagrams alone are not acceptable and will not be scored.

Hemoglobin is an iron-containing protein in red blood cells that transports oxygen from the lungs to cells throughout the body. Hemoglobin is a protein complex that is composed of four polypeptide subunits. Typically, two of the polypeptide subunits are encoded by the *HBA1* gene and are called alpha-globin subunits, while the other two polypeptide subunits are encoded by the *HBB* gene and are called beta-globin subunits. Each globin subunit contains a single iron atom capable of binding oxygen.

Prior to birth and during early infancy, expression of the *HBB* gene is low and, instead, cells produce a third polypeptide called gamma-globin. During this time, the hemoglobin complex consists of two alpha-globin subunits and two gamma-globin subunits and is referred to as fetal hemoglobin. Fetal hemoglobin is able to bind oxygen more tightly than does the typical adult hemoglobin, which enables the fetus (unborn offspring) to remove oxygen from the hemoglobin in the mother's blood. A model of the expression levels of all three types of globin polypeptides during the weeks leading up to and following birth is shown in Figure 1.

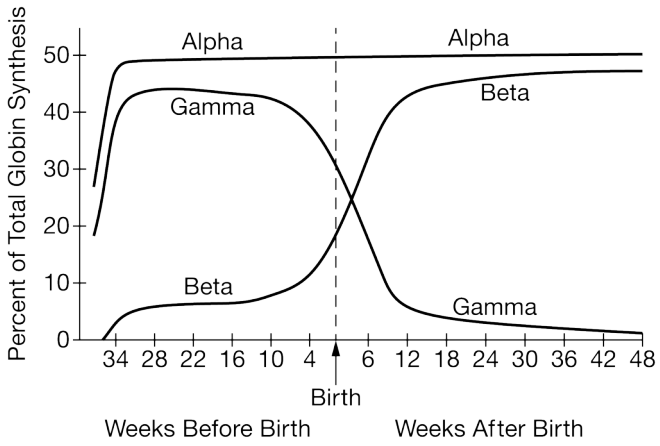
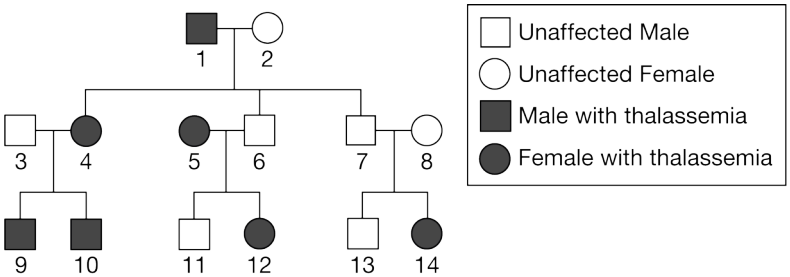


Figure 1. Model of globin expression in fetal and infant tissues

Mutations in the genes that encode globin subunits may affect the function of hemoglobin. Beta-thalassemia is an inherited disorder that results from a variety of mutations to the *HBB* gene that encodes the beta-globin subunit. The incidence of beta-thalassemia in one particular family is shown in Figure 2.



FRQ

Figure 2. Incidence of beta-thalassemia in three generations of a family

The inability to produce functional beta-globin results in an imbalance of alpha- and beta-globin subunits. The accumulation of free alpha-globin subunits causes red blood cells to lyse (burst) or undergo apoptosis (Figure 3). The extra alpha-globin subunits are unable to correctly fold into a hemoglobin protein complex and instead form structures called alpha-hemichromes. The alpha-hemichromes are unable to bind oxygen molecules and become insoluble in the cytoplasm of the red blood cells. The alpha-hemichromes are then deposited in solid form on the surface of the red blood cells, causing the red blood cells to lyse.

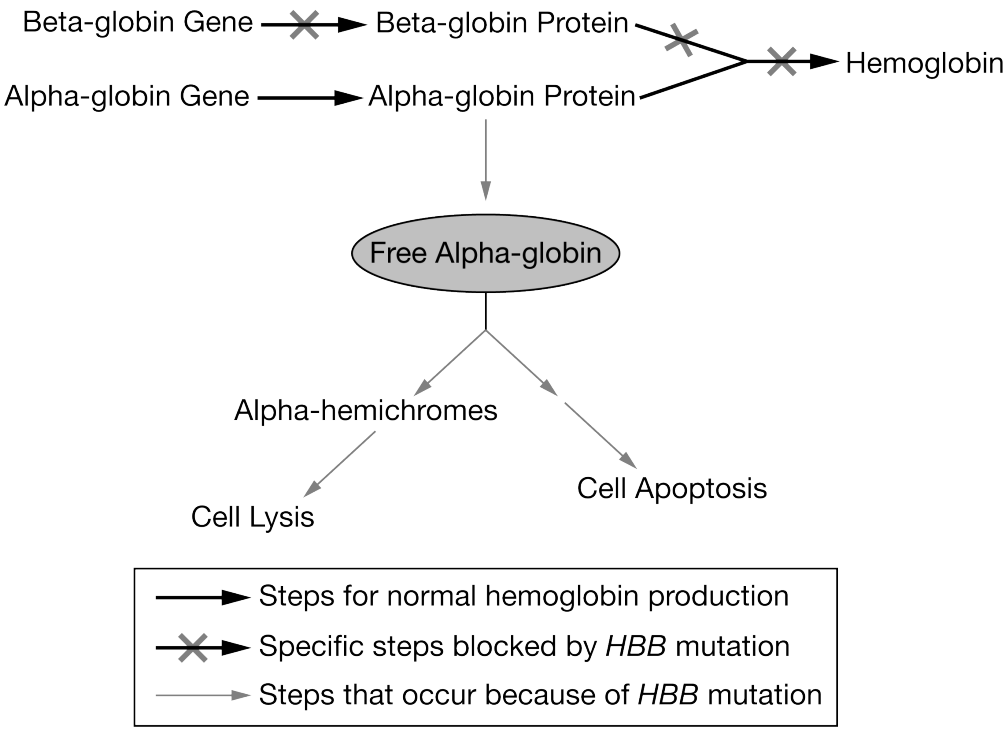


Figure 3. Model of globin synthesis and hemoglobin assembly and the effects of beta-thalassemia. Black arrows with gray Xs indicate steps in the normal hemoglobin assembly that are blocked by the HBB mutation. Gray arrows indicate steps that occur because of the HBB mutation.

(a) Based on the pedigree shown in Figure 2, **describe** the HBB alleles inherited by individual 14, including the parent from which each allele came.

(b) **Explain** how a change in the primary sequence of the amino acid chain in the beta-globin subunit could interfere with the assembly of the entire hemoglobin complex. Carbon monoxide (CO) gas binds to the oxygen-binding site of hemoglobin more tightly than O_2 binds. **Explain** why exposure to CO affects ATP production.

(c) Alpha-globin subunits can interact with either beta-globin or gamma-globin subunits. **Predict** the effect of a mutation that enables the continued expression of gamma-globin on the number of cells undergoing lysis in adults who do not produce beta-globin. **Justify** your prediction.

(d) A researcher claims that the pattern of inheritance for beta-thalassemia is autosomal recessive. Use the pedigree in Figure 2 to **support** the researcher's claim.

FRQ

Part A

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1 ✓

The response includes a description that individual 14 inherited a mutant allele from both of her parents.

Part B

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1	2 ✓

The response includes both of the following criteria.

- ☐ An explanation that a change in the primary sequence of the amino acid chain could result in an altered secondary and tertiary structure, which could prevent the beta-globin subunits from properly interacting with the alpha-globin subunits to form the hemoglobin complex
- ☐ An acceptable explanation. Acceptable explanations include the following.
  · CO interferes with the delivery of oxygen to cells, which is necessary for oxidative phosphorylation.
  · If CO is bound to hemoglobin, oxygen cannot be carried to the cells. Without oxygen to accept electrons from the electron transport chain, the cell cannot make as much ATP.

Part C

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1	2 ✓

The response includes both of the following criteria.

- ☐ A prediction that the number of cells undergoing lysis would decrease
- ☐ A justification that these adults would be able to make functional hemoglobin with alpha- and gamma-subunits, resulting in less/no free alpha-globin, the accumulation of which is responsible for cell lysis

Part D

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

FRQ

0	1 ✓

The response includes acceptable supporting information. Acceptable supporting information is as follows.

· Individuals 7 and 8 in the pedigree are unaffected but have an affected child, and individual 7 must be a carrier/ heterozygous.

· Individuals 7 and 8 in the pedigree are unaffected but have an affected child, and the trait must be autosomal because individual 5 did not pass it to individual 11.

FRQ

6. Read each question carefully. Write your response in the space provided for each part of each question. Answers must be written out in paragraph form. Outlines, bulleted lists, or diagrams alone are not acceptable and will not be scored.

In plants, a critical step in the pathway leading to the synthesis of the amino acid tryptophan includes the synthesis of an intermediate molecule, 5-enolpyruvylshikimic acid-3-phosphate (EPSP). The reactants in this step are shikimic acid-3-phosphate (S3P) and phosphoenolpyruvate (PEP), and the reaction is catalyzed by the enzyme EPSP synthase (Figure 1). Compound X, a chemical used to kill weeds, can affect EPSP synthesis by interacting with EPSP synthase.

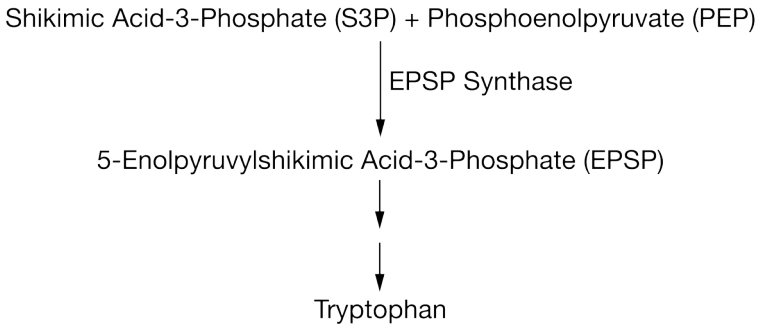


Figure 1. Reaction between shikimic acid-3-phosphate (S3P) and phosphoenolpyruvate (PEP), catalyzed by the enzyme EPSP synthase. Compound X can affect EPSP synthesis by interacting with EPSP synthase.

Scientists investigated the effect that Compound X has on EPSP synthase activity by performing an experiment to examine the relationship between EPSP synthase activity, PEP concentration, and Compound X concentration (Figure 2).

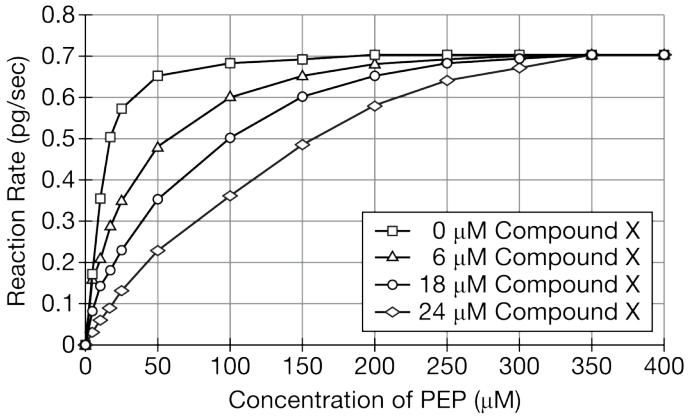


Figure 2. Rate of the reaction catalyzed by a specific amount of EPSP synthase with a specific amount of S3P, increasing concentrations of PEP, and four concentrations of Compound X

Some populations of the plant *Lolium rigidum* have evolved resistance to Compound X. To investigate the

inheritance of this resistance, researchers crossed true-breeding plants from a Compound X-susceptible (S) population with true-breeding plants from a Compound X-resistant (R) population. In half of these crosses, pollen from S plants was used to fertilize ovules in R plants. In the other half, pollen from R plants was used to fertilize ovules in S plants. Survival of parental and F_1 seedlings, differentiated by the maternal phenotype, in response to varying concentrations of Compound X is shown in Figure 3.

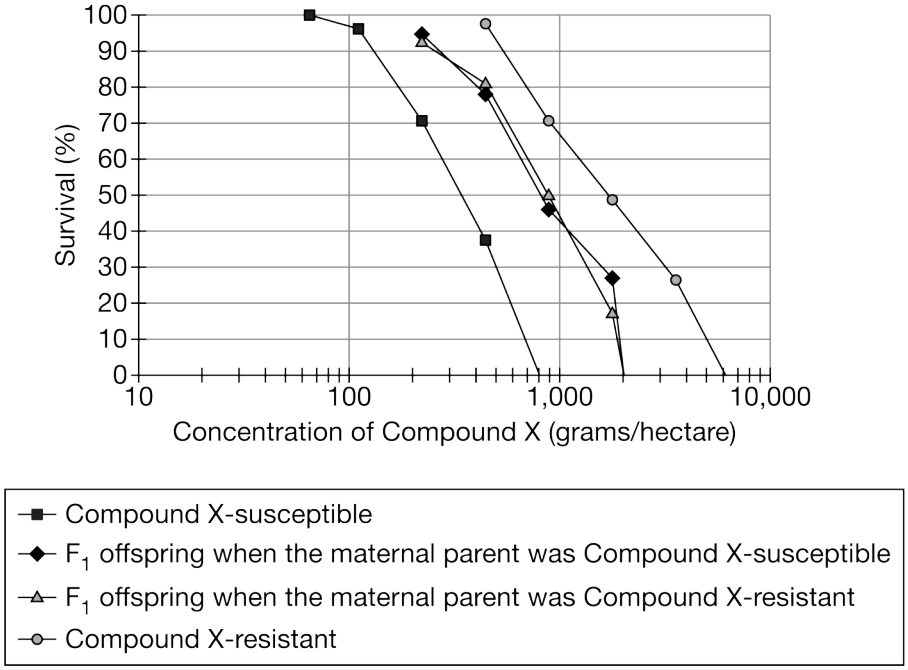


Figure 3. Percent survival of four populations of *L. rigidum* exposed to various concentrations of Compound X shown in a log scale. Data are an average from one experiment with four replicates conducted on seedlings growing in pots. $1\ \text{Hectare} = 10{,}000\ \text{m}^2$

(a) **Explain** how an increased concentration of EPSP synthase would change the rate of the reaction between S3P and PEP in the presence of a low concentration of Compound X.

(b) **Identify** one independent variable in the experiment shown in Figure 2. **Identify** the experimental group(s) in the experiment shown in Figure 3. Using the data provided in Figure 3, **identify** the concentration of Compound X, in grams/hectare, required to kill 30% of the Compound X-resistant (R) population. Using the graph in Figure 2, **calculate** the mass, in pg, of EPSP that would be produced by the reaction in one minute at a PEP concentration of $100\ \mu\text{M}$ and a Compound X concentration of $6\ \mu\text{M}$.

(c) Researchers claim that a single gene is responsible for the Compound X-resistance phenotype. In a follow-up experiment with *L. rigidum*, the researchers crossed an F_1 female with a male from the original resistant (R) population. The resulting offspring were treated with $500\ \text{grams/hectare}$ of Compound X. Based on the data presented in Figure 3, **predict** ONE genotype of the resulting offspring and the percent survival of the offspring with this genotype after treatment with Compound X. Based on the evidence in Figure 3, researchers claim that the allele for Compound X-resistance is carried on a chromosome in the nucleus. Provide reasoning to **justify** this claim. Researchers discovered a variant of the EPSP synthase gene in *L. rigidum* in which there was a mutation in the promoter region of this gene. This mutation increased the likelihood that RNA polymerase binds to the EPSP synthase gene. **Predict** the most likely effect of this mutation on the ability of *L. rigidum* plants with this

FRQ

variant of the EPSP synthase gene to produce tryptophan.

(d) Researchers have engineered Compound X-resistant corn by inserting a Compound X-resistance gene from a prokaryote into the corn genome. **Explain** why the upstream regulatory region, including the promoter, of the prokaryotic DNA would need to be altered in order to express the Compound X-resistance gene in corn. In addition to a conserved genetic code, **explain** how the prokaryotic EPSP synthase protein can be translated from identical mRNAs by both a prokaryote and a corn plant.

Part A

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1 ✓

The response includes an explanation that the reaction rate will increase because there will be more enzymes available to bind to substrates.

Part B

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1	2	3	4 ✓

The response includes all of the following criteria.

- ☐ An acceptable identification of an independent variable. Acceptable identifications include the following.
  - Concentration of PEP
  - Concentration of Compound X
- ☐ An acceptable identification of the experimental group. Acceptable identifications include the following.
  - Both F_1 generations
  - F_1 offspring when the maternal parent is Compound X-susceptible and F_1 offspring when the maternal parent is Compound X-resistant
- ☐ An identification that the concentration of Compound X, in $\text{grams} / \text{hectare}$, required to kill 30% of the Compound X-resistant (R) population is 900 ($\text{grams} / \text{hectare}$); NOTE: acceptable student responses can be between 850 and 950.
- ☐ The mass, in pg, of EPSP that would be produced by the reaction in one minute at a PEP concentration of $100\mu\text{M}$ and a Compound X concentration of $6\mu\text{M}$ is calculated to be 36 pg.

Part C

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

FRQ

0	1	2	3 ✓

The response includes all of the following criteria.

- ☐ An acceptable prediction of the percent survival of one genotype of the resulting offspring. Acceptable predications include the following.
  · 75% survival RS; NOTE: acceptable student responses can be between 72% and 78%.
  · 94% survival RR; NOTE: acceptable student responses can be between 92% and 96%.
- ☐ An acceptable justification. Acceptable justifications include the following.
  · When susceptible and resistant strains are crossed, the offspring have an intermediate phenotype. Therefore, each offspring received an allele from each parent.
  · If the allele were carried in the chloroplast or mitochondrion, then the offspring would have the same phenotype as the maternal parent.
- ☐ A prediction that the likely effect of this mutation is an increased production of tryptophan

Part D

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1	2 ✓

The response includes both of the following criteria.

- ☐ An acceptable explanation. Acceptable explanations include the following.
  · Because eukaryotic transcription factors differ from prokaryotic transcription factors, eukaryotic transcription factors will not recognize the prokaryotic promoter sequence.
  · Because eukaryotic RNA polymerase differs from prokaryotic RNA polymerase, eukaryotic RNA polymerase will not recognize the prokaryotic promoter sequence.
- ☐ An explanation that the mRNAs can be translated because both have ribosomes/transfer RNA to read the mRNA code

FRQ

7. Read each question carefully. Write your response in the space provided for each part of each question. Answers must be written out in paragraph form. Outlines, bulleted lists, or diagrams alone are not acceptable and will not be scored.

Hemoglobin is an iron-containing protein in red blood cells that transports oxygen from the lungs to cells throughout the body. Hemoglobin is a protein complex that is composed of four polypeptide subunits. Typically, two of the polypeptide subunits are encoded by the *HBA1* gene and are called alpha-globin subunits, while the other two polypeptide subunits are encoded by the *HBB* gene and are called beta-globin subunits. Each globin subunit contains a single iron atom capable of binding oxygen.

Prior to birth and during early infancy, expression of the *HBB* gene is low and, instead, cells produce a third polypeptide called gamma-globin. During this time, the hemoglobin complex consists of two alpha-globin subunits and two gamma-globin subunits and is referred to as fetal hemoglobin. Fetal hemoglobin is able to bind oxygen more tightly than does the typical adult hemoglobin, which enables the fetus (unborn offspring) to remove oxygen from the hemoglobin in the mother's blood. A model of the expression levels of all three types of globin polypeptides during the weeks leading up to and following birth is shown in Figure 1.

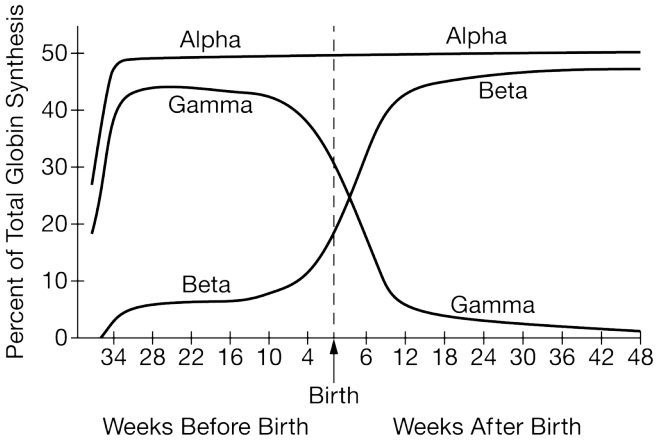
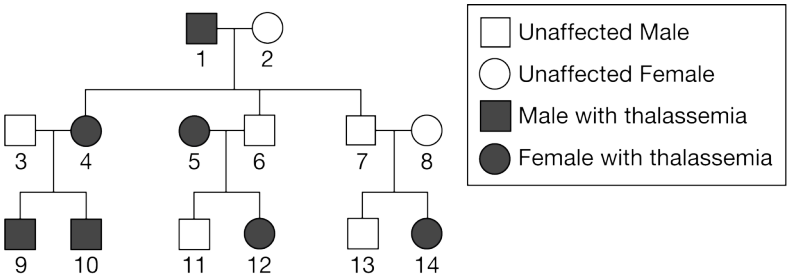


Figure 1. Model of globin expression in fetal and infant tissues

Mutations in the genes that encode globin subunits may affect the function of hemoglobin. Beta-thalassemia is an inherited disorder that results from a variety of mutations to the *HBB* gene that encodes the beta-globin subunit. The incidence of beta-thalassemia in one particular family is shown in Figure 2.



FRQ

Figure 2. Incidence of beta-thalassemia in three generations of a family

The inability to produce functional beta-globin results in an imbalance of alpha- and beta-globin subunits. The accumulation of free alpha-globin subunits causes red blood cells to lyse (burst) or undergo apoptosis (Figure 3). The extra alpha-globin subunits are unable to correctly fold into a hemoglobin protein complex and instead form structures called alpha-hemichromes. The alpha-hemichromes are unable to bind oxygen molecules and become insoluble in the cytoplasm of the red blood cells. The alpha-hemichromes are then deposited in solid form on the surface of the red blood cells, causing the red blood cells to lyse.

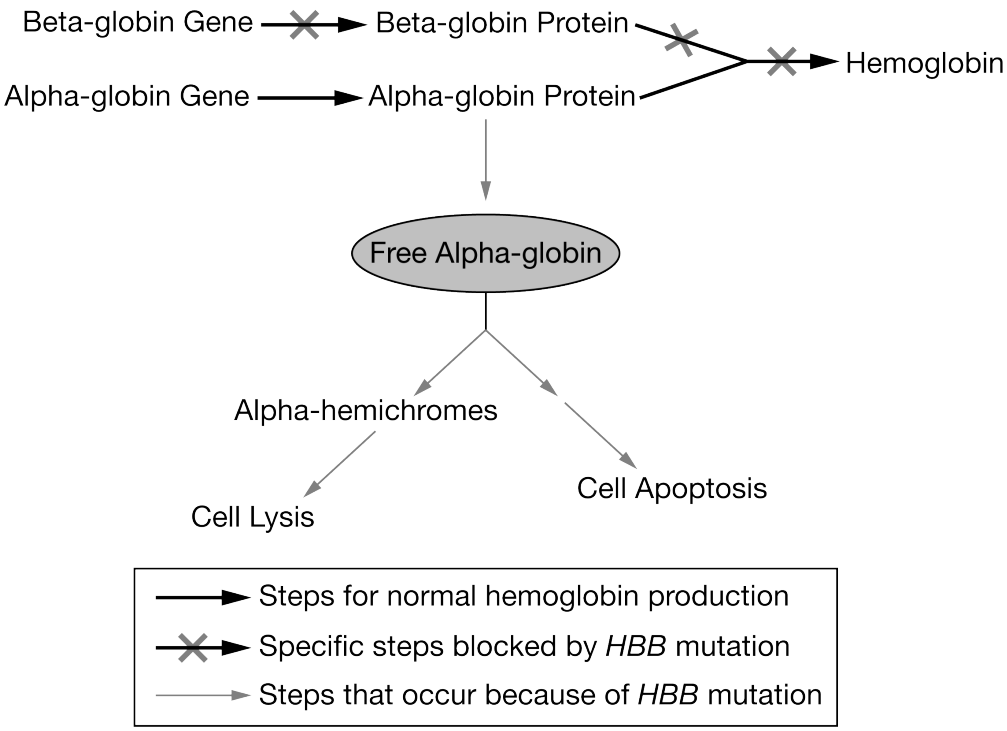


Figure 3. Model of globin synthesis and hemoglobin assembly and the effects of beta-thalassemia. Black arrows with gray Xs indicate steps in the normal hemoglobin assembly that are blocked by the HBB mutation. Gray arrows indicate steps that occur because of the HBB mutation.

(a) Based on the pedigree shown in Figure 2, **describe** the HBB alleles inherited by individual 11, including the parent from which each allele came.

(b) **Explain** how mutant hemoglobin interferes with a cell's ability to synthesize ATP. Red blood cells damaged by the accumulation of alpha-hemichromes undergo apoptosis and are taken into other cells by a process similar to endocytosis. **Explain** how these cells will digest and recycle the components of the apoptotic red blood cells.

(c) While studying the family shown in the pedigree shown in Figure 2, researchers notice that the HBB mutant allele and an allele for another trait are passed on together at a frequency greater than that which would be predicted if the genes assorted independently. **Predict** the cause of this increased frequency of inheritance. **Justify** your prediction.

(d) A researcher claims that HBB exhibits a maternal mitochondrial inheritance pattern. Use the pedigree to **refute** the researcher's claim.

FRQ

Part A

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1 ✓

The response includes a description that individual 11 inherited a normal allele from his father and a mutant allele from his mother.

Part B

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1	2 ✓

The response includes both of the following criteria.

- An explanation that oxygen will not be delivered to the cells to act as the terminal electron acceptor of the electron transport chain, and the ETC is the main source of ATP in cells/for cellular respiration
- An explanation that a vesicle containing the apoptotic red blood cell will fuse with a lysosome containing digestive enzymes (which will allow the components to be recycled)

Part C

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1	2 ✓

The response includes both of the following criteria.

- A prediction that the genes are close together on a chromosome
- An acceptable justification. Acceptable justifications include the following.
  · Genes that are located close together on the same chromosome are less likely to be separated by crossing over than are genes that are far apart from each other on the same chromosome.
  · Genes carried on separate chromosomes assort independently, but genes carried on the same chromosome tend to be inherited together unless crossing over separates them.

Part D

FRQ

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1 ✓

The response includes a refutation stating that mitochondrial genes are passed only from the maternal parent to the offspring. Therefore, all offspring should have the same phenotype as their mother, but #4/#11/#14 has a phenotype that differs from that of her/his mother.

FRQ

8. Read each question carefully. Write your response in the space provided for each part of each question. Answers must be written out in paragraph form. Outlines, bulleted lists, or diagrams alone are not acceptable and will not be scored.

In plants, photosynthetic guard cells on the surface of leaves form pores called stomata (a single pore is called a stoma). Stomata help prevent water loss and control gas exchange with the environment to maintain plant homeostasis. The turgor pressure of the guard cells on either side of the stomata determines whether the stomata are open or closed based on the transport of certain ions into and out of the guard cells (Figure 1). A proton pump in the membrane of the guard cells pumps protons into the extracellular space. This makes the interior of the cell more negatively charged than the extracellular space.

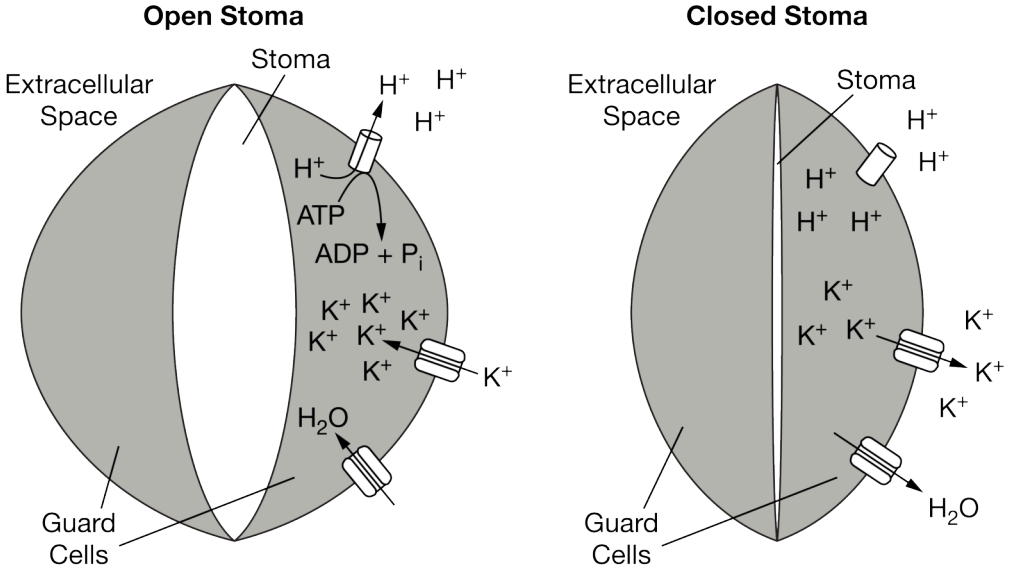


Figure 1. Models of open and closed stomata and the transport proteins in the guard cell membranes

Scientists interested in understanding drought tolerance in plants discovered a mutant variety of a plant with increased cell division, increased water content, and reduced density of leaf stomata compared with the wild type. Analysis of the mutant plant showed a mutation in the gene encoding a cyclin-dependent kinase (CDK). This mutation prevents CDK-cyclin interaction that normally results in the phosphorylation of RNA polymerase (Figure 2).

FRQ

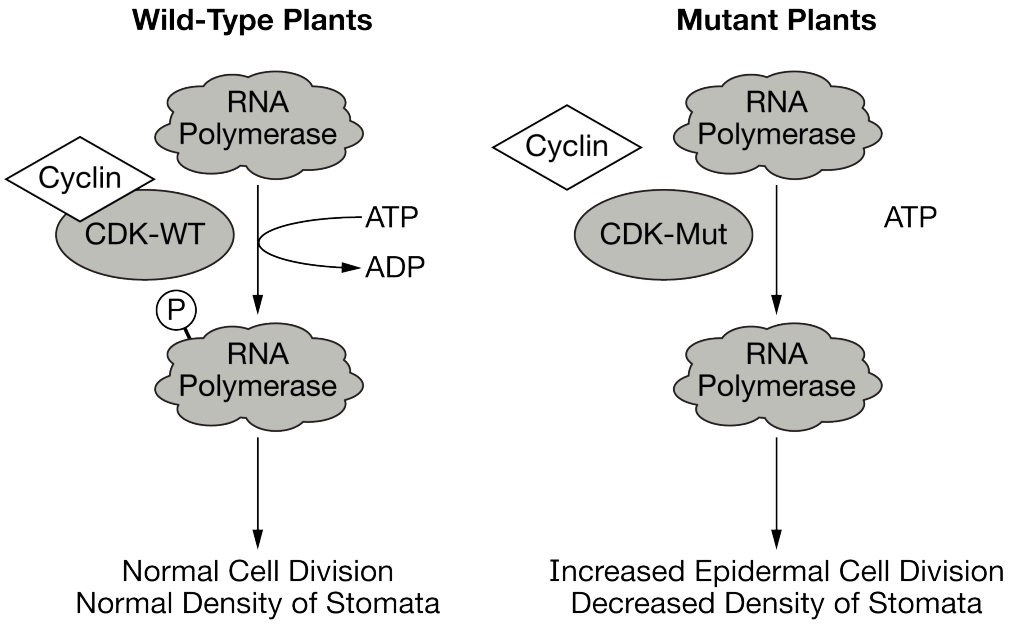


Figure 2. In wild-type plants, the wild-type CDK protein ($\mathrm{CDK\text{-}WT}$) allows interaction with a cyclin and leads to the phosphorylation of RNA polymerase. In mutant plants, the mutant CDK protein ($\mathrm{CDK\text{-}Mut}$) prevents interaction with cyclin and the phosphorylation of RNA polymerase.

(a) **Describe** the role of ATP in the model shown in Figure 2. Scientists crossed plants that were homozygous for the cyclin dependent kinase-wild type ($CDK\text{-}WT$) allele with plants that were homozygous for the cyclin dependent kinase-mutant ($CDK\text{-}Mut$) allele. Crossing offspring from the F_1 generation produced F_2 offspring with a phenotypic ratio of $3\ \text{wild-type} : 1\ \text{mutant}$. **Explain** how the F_2 generation includes offspring with the wild-type phenotype.

(b) Based on Figure 2, **explain** how the expression of the $CDK\text{-}Mut$ allele most likely causes a decrease in the density of stomata.

(c) A mutation in some guard cells inactivates the proton pump in these cells. **Predict** how the membrane potential across the membrane of these guard cells compares with the membrane potential across the membrane of guard cells with functioning proton pumps. Provide reasoning to **justify** your prediction.

(d) Drought conditions result in a decrease in turgor pressure in plant cells. Researchers claim that this reduction in turgor pressure negatively affects gas exchange in the wild-type plants but not in the mutant plants. **Refute** the researchers' claim using the information provided.

Part A

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1	2 ✓

FRQ

The response includes both of the following criteria.

- [ ] A description that ATP provides the energy and the phosphate (for the phosphorylation of RNA polymerase)
- [ ] An explanation that the F_2 generation was produced from two heterozygous parents, thus the wild-type offspring inherited at least one copy of a dominant wild-type allele

Part B

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1 ✓

The response includes an acceptable explanation. Acceptable explanations include the following.

· The mutation leads to the alteration/prevents the phosphorylation of RNA polymerase which induces the synthesis of proteins/expression of genes that encode proteins that block formation of stomata.

· The mutation leads to the alteration/prevents the phosphorylation of RNA polymerase which prevents the synthesis of proteins/expression of genes that encode proteins that promote the formation of stomata.

Part C

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1	2 ✓

The response includes both of the following criteria.

- [ ] A prediction that the membrane potential will be less in the mutated guard cells than in the normal guard cells
- [ ] A justification that the mutated proton pump will not be able to pump out positively charged protons, so the inside of the guard cell be less negative with respect to the outside of the cell when compared with guard cells without the mutation

Part D

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1 ✓

FRQ

The response includes a refutation for the claim by stating that, using the information provided, the decrease in turgor pressure in the guard cells causes the stomata to close and prevents gas exchange in both types of plants.

9. Read each question carefully. Write your response in the space provided for each part of each question. Answers must be written out in paragraph form. Outlines, bulleted lists, or diagrams alone are not acceptable and will not be scored.

Aquaporin channel proteins are found in the plasma membrane of various cell types. Scientists studied the production of aquaporins in the keratinocytes (skin cells) of healthy individuals and found that these proteins play a role in hydration of the skin. Some individuals have a skin disorder that causes the absorption of excess water by the skin of their hands and feet when they are soaked in water. Scientists determined that the keratinocytes of these individuals are characterized by the presence of abnormal aquaporins. Scientists studied a family in which some individuals have this skin disorder and constructed a pedigree (Figure 1).

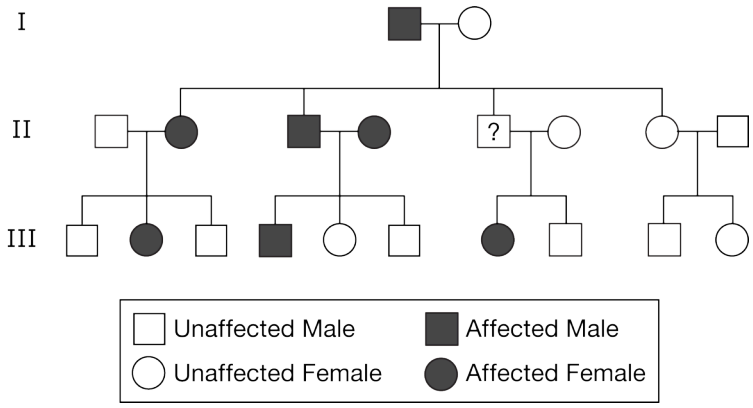


Figure 1. Pedigree of a human family in which some members have a skin disorder. An individual with an unknown phenotype is marked by a question mark.

(a) Based on the pedigree in Figure 1, **predict** the most likely genotype AND phenotype of the individual marked by "?". **Justify** your prediction.

(b) **Describe** the characteristics of the keratinocyte membrane that restrict water movement across the membrane. If isolated keratinocytes are placed in a hypotonic solution for 24 hours, **explain** why there are likely to be far fewer cells at the end of this time than there were at the start.

(c) **Explain** how the aquaporin polypeptides are transported from the Golgi complex and inserted into the plasma membrane.

(d) A student claims that individuals who have a lower than normal number of aquaporins in their skin cells will easily dehydrate. **Refute** this claim with evidence from biological concepts.

Part A

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

FRQ

0	1	2 ✓

The response includes both of the following criteria.

- ☐ A prediction that the individual marked by "?" is heterozygous (e.g., Rr) and has the disorder
- ☐ A justification that the inheritance pattern is autosomal dominant (because two affected individuals had unaffected children), and the individual marked by "?" and an unaffected female produced both affected and unaffected offspring

Part B

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1	2 ✓

The response includes both of the following criteria.

- ☐ An acceptable description. Acceptable descriptions include the following.
  · The membrane has a non-polar/hydrophobic interior.
  · The membrane includes phospholipids with fatty acid tails.
- ☐ An explanation that because of the greater solute concentration inside the keratinocytes, water will move into the keratinocytes, causing many of them to lyse by the end of the 24 hours

Part C

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1 ✓

The response includes an acceptable explanation. Acceptable explanations include the following.

· A portion of the Golgi complex containing the aquaporin polypeptide buds off as a vesicle that fuses with the plasma membrane (allowing the aquaporin to embed in the membrane).

· The aquaporin polypeptide is carried by a vesicle that fuses with the plasma membrane.

Part D

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

FRQ

0	1 ✓

The response includes a refutation of the claim by stating that fewer aquaporin channels will result in less/slower water loss compared with water loss from cells with a normal number of aquaporins.

FRQ

10. Read each question carefully. Write your response in the space provided for each part of each question. Answers must be written out in paragraph form. Outlines, bulleted lists, or diagrams alone are not acceptable and will not be scored.

Increased expression and activity of the ligand-gated ion channel TRPV1 has been shown to block cell division and induce apoptosis in human cells. The structure of the TRPV1 protein consists of several transmembrane domains that are embedded in the membrane as well as a carboxy-terminus and an amino-terminus that are located inside the cell (Figure 1).

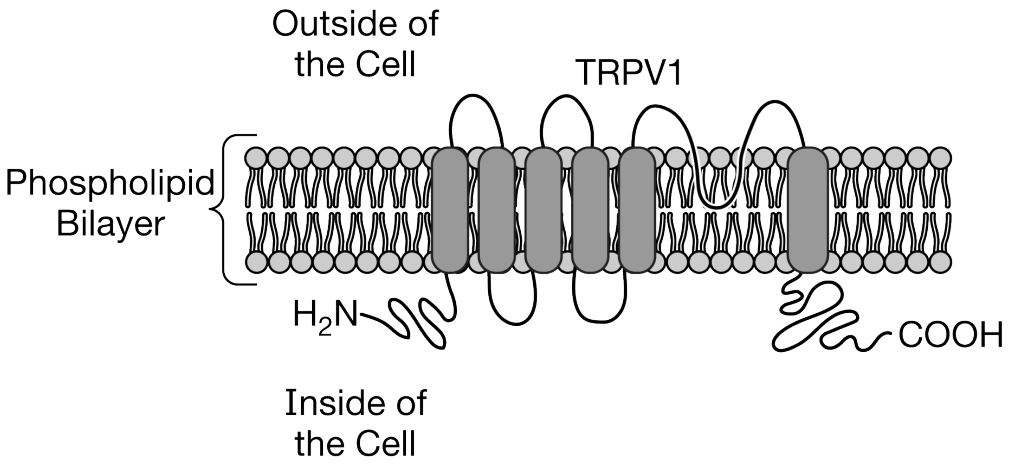


Figure 1. Structure of the TRPV1 protein channel in the membrane

TRPV1 is activated by several stimuli, including the binding of capsaicin, a chemical found in chili peppers. Capsaicin enters the cell by simple diffusion and then binds to one of the transmembrane domains of TRPV1, which opens the ion channel.

The opening of the TRPV1 channel allows Ca^{2+} ions to enter the cell, leading to the activation of the enzyme calcineurin. Calcineurin removes phosphate groups from the phosphorylated form of the transcription factor NFAT2, which is typically found in the cytoplasm. Once dephosphorylated, NFAT2 moves into the nucleus, where it blocks the transcription of another protein, ATF3. ATF3, when active, prevents the cell cycle regulatory protein pGoG from inducing apoptosis and blocking cell division (Figure 2).

FRQ

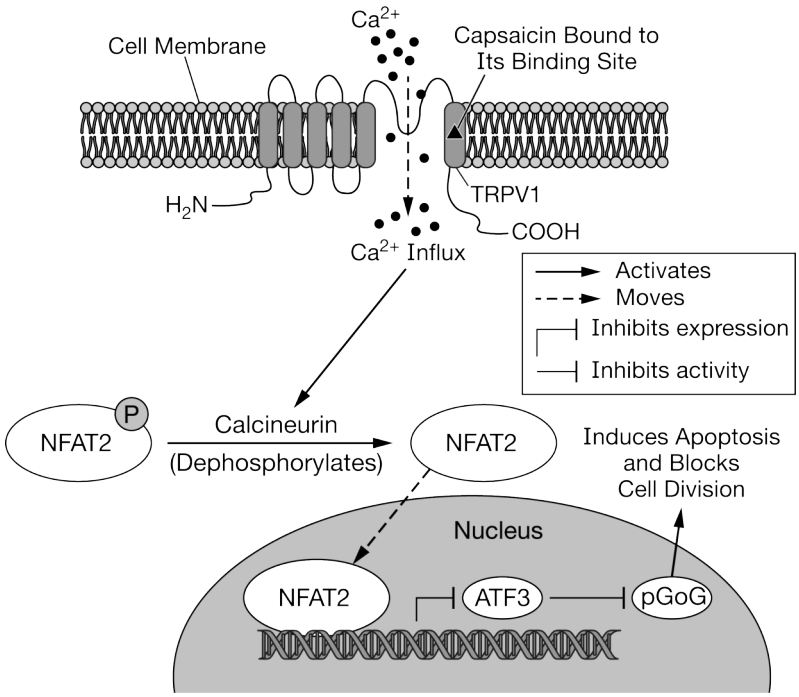


Figure 2. Capsaicin-activated TRPV1 signaling pathway in human cells

(a) **Describe** the differences between the R groups of the amino acids located in the transmembrane domains and those located in the amino- or carboxy-terminus of TRPV1. **Describe** the specific location in the cell where TRPV1 protein is synthesized.

(b) The pGoG protein leads to the inhibition of cyclin activity. **Explain** how this interferes with cell cycle progression.

(c) Researchers discover a mutation that prevents NFAT2 from being phosphorylated. **Predict** the effect of this mutation on apoptosis by the pathway shown in Figure 2. Provide reasoning to **justify** your prediction.

(d) The drug cyclosporin A is known to inhibit calcineurin. Researchers claim that cyclosporin A should not be given to patients with cancer. **Support** the researchers' claim based on the pathway shown in Figure 2.

Part A

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1	2 ✓

The response includes both of the following criteria.

FRQ

- [ ] A description that R groups in the transmembrane domain should be hydrophobic/nonpolar, while R groups in either terminus should be hydrophilic/polar/charged
- [ ] A description that TRPV1 protein is synthesized on the ribosomes attached to the endoplasmic reticulum/ ER /rough ER

Part B

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1 ✓

The response includes an explanation that cyclins cannot activate cyclin-dependent kinases which normally (phosphorylate other proteins to) move the cell through a cell cycle checkpoint.

Part C

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1	2 ✓

The response includes both of the following criteria.

- [ ] A prediction that apoptosis will be induced
- [ ] A justification that if NFAT2 cannot be phosphorylated, it will be able to (move to the nucleus and) inhibit ATF3 expression, which enables activity of pGoG which will induce apoptosis

Part D

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1 ✓

The response includes support for the claim by stating that if calcineurin is inhibited, then the steps in this pathway that follow from calcineurin activation will be unable to block cell division/induce apoptosis. Since cancer is characterized by excessive cell division, this drug would be an inappropriate treatment for cancer.

FRQ

11. Read each question carefully. Write your response in the space provided for each part of each question. Answers must be written out in paragraph form. Outlines, bulleted lists, or diagrams alone are not acceptable and will not be scored.

Gitelman syndrome (GS) is a heritable disorder caused by a mutation in the *SLC* gene that encodes a Na^+/Cl^- cotransporter protein (A in Figure 1) that is expressed in kidney cells. When molecules or ions are moved by cotransporters, the molecules or ions must move across a membrane together, either in the same direction or in opposite directions. The Na^+/Cl^- cotransporter simultaneously transports Na^+ and Cl^- from one region of the kidney (Region I) into the cell. Other transporters in kidney cell membranes include the Na^+/K^+ ATPase (B in Figure 1) and a Na^+/Ca^{2+} cotransporter (D in Figure 1) that allow ions to move across the membrane between the kidney cell and a second region of the kidney (Region II). Kidney cells also have channel proteins like the Ca^{2+} channel (C in Figure 1) that transports Ca^{2+} from Region I into the kidney cell. A model representing the exchange of various ions between the kidney cell and the two extracellular regions is shown in Figure 1. The size of an ion's symbol indicates the relative concentration of that ion in a given area. The larger symbol indicates the area of higher concentration, and the smaller symbol indicates the area of lower concentration.

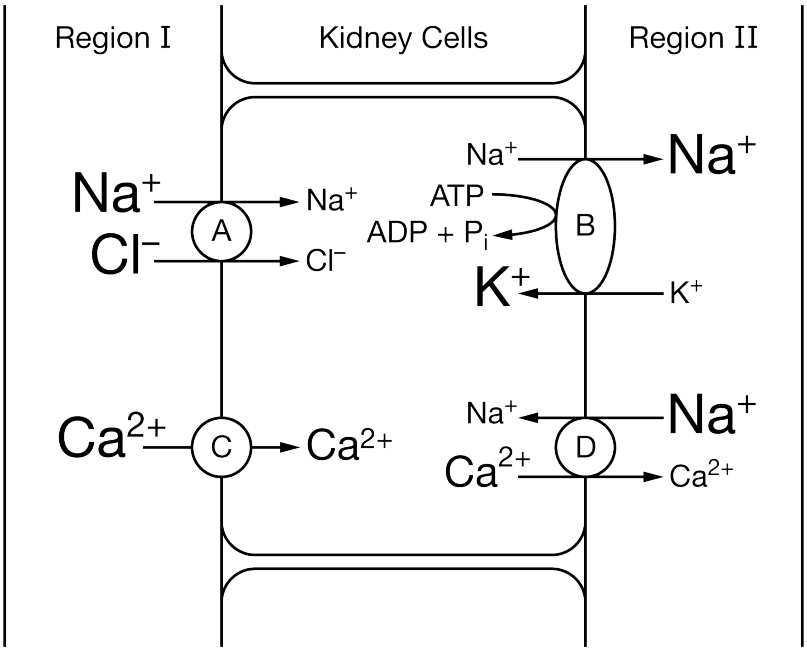


Figure 1. Model representing the exchange of ions between a kidney cell and two isolated extracellular regions. The larger symbols indicate the areas of higher ion concentration, and the smaller symbols indicate the areas of lower ion concentration.

Researchers have determined that there are a number of mutations to the SLC wild-type allele (*SLC*) that result in GS. One mutant allele, *SLC-gs1*, results from a single nucleotide deletion that creates a stop codon midway through the coding region of the gene. A second mutant allele, *SLC-gs2*, results from a mutation that disrupts the sequence of a splicing site between an intron and an exon, preventing the removal of an intron. To better understand how GS is inherited, researchers studied a number of families with a history of GS. The pedigree of

FRQ

one of these families is shown in Figure 2.

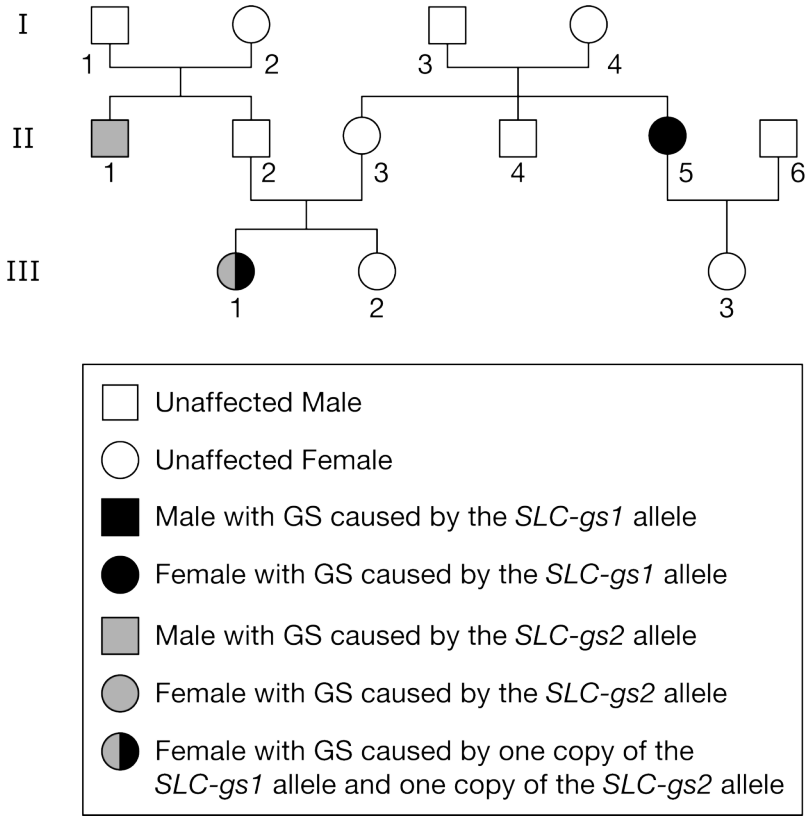


Figure 2. Pedigree of a family with a history of GS caused by two different mutant alleles

(a) **Describe** the characteristics of the Ca^{2+} ions and the plasma membrane of the kidney cell that prevent the Ca^{2+} ions from moving from Region I into the kidney cell without the channel labeled C.

(b) **Explain** the effect of the mutation in the *SLC-gs2* allele on the $\mathrm{Na}^{+}/\mathrm{Cl}^{-}$ cotransporter protein produced by the cell. **Explain** why the transporter labeled B in Figure 1 requires ATP to move ions between Region II and the kidney cells.

(c) **Predict** the immediate effect on the movement of Na^{+} AND Ca^{2+} by the cotransporter labeled D in Figure 1 if the $\mathrm{Na}^{+}/\mathrm{Cl}^{-}$ cotransporter (A in Figure 1) is nonfunctional. Based on Figure 1, provide reasoning to **justify** your prediction.

(d) Based on the information provided in Figure 2, **predict** the two alleles of the SLC gene that make up the genotype of individual $\mathrm{II\text{-}2}$ AND the two alleles of the SLC gene that make up the genotype of individual $\mathrm{II\text{-}3}$.

Part A

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

FRQ

0	1 ✓

The response includes an explanation that Ca^{2+} ions are charged/hydrophilic and the interior of the membrane is nonpolar/hydrophobic.

Part B

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1	2 ✓

The response includes both of the following criteria.

- ☐ An acceptable explanation. Acceptable explanations include the following.
  · The sequence of the intron will be translated and (as long as a stop codon is not introduced) this will result in a longer protein.
  · The sequence of the intron will be translated, which could introduce a stop codon before the existing stop codon, resulting in a shorter protein.
  · The sequence of the intron will be translated resulting in a change in the structure/function of the protein.
- ☐ An explanation that energy (from ATP) is required because the transporter is moving both sodium and potassium ions against their concentration gradient/from low concentration to high concentration

Part C

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1	2 ✓

The response includes both of the following criteria.

- ☐ An acceptable prediction. Acceptable predictions include the following.
  · Na^{+} will continue to enter the cell and Ca^{2+} will continue to leave the cell.
  · The movement of Na^{+} AND Ca^{2+} by D will not change.
- ☐ A justification that (with A not functioning) there is still a concentration gradient because transporter B/C still functions, so Na^{+} and Ca^{2+} continue to be transported through transporter D

Part D

FRQ

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1 ✓

The response includes a prediction that that the two alleles that make up the genotype of individual II-2 are SLC and $SLC-gs2$, AND the two alleles that make up the genotype of individual II-3 are SLC, $SLC-gs1$.

FRQ

12. Read each question carefully. Write your response in the space provided for each part of each question. Answers must be written out in paragraph form. Outlines, bulleted lists, or diagrams alone are not acceptable and will not be scored.

Increased expression and activity of the ligand-gated ion channel TRPV1 has been shown to block cell division and induce apoptosis in human cells. The structure of the TRPV1 protein consists of several transmembrane domains that are embedded in the membrane as well as a carboxy-terminus and an amino-terminus that are located inside the cell (Figure 1).

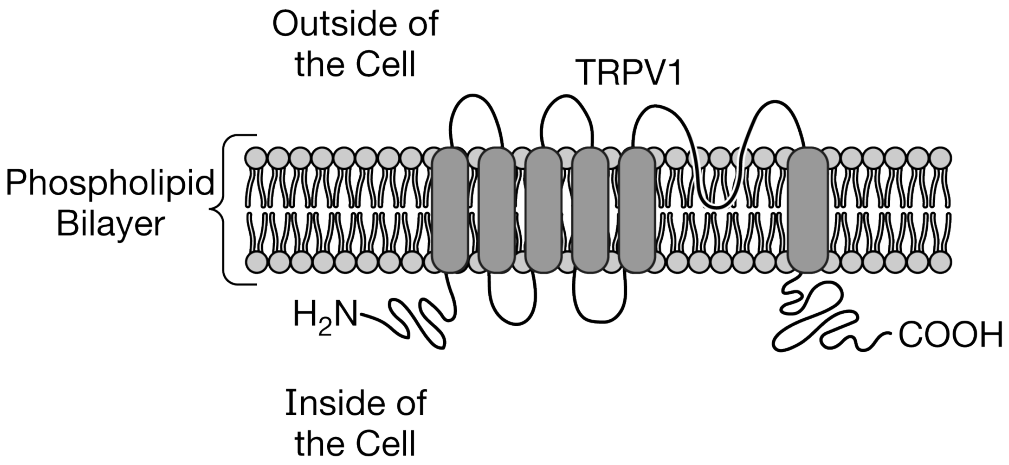


Figure 1. Structure of the TRPV1 protein channel in the membrane

TRPV1 is activated by several stimuli, including the binding of capsaicin, a chemical found in chili peppers. Capsaicin enters the cell by simple diffusion and then binds to one of the transmembrane domains of TRPV1, which opens the ion channel.

The opening of the TRPV1 channel allows Ca^{2+} ions to enter the cell, leading to the activation of the enzyme calcineurin. Calcineurin removes phosphate groups from the phosphorylated form of the transcription factor NFAT2, which is typically found in the cytoplasm. Once dephosphorylated, NFAT2 moves into the nucleus, where it blocks the transcription of another protein, ATF3. ATF3, when active, prevents the cell cycle regulatory protein pGoG from inducing apoptosis and blocking cell division (Figure 2).

FRQ

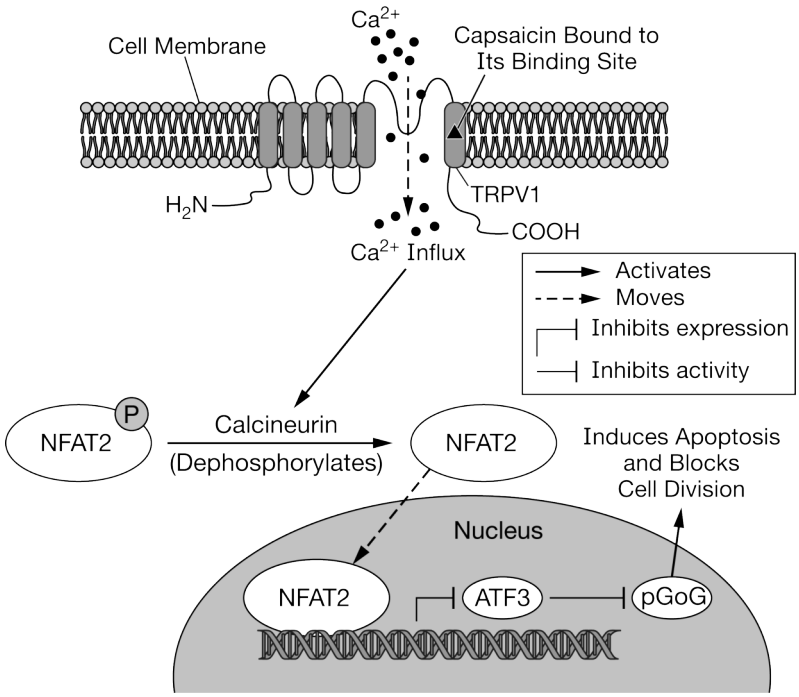


Figure 2. Capsaicin-activated TRPV1 signaling pathway in human cells

(a) **Describe** the characteristics of capsaicin that allow it to enter the cell before binding to a transmembrane domain of TRPV1 . **Describe** what prevents NFAT2 from inhibiting ATF3 expression in the absence of calcineurin.

(b) The pGoG protein is known to indirectly interfere with the interaction between cyclins and cyclin-dependent kinases. **Explain** why this prevents DNA replication from happening in a cell.

(c) Researchers discover a mutation that results in the constant expression of ATF3 regardless of the presence of NFAT2. **Predict** the effect of this mutation on the cell cycle of cells treated with capsaicin. Provide reasoning to **justify** your prediction.

(d) Researchers claim that capsaicin could be used as a potential treatment for melanoma, a type of skin cancer. **Support** the researchers' claim.

Part A

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1	2 ✓

The response includes both of the following criteria.

FRQ

- [ ] A description that capsaicin is relatively small and nonpolar
- [ ] A description that NFAT2 stays phosphorylated and remains in the cytoplasm/does not enter the nucleus

Part B

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1 ✓

The response includes an explanation that the indirect interference by pGoG would prevent entrance into S phase because the cyclins are not able to activate the CDKs to move the cell cycle forward.

Part C

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1	2 ✓

The response includes both of the following criteria.

- [ ] A prediction that the cell cycle will continue
- [ ] A justification that (since ATF3 will always be expressed) ATF3 will always block activity of pGoG, and pGoG will be unable to block the cell cycle and/or induce apoptosis

Part D

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1 ✓

The response includes support for the claim by stating that since cancer is characterized by excessive cell division, capsaicin will be an effective treatment because it leads to the activation of pGoG, which decreases cell division and/or induces apoptosis.

FRQ

13. Read each question carefully. Write your response in the space provided for each part of each question. Answers must be written out in paragraph form. Outlines, bulleted lists, or diagrams alone are not acceptable and will not be scored.

In plants, photosynthetic guard cells on the surface of leaves form pores called stomata (a single pore is called a stoma). Stomata help prevent water loss and control gas exchange with the environment to maintain plant homeostasis. The turgor pressure of the guard cells on either side of the stomata determines whether the stomata are open or closed based on the transport of certain ions into and out of the guard cells (Figure 1). A proton pump in the membrane of the guard cells pumps protons into the extracellular space. This makes the interior of the cell more negatively charged than the extracellular space.

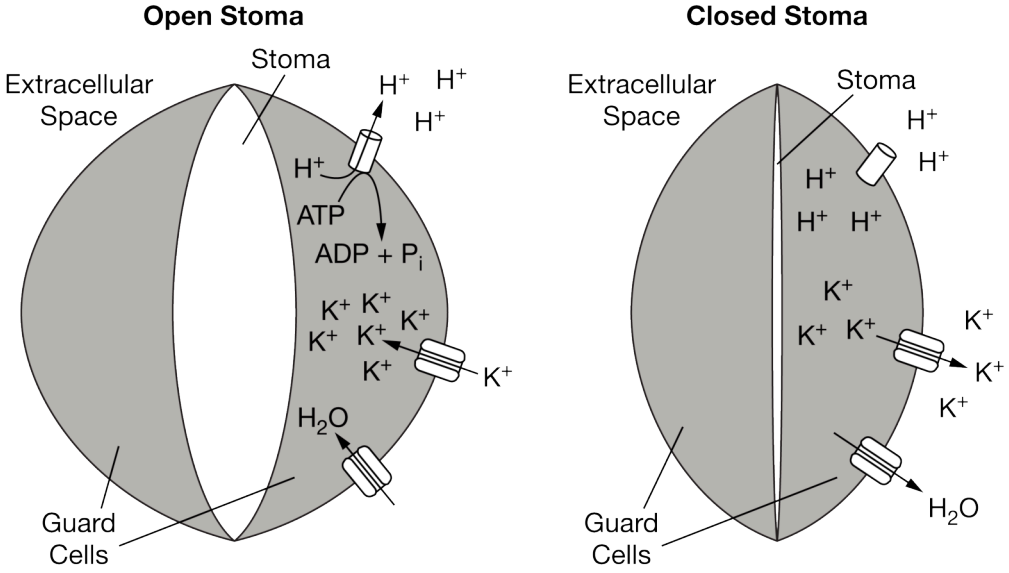


Figure 1. Models of open and closed stomata and the transport proteins in the guard cell membranes

Scientists interested in understanding drought tolerance in plants discovered a mutant variety of a plant with increased cell division, increased water content, and reduced density of leaf stomata compared with the wild type. Analysis of the mutant plant showed a mutation in the gene encoding a cyclin-dependent kinase (CDK). This mutation prevents CDK-cyclin interaction that normally results in the phosphorylation of RNA polymerase (Figure 2).

FRQ

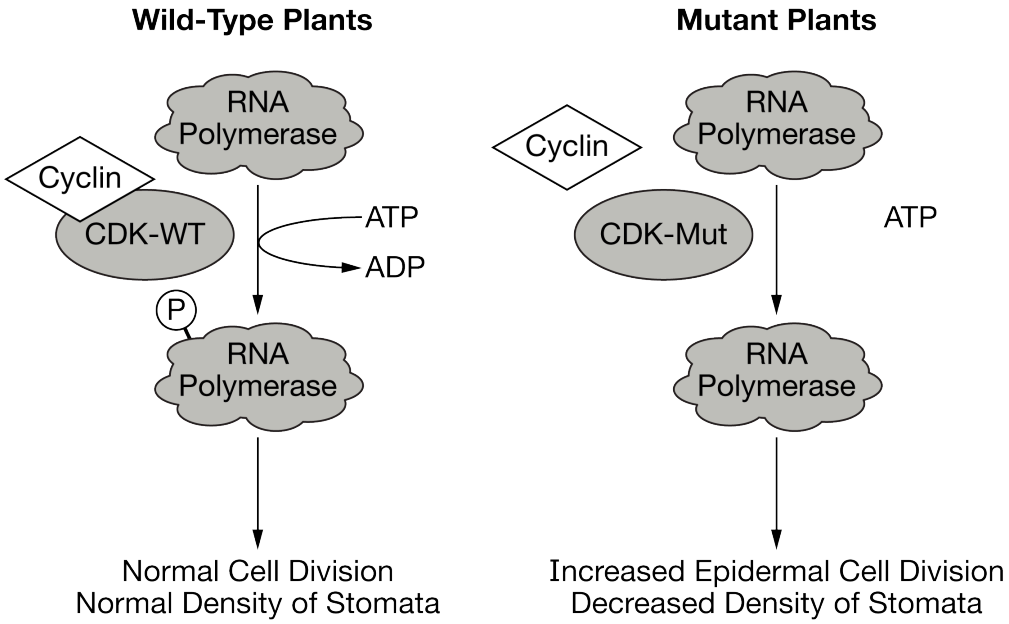


Figure 2. In wild-type plants, the wild-type CDK protein ($\mathrm{CDK\text{-}WT}$) allows interaction with a cyclin and leads to the phosphorylation of RNA polymerase. In mutant plants, the mutant CDK protein ($\mathrm{CDK\text{-}Mut}$) prevents interaction with cyclin and the phosphorylation of RNA polymerase.

(a) **Describe** the movement of K^{+} ions into the guard cell based on the model in Figure 1. Scientists crossed plants that were homozygous for the cyclin dependent kinase-wild type ($CDK\text{-}WT$) allele with plants that were homozygous for the cyclin dependent kinase-mutant ($CDK\text{-}Mut$) allele. The F_1 offspring of this cross were then crossed, producing F_2 offspring. **Explain** how the ratio of phenotypes in the F_2 generation would demonstrate that the $CDK\text{-}Mut$ allele is recessive.

(b) **Explain** the relationship between the turgor pressure of the guard cell and the pH of the extracellular space.

(c) In another strain of the same species of plant, a deletion has removed a portion of the promoter region of the gene encoding a polypeptide of the proton pump shown in Figure 1. **Predict** the effect of this mutation on the expression of the proton pump. Provide reasoning to **justify** your prediction.

(d) Researchers claim that the $CDK\text{-}Mut$ allele has a positive effect on plant survival in drought conditions. Use the information provided to **support** the researchers' claim.

Part A

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1	2 ✓

The response includes both of the following criteria.

FRQ

- [ ] An acceptable description. Acceptable descriptions include the following.
  · K^+ ions move against their concentration gradient through a transmembrane channel.
  · K^+ ions move as a result of the membrane potential established by the pumping of the protons.
- [ ] An explanation that there will be a phenotypic ratio of 3 wild-type : 1 mutant/25% mutant in the F_2 generation, because the F_1 parents are heterozygous

Part B

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1 ✓

The response includes an acceptable explanation. Acceptable explanations include the following.

· As the pH of the extracellular space decreases, turgor pressure increases because protons are being pumped into the extracellular space.

· As protons are pumped into the extracellular space, the guard cells fill up with water, resulting in a lower extracellular pH and a higher turgor pressure.

· There is an inverse relationship because as pH decreases in the extracellular space, turgor pressure increases in guard cells.

Part C

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1	2 ✓

The response includes both of the following criteria.

- [ ] A prediction that there will be a decrease/no expression (of the proton pump)
- [ ] A justification that transcription factors and RNA polymerase must bind to the promoter to initiate transcription

Part D

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1 ✓

FRQ

The response includes support for the claim by stating that, based on the information provided, the reduction in density of stomata in the mutant plants reduces water loss from the plants.

14. Read each question carefully. Write your response in the space provided for each part of each question. Answers must be written out in paragraph form. Outlines, bulleted lists, or diagrams alone are not acceptable and will not be scored.

Aquaporin channel proteins are found in the plasma membrane of various cell types. Scientists studied the production of aquaporins in the keratinocytes (skin cells) of healthy individuals and found that these proteins play a role in hydration of the skin. Some individuals have a skin disorder that causes the absorption of excess water by the skin of their hands and feet when they are soaked in water. Scientists determined that the keratinocytes of these individuals are characterized by the presence of abnormal aquaporins. Scientists studied a family in which some individuals have this skin disorder and constructed a pedigree (Figure 1).

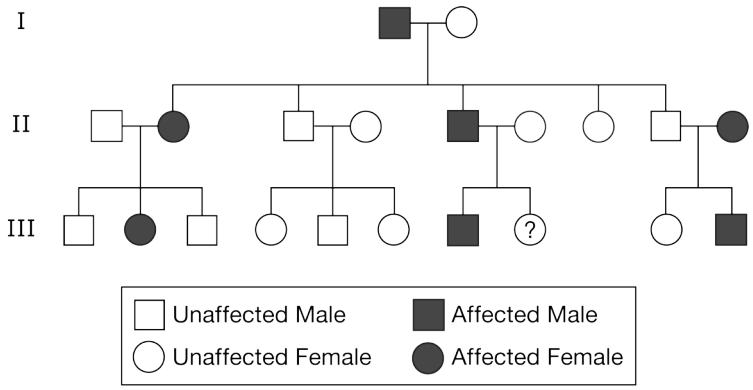


Figure 1. Pedigree of a human family in which some members have a skin disorder. An individual with an unknown phenotype is marked by a question mark.

(a) Based on the pedigree in Figure 1, **predict** the probability that the individual marked by the "?" will have the skin disorder. **Justify** your prediction.

(b) **Describe** the characteristics of the interior of the aquaporin channel protein that allow water to flow through AND exclude glucose. If isolated keratinocytes are placed in a hypotonic solution for 24 hours, **explain** the net direction of water movement AND why the rate of water movement slows down over time.

(c) **Explain** how specific membrane-bound organelles of the keratinocytes are used to produce, transport, and modify aquaporin polypeptides from mRNA.

(d) A scientist claims that the presence of aquaporins in chloroplast membranes supports the endosymbiotic theory. **Support** the claim with evidence from biological concepts.

Part A

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

FRQ

0	1	2 ✓

The response includes both of the following criteria.

- ☐ A prediction that the probability that the individual marked by the "?" will have the skin disorder is 0.5 OR $1/2$ OR 50%
- ☐ An acceptable justification. Acceptable justifications include the following.
  · Based on the autosomal dominant inheritance pattern, the mother of "?" must be homozygous recessive, and the father must be heterozygous because his mother is unaffected.
  · Based on the autosomal recessive inheritance pattern, the father of "?" must be homozygous recessive, and the mother must be heterozygous because the sibling is affected.

Part B

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1	2 ✓

The response includes both of the following criteria.

- ☐ A description that the interior of the aquaporin channel is hydrophilic/polar and is large enough for water to pass through but too small for glucose to do so
- ☐ An explanation that because of the greater solute concentration inside the keratinocyte, the water will move quickly into the keratinocyte, and then the rate will decrease as equilibrium is reached (if the cells do not lyse)

Part C

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1 ✓

The response includes an explanation that the aquaporin polypeptide is synthesized by the rough endoplasmic reticulum, transported by the (rough) endoplasmic reticulum/a vesicle, and then modified by the rough endoplasmic reticulum/Golgi complex.

Part D

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

FRQ

0	1 ✓

The response includes supporting evidence that chloroplasts evolved when an ancestral cell engulfed a (free-living prokaryotic) cell. Aquaporins in chloroplasts are likely derived from aquaporins in (the plasma membrane of) the engulfed cell.

FRQ

15. Read each question carefully. Write your response in the space provided for each part of each question. Answers must be written out in paragraph form. Outlines, bulleted lists, or diagrams alone are not acceptable and will not be scored.

Once thought of as only a toxic by-product of cellular metabolism, the chemical hydrogen peroxide (H_2O_2) is now known to play an integral role in maintaining cell homeostasis. Recent research suggests that H_2O_2 is one of a small group of molecules that can be transported across the plasma membrane through aquaporins, where it can affect cell signaling. To investigate the transport of H_2O_2 through aquaporins, researchers made four different groups of liposomes (spherical vesicles surrounded by a phospholipid bilayer). One group of liposomes was constructed as a simple vesicle of phospholipids (group A). Into the phospholipid bilayer of the three other groups of liposomes, researchers embedded either a type of human aquaporin (group B) or one of two types of plant aquaporin (groups C and D). The rate at which H_2O_2 was transported into the liposomes of each group was then determined (Table 1).

TABLE 1. RATE OF HYDROGEN PEROXIDE TRANSPORT INTO LIPOSOMES

Group	Type of Aquaporin Embedded in Liposome Phospholipid Bilayer	Average Rate of H_2O_2 Transport ($\mu L/s \pm 2SE_{\bar{x}}$)
A	None	0.9 ± 0.1
B	Human	1.4 ± 0.1
C	Plant I	1.7 ± 0.1
D	Plant II	1.8 ± 0.1

Figure 1 shows a simplified model of the relationship between H_2O_2 concentration and an intracellular signaling pathway. When the concentration of H_2O_2 is low, the enzyme Tpx 1 is active, leading to the activation of the transcription factor Pap 1 in a reaction that consumes H_2O_2. Once activated, Pap 1 moves to the nucleus, where it can activate transcription of a group of Pap 1-regulated genes.

As the concentration of H_2O_2 starts to increase, Tpx 1 becomes inactivated, allowing for even more H_2O_2 to accumulate, because it is no longer consumed in the activation of Pap 1. The accumulation of H_2O_2 leads to the activation of the enzyme Sty 1, which then activates transcription factor Atf 1. Atf 1 moves to the nucleus and activates transcription of a group of Atf 1-regulated genes.

FRQ

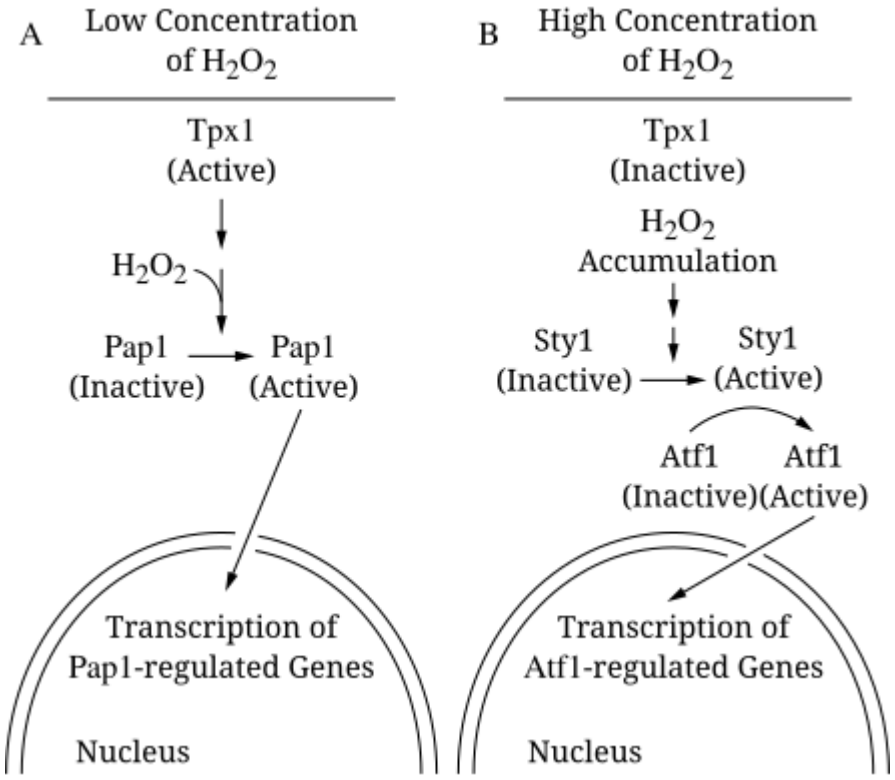
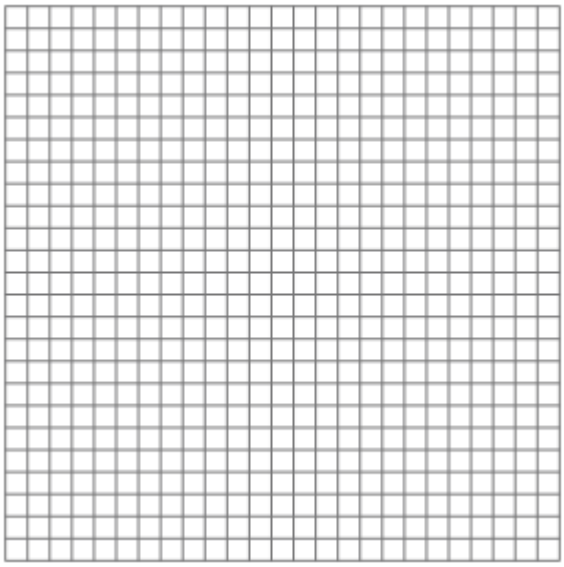


Figure 1. Simplified model of a signaling pathway involving H_2O_2 at both low (A) and high (B) intracellular concentrations of H_2O_2

(a) **Describe** the advantage to a cell of having aquaporins when the substances that pass directly through aquaporins can also pass through the phospholipid bilayer.

(b) Using the template in the space provided for your response, **construct** an appropriately labeled graph that represents the data in Table 1. **Determine** the type(s) of aquaporins that transport(s) H_2O_2 at a rate that is different from the rate at which the plant type II aquaporins transport H_2O_2.

FRQ



(c) Based on the data in Table 1, **identify** the rate in $\mu L/s$ at which H_2O_2 can be transported by simple diffusion through the phospholipid bilayer. **Calculate** the difference between the average microliters of H_2O_2 transported into group A liposomes in five minutes and the average microliters of H_2O_2 transported into group C liposomes in five minutes.

(d) An enzyme called $Srx\ 1$ is required for the activation of $Tpx\ 1$. Researchers identified a mutation that results in a nonfunctional form of $Srx\ 1$. Based on the model in Figure 1, **predict** the effect of this nonfunctional $Srx\ 1$ on the expression of $Pap\ 1$-regulated genes AND $Atf\ 1$-regulated genes. Provide reasoning to **justify** your prediction.

Part A

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1 ✓

The response includes a description that aquaporins are able to transport these substances at a faster rate/in large quantities.

Part B

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1	2	3 ✓

FRQ

The response includes three of the following criteria.

- [ ] Appropriate labeling
- [ ] Data graphed on a bar/modified bar graph
- [ ] Accurately plotted data points and error bars
- [ ] An acceptable determination. Acceptable determinations include the following.
    - The human aquaporin/Group B
    - None and the human aquaporin/Groups A and B

Part C

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1	2 ✓

The response includes all of the following criteria.

- [ ] An acceptable identification of the rate. Acceptable identifications include the following.
    - $0.9\ (\pm\ 0.1)$
        - $0.8 - 0.1$
- [ ] A calculation that the difference in microliters transported equals 240

Part D

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1	2 ✓

The response includes all of the following criteria.

- [ ] An acceptable prediction. Acceptable predictions include the following.
    - Expression of $\mathrm{Pap1}$-regulated genes will not occur/will decrease, while expression of $\mathrm{Atf1}$ -regulated genes will stay the same/increase.
    - The expression levels of $\mathrm{Pap1}$ - and $\mathrm{Atf1}$-regulated genes will be similar to expression levels

under a high concentration of H_2O_2.

- A justification that (if Tpx1 cannot be activated) H_2O_2 will accumulate/Sty1 will be activated, resulting in the activation of Atf1 but not Pap1

FRQ

16. Read each question carefully. Write your response in the space provided for each part of each question. Answers must be written out in paragraph form. Outlines, bulleted lists, or diagrams alone are not acceptable and will not be scored.

Chemical Z blocks the cell cycle of T cells, cells that divide in order to respond to pathogens to prevent infectious diseases. Scientists activated T cells that were in cell cycle phase G_0 to make them reenter the cell cycle. The scientists treated the activated T cells with chemicals A, B, C, or Z. Chemicals A, B, and C are known to block cell cycle progression at certain phases of interphase (Figure 1).

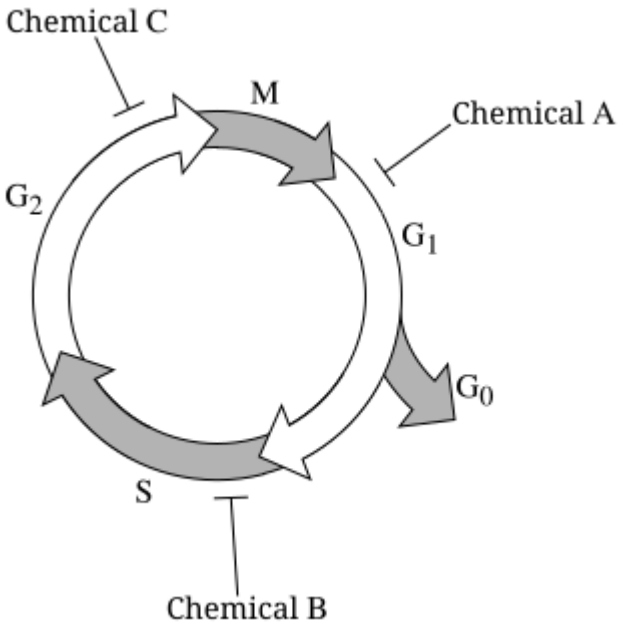


Figure 1. A model of the eukaryotic cell cycle showing the points (marked by bars, ⊥) at which chemicals A, B, and C block progression through interphase

By measuring the amount of DNA in the T cells in each sample, the scientists were able to determine the percent of cells in each phase of interphase (Figure 2).

FRQ

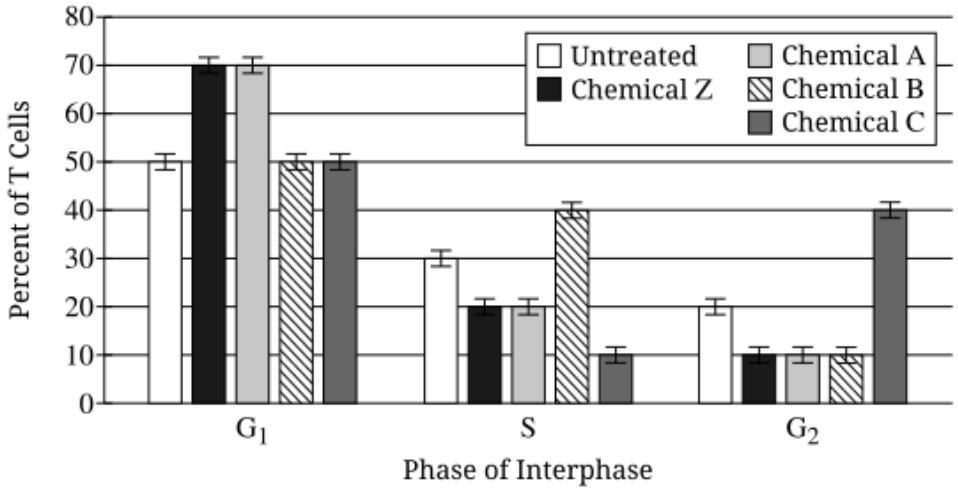


Figure 2. Percent of activated T cells in each phase of interphase in untreated cells and cells treated with chemicals A, B, C, and Z. Error bars represent $\pm 2\mathrm{SE}_{\bar{x}}$.

(a) Chemical Z is a large, polar molecule. **Describe** the process by which chemical Z moves across the plasma membrane without an input of cellular energy. **Explain** why the amount of DNA in a cell indicates whether the cell is in G_1, S, or G_2.

(b) **Identify** a dependent variable in the experiment. **Justify** the use of untreated cells in the experiment. **Justify** the use of both chemicals B and C in the experiment.

(c) Based on the data in Figure 2, **identify** the phase in interphase at which chemical Z treatment blocks the cell cycle. **Describe** the trends in the data that support your identification. Based on the mean values shown in Figure 2, **calculate** the difference in the number of cells in G_1 between the sample treated with chemical Z and the untreated sample if a total of 600 cells are counted for each sample.

(d) Cancerous T cells result from genetic mutations and lack normal cell cycle controls. Scientists tested whether chemical Z blocks the cell cycle of cancerous T cells. **State** the null hypothesis for their experiment. Individuals with cancerous T cells also have normal T cells. Scientists find that chemical Z is an effective treatment against cancerous T cells. However, they claim that using chemical Z can have a negative effect on a patient's ability to fight off an infection. Provide reasoning to **justify** their claim.

Part A

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1	2 ✓

The response includes both of the following criteria.

FRQ

- An acceptable description. Acceptable descriptions include the following.
  · Chemical Z requires a transport protein and moves from a high to a low concentration.
  · Chemical Z moves by facilitated diffusion.
- An explanation that there will be twice as much DNA per cell in cells that are (at the end of S and) in G_2 than in cells that are in G_1 because DNA replication occurs during S phase

Part B

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1	2	3 ✓

The response includes all of the following criteria.

- An acceptable identification of a dependent variable in the experiment. Acceptable identifications include the following.
  · Percent of T cells (in each phase of the cell cycle/interphase)
  · Amount of DNA in the T cells (in each phase of the cell cycle/interphase)
- A justification that the use of untreated cells in the experiment provides a control to see the percent of cells in each phase of the cell cycle in the/effects of an absence of chemicals/chemical Z
- A justification that the use of both Chemicals B and C in the experiment shows the percent of cells in/effect on each phase when blocked by chemical B/in S and by chemical C /in G_2 in order to help interpret the data from cells treated with chemical Z

Part C

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1	2	3 ✓

The response includes all of the following criteria.

- An identification that the phase in interphase at which chemical treatment blocks the cell cycle is G_1
- An acceptable description. Acceptable descriptions include the following.
  · In comparison with untreated T cells, a greater percent of the chemical Z-treated cells is in G_1 .
  · Chemical A blocks the cell cycle in G_1, and chemical Z-treated cells have the same pattern as chemical A-treated cells.
- A calculation that the difference in the number of cells in G_1 between the sample treated with chemical and the untreated sample if a total of cells are counted for each sample is 120 (Treated: $600 \times 0.7 = 420$; untreated: $600 \times 0.5 = 300$; $420 - 300 = 120$)

Part D

FRQ

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1	2 ✓

The response includes both of the following criteria.

- ☐ An acceptable null hypothesis. Acceptable null hypotheses include the following.
  - · There will be no difference between the cell cycle of untreated cancerous T cells and the cell cycle of (cancerous) T cells treated with chemical Z.
  - · Chemical Z has no effect on the cell cycle of cancerous T cells.
- ☐ An acceptable justification that chemical Z can have a negative effect because chemical Z will prevent cell division/block the cell cycle of normal T cells (in response to an infection)

FRQ

17. Read each question carefully. Write your response in the space provided for each part of each question. Answers must be written out in paragraph form. Outlines, bulleted lists, or diagrams alone are not acceptable and will not be scored.

The red king crab, *Paralithodes camtschaticus*, inhabits shallow coastal waters in the northern Pacific Ocean that can vary in temperature from $1^{\circ}C$ to $13^{\circ}C$. The body temperature of the crab is typically similar to that of the surrounding water. In order to study the effect of temperature on crab metabolism, scientists purified the enzyme phosphofructokinase (PFK) from red king crabs. PFK catalyzes a reaction early in the glycolysis pathway (Figure 1).

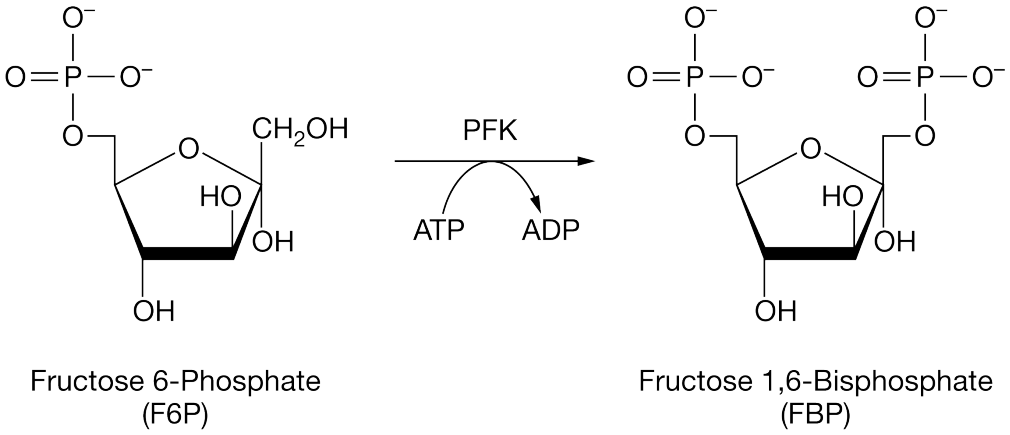


Figure 1. The conversion of fructose 6-phosphate ($F6P$) to fructose $1,6$-bisphosphate (FBP) by PFK

The scientists determined the rate of the PFK-catalyzed reaction using a constant amount of PFK and increasing concentrations of the substrate $F6P$ at $15^{\circ}C$ and $25^{\circ}C$. To determine whether the reaction rate is affected by the nucleotide AMP, the analyses were additionally performed in the presence of $0.5\ mM\ AMP$ (Figure 2).

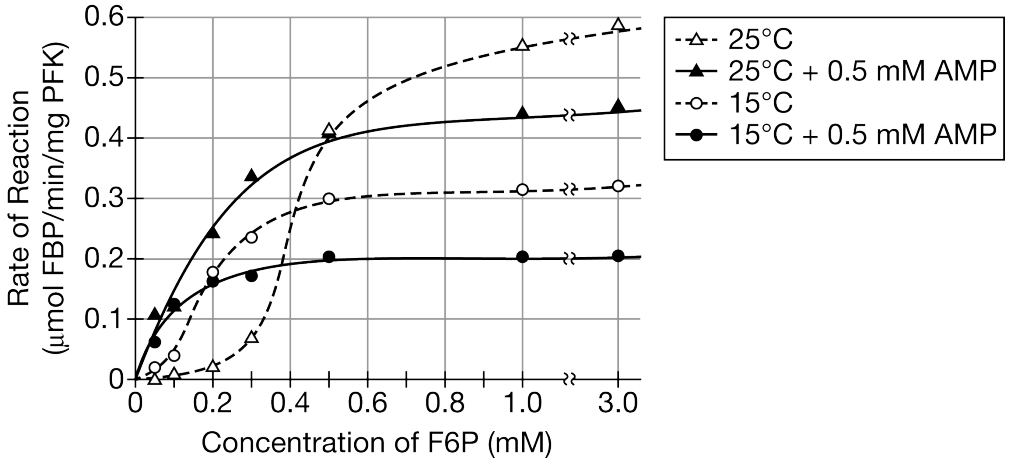


Figure 2. Effect of temperature and AMP on the rate of the reaction catalyzed by PFK

FRQ

In a second analysis of PFK activity at different temperatures, the scientists determined how the concentration of citric acid in the reaction mixture affects the rate of the reaction at 5°C, 15°C, and 25°C (Figure 3) in the presence of a constant amount of PFK. Citric acid is the first metabolic intermediate produced by the Krebs cycle. All reactions were performed in the presence of 1 mM F6P and 0.5 mM ATP.

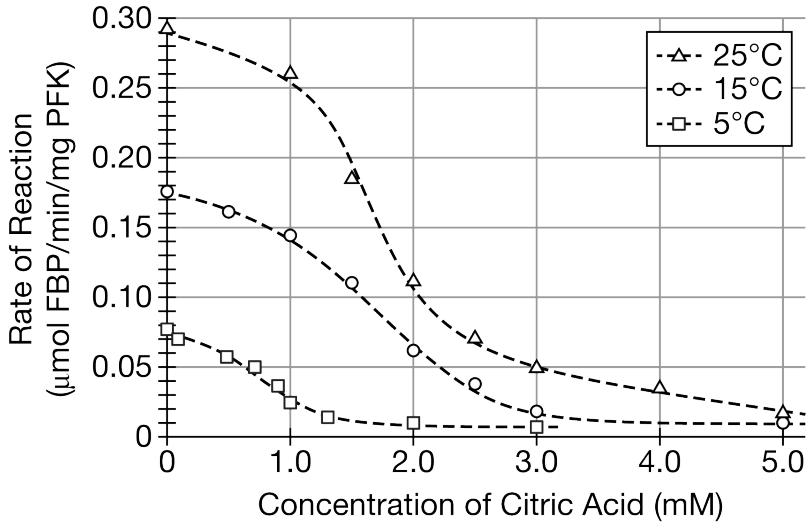


Figure 3. Effect of citric acid on the rate of the reaction catalyzed by PFK. All reactions were carried out in the presence of 1 mM F6P and 0.5 mM ATP.

(a) **Describe** the role of ATP in the reaction shown in Figure 1.

(b) **Identify** an independent variable in the experiment shown in Figure 2. **Justify** the use of increasing concentrations of the substrate F6P when the scientists performed the experiments shown in Figure 2. Based on Figure 2, **describe** the effect of adding AMP to the reactions carried out at 15°C. Based on Figure 3, **calculate** the fold increase in the reaction rate (how many times greater the reaction rate is) between the rate at 5°C and at 25°C in the presence of 2 mM citric acid.

(c) **State** the alternative hypothesis for the experiments shown in Figure 3. The scientists claim that, in the absence of AMP, PFK of red king crabs functions efficiently at the crabs' typical low environmental temperatures because the active site of the enzyme can bind substrate more tightly at the low temperatures than it does at higher temperatures. **Support** the scientists' claim based on the data provided. **Predict** the effect on the rate of the reaction catalyzed by PFK in the experiment shown in Figure 2 if increasing concentrations of FBP are added to the reaction.

(d) When crustaceans such as crabs are kept in seawater with a lower-than-normal pH, ATP-requiring pumps that maintain ion gradients in the animals' bodies significantly increase their activity in comparison with normal levels. **Explain** how the increased use of ATP by these pumps will most likely affect the ability of the animals to carry out other metabolic reactions such as that catalyzed by PFK. Hemolymph is the fluid in the red king crab's body that surrounds its cells and is typically slightly hypertonic to the seawater in which the crabs live. When adult crabs were removed from their native environment and placed into seawater solutions with gradually decreasing salt concentrations, their body weight increased. **Explain** how the crabs gained weight during this procedure.

Part A

FRQ

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1 ✓

The response includes a description that ATP provides both the phosphate group and the energy required to convert $\mathrm{F6P}$ to FBP.

Part B

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1	2	3	4 ✓

The response includes all of the following criteria.

- ☐ An acceptable identification of an independent variable. Acceptable identifications include the following.
  - · Temperature at which reaction was performed
  - · Concentration of $\mathrm{F6P}$ in the reaction mixture
  - · Presence or absence of AMP in the reaction mixture
- ☐ A justification that the scientists increased substrate concentration to determine the substrate concentration at which the reaction rate would reach a maximum/level off
- ☐ A description that at any given $\mathrm{F6P}$ concentration below $0.15\ \mathrm{mM}$ (accept $0.1\ \mathrm{mM}\ -\ 0.2\ \mathrm{mM}$), the reaction rate is more rapid in the presence of AMP, whereas at any given $\mathrm{F6P}$ concentration above $0.15\ \mathrm{mM}$, the reaction rate is slower in the presence of AMP
- ☐ The fold increase in the reaction rate is calculated to be 11.

Part C

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1	2	3 ✓

The response includes all of the following criteria.

- ☐ An acceptable stated alternative hypothesis. Acceptable statements include the following.
  - · Increasing concentration of citric acid causes a change (increase/decrease) in reaction rate.
  - · The reaction rate changes (increases/decreases) at different temperatures.
- ☐ Support for the claim based on data that at low concentrations of substrate, the reaction rate is greater at $15^{\circ}\mathrm{C}$ (than at $25^{\circ}\mathrm{C}$ in the absence of AMP)

FRQ

- [ ] A prediction that the reaction rate will decrease

Part D

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1	2 ✓

The response includes both of the following criteria.

- [ ] An explanation that the rate of other metabolic reactions is likely to decrease because the cells have only so much ATP to use
- [ ] An explanation that (more) water moved from the dilute seawater into the hypertonic hemolymph of the crabs (than normally moves into the hemolymph)

FRQ

18. Read each question carefully. Write your response in the space provided for each part of each question. Answers must be written out in paragraph form. Outlines, bulleted lists, or diagrams alone are not acceptable and will not be scored.

ATCase is a complex enzyme involved in the production of the nucleotide CTP. ATCase is composed of many subunits that are grouped into two functional domains. The catalytic domain of ATCase (C in Figure 1) contains an active site which binds two substrates (S1 and S2) and catalyzes a chemical reaction, resulting in the formation of two products (P1 and P2). P2 then acts as a substrate in the pathway leading to CTP production. The regulatory domain of ATCase (R in Figure 1) contains an allosteric site capable of binding with the nucleotides CTP and ATP.

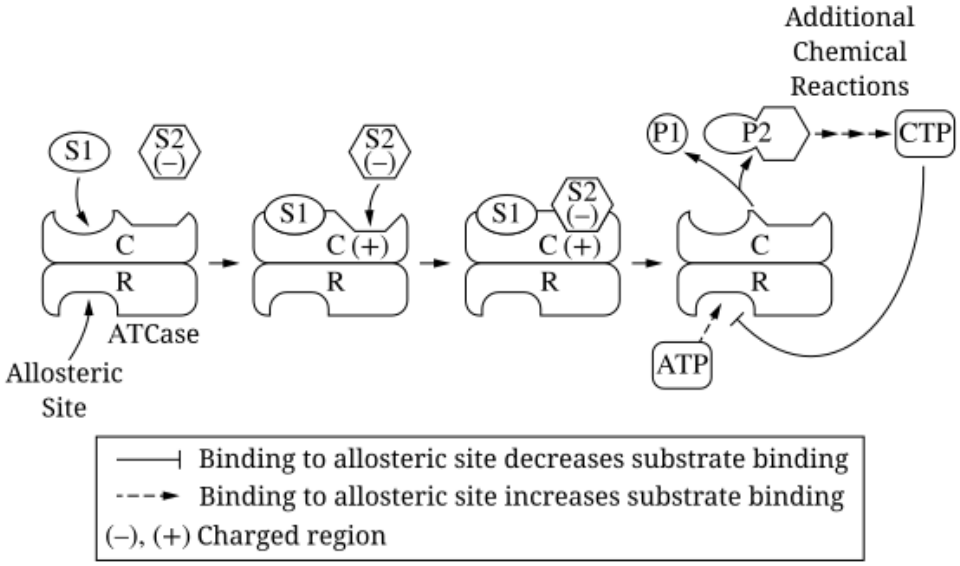


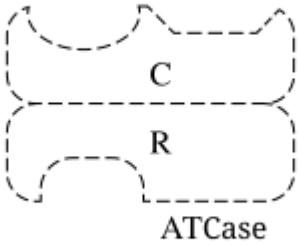
Figure 1. Simplified model of the regulation of an ATCase-catalyzed reaction

(a) **Describe** the effect that the binding of a noncompetitive inhibitor has on the structure of an enzyme.

(b) Based on Figure 1, **explain** how the binding of S1 increases the likelihood that S2 will bind.

(c) Researchers studying ATCase observed that, when ATCase is heated at 60°C for 4 minutes, it is still able to form P1 and P2, but the presence of regulatory molecules has no effect on substrate binding. Using the template in the space provided for your response, **represent** the effect of the increased temperature on ATCase structure by modifying the model of ATCase provided.

FRQ



(d) **Explain** how the concentration of CTP in a cell contributes to the regulation of CTP production.

Part A

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1 ✓

The response includes the description that the binding of a noncompetitive inhibitor changes the conformation/shape/structure of the enzyme/active site.

Part B

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1 ✓

The response includes an acceptable explanation. Acceptable explanations include the following.

- It changes the conformation/shape of the active site to better match the shape of S2.
- The active site becomes more positively charged, which attracts negatively charged S2.

Part C

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

FRQ

0	1 ✓

The response includes a representation that shows an unchanged active site and a modified allosteric site that is not compatible with the regulators.

Part D

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1 ✓

The response includes an acceptable explanation. Acceptable explanations include the following.

- The concentration of CTP determines whether ATP or CTP is more likely to bind to the allosteric site, which affects ATCase activity.
- Competition between ATP and CTP for the allosteric site of ATCase affects ATCase activity.
- When CTP levels are elevated, CTP binds to the allosteric site, decreasing the production of additional CTP (a negative feedback loop).
- When CTP levels are low, ATP binds to the allosteric site, increasing the production of CTP.

FRQ

19. Read each question carefully. Write your response in the space provided for each part of each question. Answers must be written out in paragraph form. Outlines, bulleted lists, or diagrams alone are not acceptable and will not be scored.

Glycogen is a complex polymer of glucose molecules involved in the storage of energy in many organisms. Depending on the signals received by a cell, different enzymes are activated, which will lead to either the synthesis or breakdown of glycogen. The product of glycogen breakdown, glucose 1-phosphate, can be modified into glucose 6-phosphate, an intermediate of glycolysis (Figure 1).

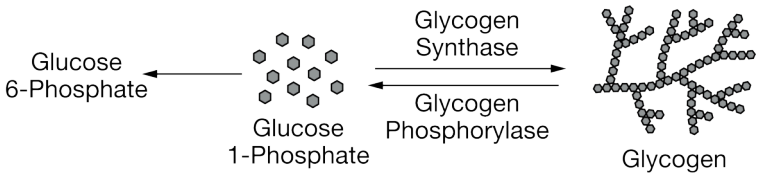
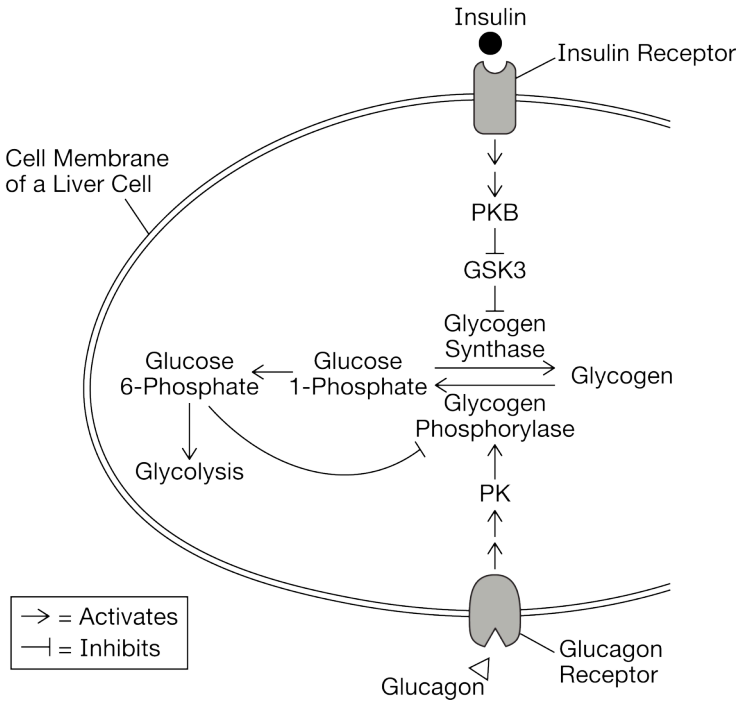


Figure 1. Glycogen synthesis and breakdown

In the liver, glycogen synthase and glycogen phosphorylase are regulated by kinases. These kinases transfer phosphate groups from ATP to amino acids in glycogen synthase and glycogen phosphorylase. Phosphorylase kinase B (PKB) is a kinase that regulates glycogen synthase kinase 3 (GSK3) activity. GSK3 regulates the activity of glycogen synthase. Phosphorylase kinase (PK) regulates the activity of glycogen phosphorylase. The activity of these kinases depends on the activation of signaling pathways in response to the binding of specific cell signals, including insulin and glucagon, to their receptors (Figure 2).



FRQ

Figure 2. A simplified model of signaling pathways in a liver cell that are activated by insulin and glucagon

(a) A cell has a mutation that results in the permanent activation of the glucagon receptor, even in the absence of glucagon. Based on the information in Figure 2, **predict** the effect of this mutation on the activity of glycogen phosphorylase. **Justify** your prediction.

(b) **Describe** the role of ATP in the regulation of glycogen synthase activity. **Explain** how the addition of a phosphate group to certain amino acids of glycogen synthase can result in the inhibition of glycogen synthase.

(c) **Explain** why an increase in glycogen phosphorylase activity might result in an increase in pyruvate production in the cell.

(d) Diabetes often results in the production of reduced levels of insulin. Researchers claim that a drug that binds to and blocks the active site of GSK3 might mimic the role of insulin in the pathway shown in Figure 2. Provide reasoning to **justify** the researchers' claim.

Part A

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1	2 ✓

The response includes both of the following criteria.

- ☐ An acceptable prediction. Acceptable predictions include the following.
  - · Glycogen phosphorylase will be active.
  - · Glycogen phosphorylase will have the same activity as normal.
- ☐ An acceptable justification. Acceptable justifications include the following.
  - · The activation of the glucagon receptor leads to the activation of glycogen phosphorylase, provided there is not enough glucose 6-phosphate to inhibit glycogen phosphorylase.
  - · Glucose 6-phosphate will regulate the activity of glycogen phosphorylase even if the glucagon receptor is permanently active.

Part B

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1	2 ✓

The response includes both of the following criteria.

FRQ

- [ ] A description that ATP provides the energy to form a (covalent) bond between a phosphate group and glycogen synthase
- [ ] An explanation that the addition of a phosphate group to glycogen synthase could alter the structure and/or charge of glycogen synthase such that the substrate can no longer interact with the active site/enzyme can no longer catalyze the reaction

Part C

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1 ✓

The response includes the explanation that an increase in glycogen phosphorylase activity can lead to an increase in glycolysis which produces pyruvate.

Part D

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1 ✓

The response includes an acceptable justification. Acceptable justifications include the following.

· Both insulin and this drug will prevent the binding of substrate to GSK3, effectively inhibiting the enzyme.

· The drug mimics the activity of PKB, which is activated by insulin.

FRQ

20. Read each question carefully. Write your response in the space provided for each part of each question. Answers must be written out in paragraph form. Outlines, bulleted lists, or diagrams alone are not acceptable and will not be scored.

Solid tumors are clusters of cancer cells and often contain blood vessels. When molecule B binds to the wild-type Brec protein in the plasma membrane of certain solid tumor cancer cells (Figure 1), the cancer cells express the membrane protein A and sometimes stimulate increased growth of blood vessels into the tumors.

Cells with a particular mutation in the *Brec* gene (Brec-MUT cells) have much increased expression levels of A and stimulate greater growth of blood vessels than do cancer cells with the wild-type Brec (Brec-WT cells); the cells with the mutant Brec can trigger intracellular signaling in the absence of B.

Researchers proposed that the signaling pathway modeled in Figure 1 is triggered by activation of the wild-type Brec and is associated with phosphorylation and activation of kinase D, expression of A, and the ability of the cancer cells to stimulate blood vessel growth.

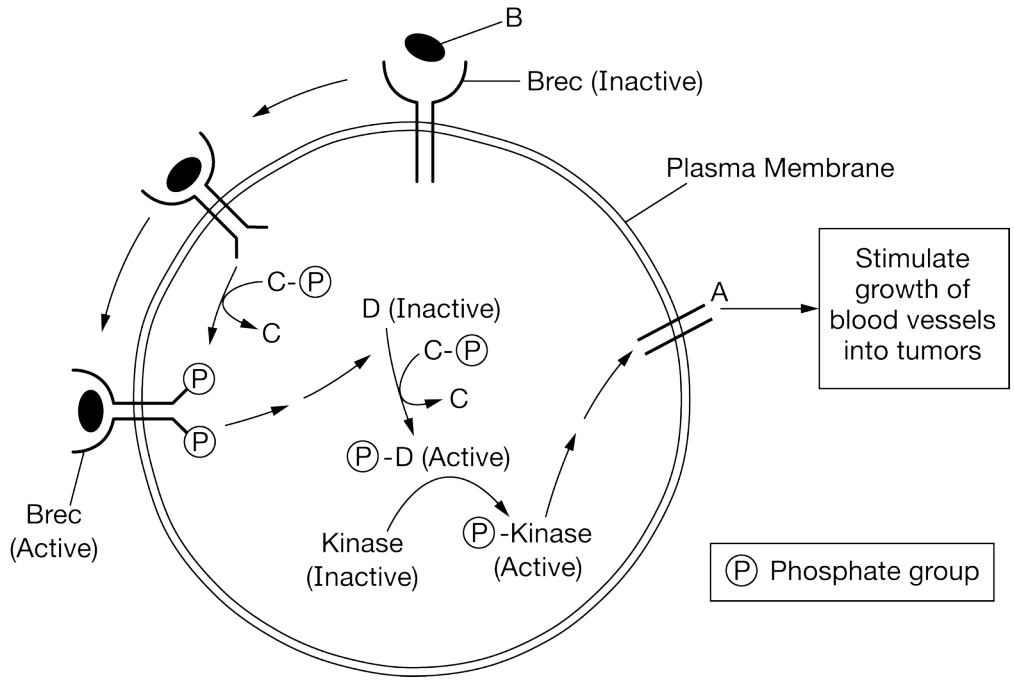


Figure 1. A simplified model of the normal signaling pathway hypothesized to play a role in certain cancer cells expressing A and stimulating blood vessel growth into solid tumors

(a) Based on the signaling model shown in Figure 1, **describe** the role of molecule B.

(b) **Explain** how the addition of a phosphate group to certain amino acids of Brec likely affects the tertiary structure and function of Brec. **Explain** how the addition of a small RNA could prevent the activation of kinase D.

(c) Based on the proposed signaling pathway, **predict** the relative amount of phosphorylated to unphosphorylated

FRQ

kinase D in the Brec-WT cells when the cells are grown in nutrient broth lacking B. **Justify** your prediction.

(d) Cells are treated with a compound that prevents the inactivation of kinase D. Researchers claim that Brec-WT cells treated with this compound will stimulate blood vessel-forming cell division to the same degree as Brec-MUT cells do in the absence of B. Based on the information provided, provide reasoning to **support** their claim.

Part A

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1 ✓

The response includes a description that molecule B is the ligand/binds to the receptor and activates the receptor/the signaling pathway.

Part B

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1	2 ✓

The response includes both of the following criteria.

- ☐ An explanation that the addition of a phosphate group could alter the structure and/or charge of Brec so that Brec can now interact with other signaling molecules in the cell
- ☐ An acceptable explanation. Acceptable explanations include the following.
  · The small RNA could inhibit translation of an mRNA encoding one of the proteins in the pathway leading to kinase D activation.
  · The small RNA could lead to the degradation of an mRNA encoding a protein in the pathway leading to kinase D activation.

Part C

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1	2 ✓

The response includes both of the following criteria.

FRQ

- [ ] A prediction that there will be more unphosphorylated than phosphorylated kinase D
- [ ] A justification that because the wild-type Brec is inactive in the absence of B, the receptor will not stimulate intracellular signaling, and kinase D will remain unphosphorylated

Part D

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1 ✓

The response includes support for the claim by stating that, based on the information provided, both cell types will now have continuously active kinase D, which leads to the expression of A, and the continuous expression of A will stimulate blood vessel-forming cell division by both cell types.

FRQ

1. Read each question carefully. Write your response in the space provided for each part of each question. Answers must be written out in paragraph form. Outlines, bulleted lists, or diagrams alone are not acceptable and will not be scored.

In response to an increase in the frequency of thunderstorms that produce damaging hail, some apple growers have started to cover their apple orchards with large nets. While these nets protect the apple trees from hail damage, the nets also decrease the amount of light that reaches the trees. In an attempt to reduce the impact of shading by the nets, researchers investigated the ability of different ground covers placed around the base of the trees to reflect light back to the apple trees. An apple orchard with genetically identical apple trees was divided in half. A reflective ground cover was applied to one half of the orchard, while grass was left as the ground cover for the other half. A sample of apples was taken from every tree. The percent of light reflected by the two ground covers was measured at three distances above the ground (Figure 1).

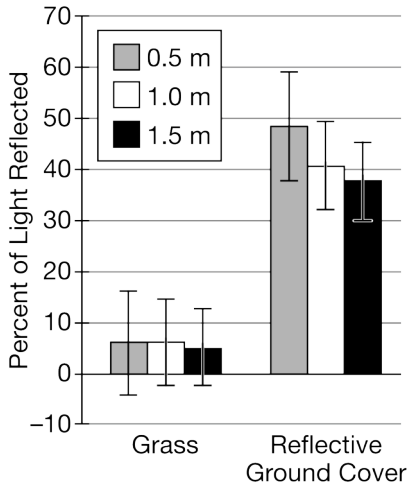


Figure 1. Percent of light reflected by ground covers, measured at three distances above the ground. Error bars equal $\pm 2\mathrm{SE}_{\bar{x}}$.

The amount of light that apple trees receive changes the expression of a certain transcription factor. This transcription factor affects the transcription rate of the gene encoding the enzyme phenylalanine ammonium-lyase (PAL). The PAL enzyme is required for the production of two types of pigments, anthocyanins and flavonoids, which contribute to the coloration of apples. Light also affects the activity of the PAL enzyme by a signal transduction pathway (Figure 2).

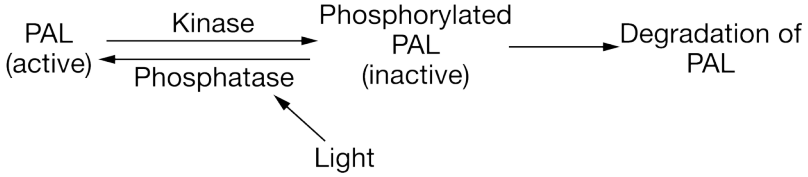


Figure 2. Simplified model of the signal transduction pathway leading to PAL activation and breakdown

FRQ

The researchers measured the average concentrations of anthocyanins and flavonoids in apples from trees grown with both types of ground cover (Table 1).

TABLE 1. CONCENTRATION OF ANTHOCYANINS AND FLAVONOIDS IN APPLES

Type of Pigment	Concentration of Pigment (mean $\pm 2\mathrm{SE}_{\bar{x}}$ in $\mathrm{nmol\ cm^{-2}}$) in Apples Grown with Grass Ground Cover	Concentration of Pigment (mean $\pm 2\mathrm{SE}_{\bar{x}}$ in $\mathrm{nmol\ cm^{-2}}$) in Apples Grown with Reflective Ground Cover
Anthocyanins	14.51 ± 1.9	24.10 ± 3.0
Flavonoids	29.24 ± 3.6	44.50 ± 5.7

(a) Using the information provided in Figure 2, **explain** the role of light in the activation of the PAL enzyme.

(b) **Identify** the variable in the researchers' experiment that is dependent on the percent of reflected light the apple trees receive. **Justify** the use of genetically identical apple trees as a controlled factor in the experiment. Using the data provided, **describe** the effect of the different ground covers on the percent of light reflected. **Calculate** the percent change in the average concentration of flavonoids in apples grown with the reflective ground cover compared with the apples grown with the grass ground cover.

(c) Researchers also study apple trees that have been genetically modified so that the DNA binding domain of the transcription factor that stimulates expression of the PAL enzyme has been deleted. The researchers repeat the ground cover experiment with the genetically modified apple trees. **Predict** the effect of the two ground covers on pigment concentrations in the genetically modified apples. Provide reasoning to **justify** your prediction.

(d) Based on the information provided in Figure 2, **predict** the effect of a constantly active kinase on flavonoid pigment production. In leaves, the PAL enzyme is also required for the production of other pigment molecules that are capable of absorbing UV and other radiation from the sun. **Explain** how the relationship between light and pigment production might serve as a protective mechanism for the cells of the apple tree. **Explain** how the PAL enzyme interacts with its substrate in the absence of light.

Part A

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1 ✓

FRQ

The response includes an explanation that the role of light in the activation of the PAL enzyme is to activate phosphatase which removes the phosphate group from the PAL enzyme, thereby activating the enzyme.

Part B

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1	2	3	4 ✓

The response includes all of the following criteria.

- ☐ An identification of the concentration of pigments (anthocyanins and flavonoids) in the apples as the variable dependent on the percent of reflected light the apple trees receive
- ☐ A justification that, because the apple trees are genetically identical, any differences in the concentration of pigments can be attributed to the differences in the experimental treatment (type of ground cover/ amount of reflected light)
- ☐ A description that there is (statistically) more light reflected from the reflective ground cover
- ☐ The percent change in the average concentration of flavonoids calculated to be 52.19%.

Part C

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1	2 ✓

The response includes both of the following criteria.

- ☐ An acceptable prediction. Acceptable predictions include the following.
  · These genetically modified trees will have low levels of/no pigments regardless of the type of ground cover.
  · The type of ground cover will have no effect on pigment concentrations in these genetically modified trees.
- ☐ An acceptable justification. Acceptable justifications include the following.
  · If the transcription factor cannot bind to (the regulatory region of) the DNA encoding the PAL enzyme, the gene cannot be transcribed (to mRNA), and no PAL enzyme will be produced.
  · If PAL cannot be produced, the amount of light (reflected by the ground cover) will not affect the production or activation of PAL.

Part D

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

FRQ

0	1	2	3 ✓

The response includes all of the following criteria.

- [ ] A prediction that flavonoid production will decrease/there will be no pigment production
- [ ] An explanation that, by increasing the amount of pigment molecules in the leaves in response to an increase in sunlight, plants reduce radiation damage to DNA
- [ ] An explanation that, in the absence of light, the PAL enzyme and substrate would not interact because the phosphate group would still be attached, which would lead to the degradation of PAL/an inactive shape of PAL

FRQ

2. Read each question carefully. Write your response in the space provided for each part of each question. Answers must be written out in paragraph form. Outlines, bulleted lists, or diagrams alone are not acceptable and will not be scored.

In plants, a critical step in the pathway leading to the synthesis of the amino acid tryptophan includes the synthesis of an intermediate molecule, 5-enolpyruvylshikimic acid-3-phosphate (EPSP). The reactants in this step are shikimic acid-3-phosphate (S3P) and phosphoenolpyruvate (PEP), and the reaction is catalyzed by the enzyme EPSP synthase (Figure 1). Compound X, a chemical used to kill weeds, can affect EPSP synthesis by interacting with EPSP synthase.

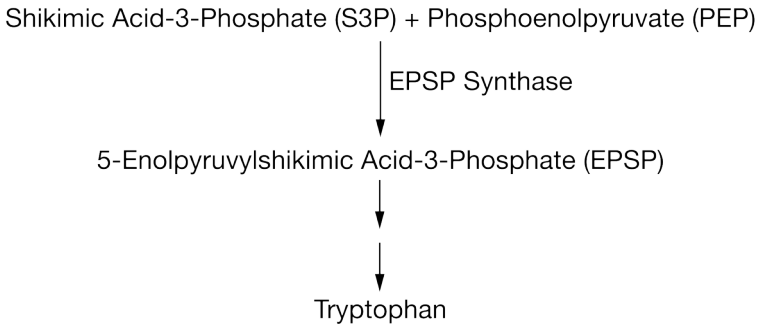


Figure 1. Reaction between shikimic acid-3-phosphate (S3P) and phosphoenolpyruvate (PEP), catalyzed by the enzyme EPSP synthase. Compound X can affect EPSP synthesis by interacting with EPSP synthase.

Scientists investigated the effect that Compound X has on EPSP synthase activity by performing an experiment to examine the relationship between EPSP synthase activity, PEP concentration, and Compound X concentration (Figure 2).

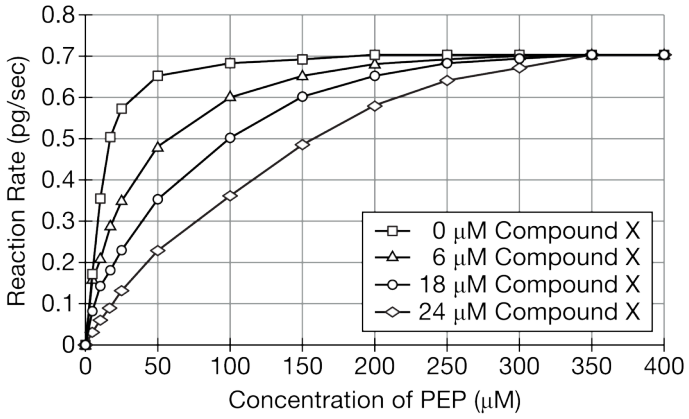


Figure 2. Rate of the reaction catalyzed by a specific amount of EPSP synthase with a specific amount of S3P, increasing concentrations of PEP, and four concentrations of Compound X

Some populations of the plant *Lolium rigidum* have evolved resistance to Compound X. To investigate the

inheritance of this resistance, researchers crossed true-breeding plants from a Compound X-susceptible (S) population with true-breeding plants from a Compound X-resistant (R) population. In half of these crosses, pollen from S plants was used to fertilize ovules in R plants. In the other half, pollen from R plants was used to fertilize ovules in S plants. Survival of parental and F_1 seedlings, differentiated by the maternal phenotype, in response to varying concentrations of Compound X is shown in Figure 3.

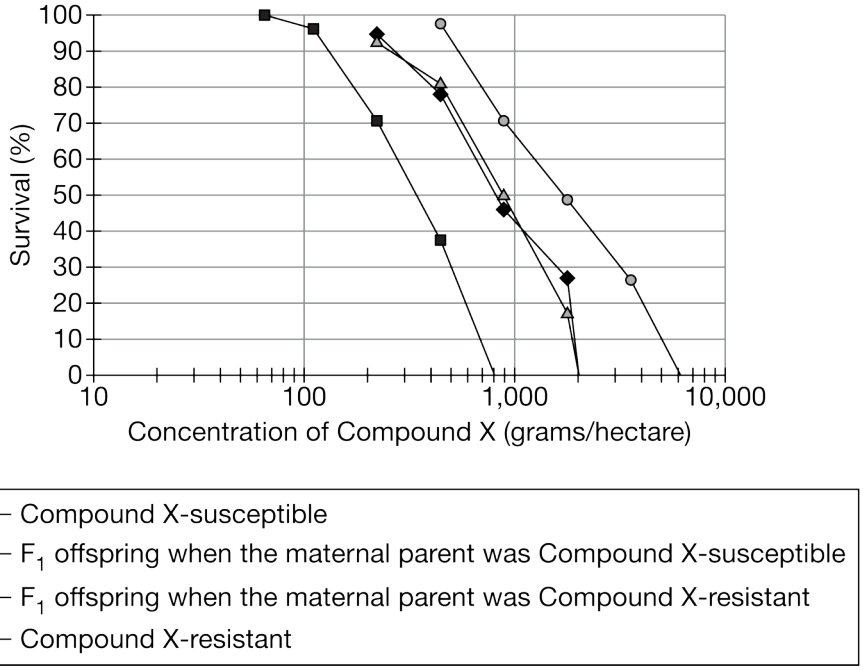


Figure 3. Percent survival of four populations of *L. rigidum* exposed to various concentrations of Compound X shown in a log scale. Data are an average from one experiment with four replicates conducted on seedlings growing in pots. $1\ \text{Hectare} = 10{,}000\ \text{m}^2$

(a) Compound X competes with PEP for binding to the active site of EPSP synthase. **Explain** how increasing the concentration of PEP can reverse the inhibitory effect of Compound X on the reaction catalyzed by EPSP synthase.

(b) **Identify** a control group in the experiment shown in Figure 2. **Identify** the dependent variable in the experiment shown in Figure 3. Using the data provided in Figure 3, **identify** the concentration of Compound X, in grams/hectare required to kill 60% of F_1 offspring when the maternal parent was Compound X-susceptible. Using the graph in Figure 2, **calculate** the mass, in pg, of EPSP that would be produced by this reaction in one minute at a PEP concentration of $200\ \mu\text{M}$ and a Compound X concentration of $0\ \mu\text{M}$.

(c) Researchers claim that a single gene is responsible for the Compound X-resistance phenotype. In a follow-up experiment with *L. rigidum*, the researchers crossed an F_1 female with a male from the original susceptible (S) population. The resulting offspring were treated with $300\ \text{grams/hectare}$ of Compound X. Based on the data presented in Figure 3, **predict** ONE genotype of the resulting offspring and the percent survival of the offspring with this genotype after treatment with Compound X. A student claims that the allele for the Compound X -resistance gene exhibits incomplete dominance over the Compound X-susceptible allele. Based on data provided in Figure 3, provide reasoning to **support** this claim. Researchers discovered a variant of the Compound X -resistance gene in *L. rigidum* in which there was a two base pair deletion early in the coding region of both copies

FRQ

of the gene. **Predict** the most likely effect on the phenotype of the plant.

(d) Researchers have engineered Compound X-resistant corn by inserting a Compound X-resistance gene from a prokaryote into the corn genome. **Explain** how scientists could ensure that the gene would be reliably transmitted to the offspring by either the pollen or the ovule. **Explain** why the *EPSP* synthase gene and the pathway it regulates can exist in both prokaryotes and corn plants.

Part A

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1 ✓

The response includes an explanation that as the concentration of PEP/ratio of PEP to Compound X increases, the chance of PEP instead of Compound X binding to the active site of EPSP synthase increases.

Part B

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1	2	3	4 ✓

The response includes all of the following criteria.

- ☐ An acceptable identification of a control group. Acceptable identifications include the following.
  - 0 µM Compound X
  - 0 µM PEP
- ☐ An identification that the dependent variable is percent survival
- ☐ An identification that the concentration of Compound X in grams / hectare, required to kill 60 % of F_1 offspring when the maternal parent was Compound X-susceptible is 1100 (grams / hectare); NOTE: acceptable student responses can be between 1050 and 1200.
- ☐ The mass, in pg, of EPSP that would be produced by this reaction in one minute at a PEP concentration of 200μM and a Compound X concentration of 0 µM is calculated to be 42 pg.

Part C

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

FRQ

0	1	2	3 ✓

The response includes all of the following criteria.

- ☐ An acceptable prediction of the percent survival of one genotype of the resulting offspring. Acceptable predictions include the following.
  - 87% survival RS; NOTE: acceptable student responses can be between 85% and 90%.
  - 55% survival SS (susceptible); NOTE: acceptable student responses can be between 52% and 58%.
- ☐ The reasoning that heterozygous F_1 offspring are more resistant than the original susceptible population is but not as resistant as is the original resistant population
- ☐ An acceptable prediction of the most likely effect on the genotype of the plant. Acceptable predictions include the following.
  - No Compound X resistance
  - Compound X-susceptibility
  - Death following exposure to Compound X

Part D

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1	2 ✓

The response includes both of the following criteria.

- ☐ An explanation that the gene would have to be placed in a nuclear chromosome because nuclear chromosomes are transmitted in both ovules and pollen
- ☐ An acceptable explanation. Acceptable explanations include the following.
  - Corn and prokaryotes share a common ancestor.
  - Core metabolic pathways are conserved in multiple domains of life.

FRQ

3. Read each question carefully. Write your response in the space provided for each part of each question. Answers must be written out in paragraph form. Outlines, bulleted lists, or diagrams alone are not acceptable and will not be scored.

Saccharomyces cerevisiae, also known as baker's yeast, is a single-celled eukaryote that can make ATP both aerobically and anaerobically. To obtain a readily accessible form of chemical energy for cellular activities, many strains of yeast convert glucose into ethyl alcohol and carbon dioxide.

Researchers conducted an experiment to determine the effect of temperature on the rate of glucose conversion to ethyl alcohol and carbon dioxide. They added an equal mass of yeast cells to equal volumes of a glucose solution in a series of sealed test tubes. Groups of tubes were incubated at temperatures ranging from $28^{\circ}\mathrm{C}$ to $40^{\circ}\mathrm{C}$, the optimal range for yeast growth. The researchers measured carbon dioxide production in the tubes.

(a) **Describe** the importance of fermentation to eukaryotic cells such as yeast.

(b) **Identify** the independent variable in the experiment.

(c) **Predict** the relative rates of glucose breakdown at $4^{\circ}\mathrm{C}$ AND at $56^{\circ}\mathrm{C}$ compared with the rate at $37^{\circ}\mathrm{C}$.

(d) **Provide reasoning** to support your prediction in part (c).

Part A

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1 ✓

The response includes a description that the importance of fermentation to cells such as yeast is that it (fermentation) enables the yeast cells to regenerate NAD^{+}/make ATP/obtain energy without oxygen.

Part B

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1 ✓

The response identifies the independent variable as temperature.

Part C

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

FRQ

0	1 ✓

The response predicts that the rates [of glucose breakdown] will be lower at both $4^{\circ}C$ and $56^{\circ}C$ (than at $37^{\circ}C$).

Part D

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1 ✓

The response supports the prediction in part (c) because it provides reasoning that at $4^{\circ}C$, molecules move more slowly, so there are fewer collisions between enzymes and substrates. At $56^{\circ}C$, enzymes denature/change their structure and function less efficiently.

FRQ

4. Read each question carefully. Write your response in the space provided for each part of each question. Answers must be written out in paragraph form. Outlines, bulleted lists, or diagrams alone are not acceptable and will not be scored.

In plants, a critical step in the pathway leading to the synthesis of the amino acid tryptophan includes the synthesis of an intermediate molecule, 5-enolpyruvylshikimic acid-3-phosphate (EPSP). The reactants in this step are shikimic acid-3-phosphate (S3P) and phosphoenolpyruvate (PEP), and the reaction is catalyzed by the enzyme EPSP synthase (Figure 1). Compound X, a chemical used to kill weeds, can affect EPSP synthesis by interacting with EPSP synthase.

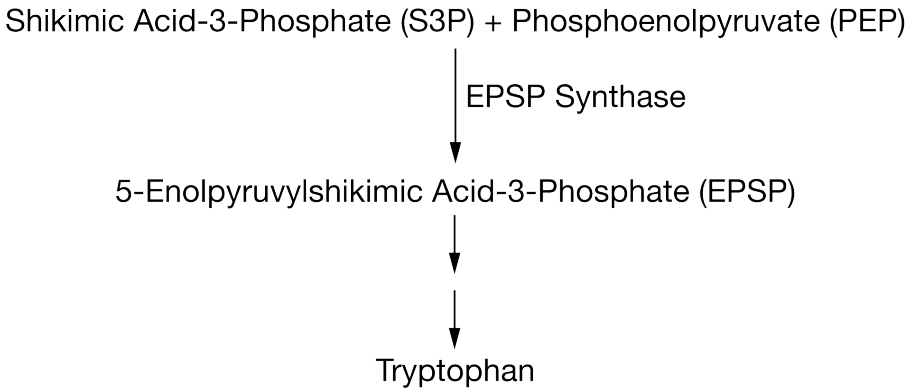
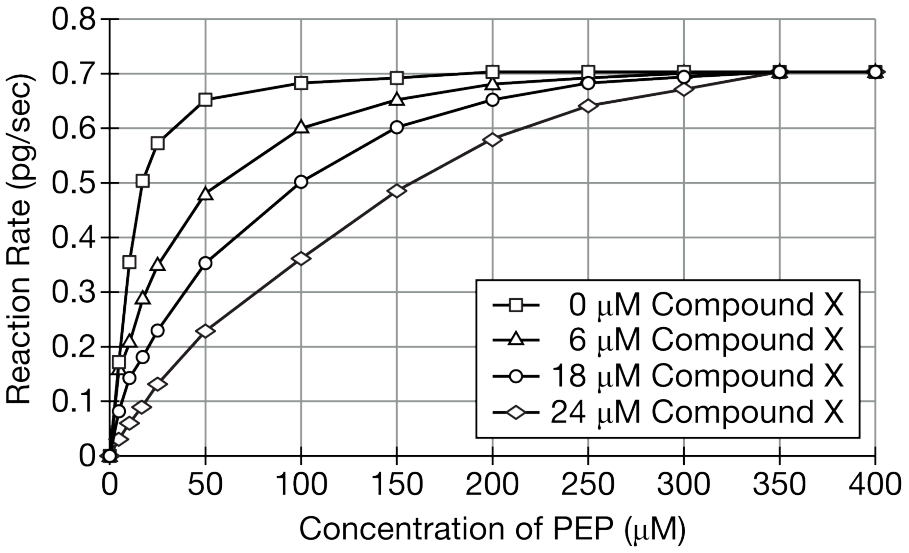


Figure 1. Reaction between shikimic acid-3-phosphate (S3P) and phosphoenolpyruvate (PEP), catalyzed by the enzyme EPSP synthase. Compound X can affect EPSP synthesis by interacting with EPSP synthase.

Scientists investigated the effect that Compound X has on EPSP synthase activity by performing an experiment to examine the relationship between EPSP synthase activity, PEP concentration, and Compound X concentration (Figure 2).



FRQ

Figure 2. Rate of the reaction catalyzed by a specific amount of EPSP synthase with a specific amount of S3P, increasing concentrations of PEP, and four concentrations of Compound X

Some populations of the plant *Lolium rigidum* have evolved resistance to Compound X. To investigate the inheritance of this resistance, researchers crossed true-breeding plants from a Compound X-susceptible (S) population with true-breeding plants from a Compound X-resistant (R) population. In half of these crosses, pollen from S plants was used to fertilize ovules in R plants. In the other half, pollen from R plants was used to fertilize ovules in S plants. Survival of parental and F_1 seedlings, differentiated by the maternal phenotype, in response to varying concentrations of Compound X is shown in Figure 3.

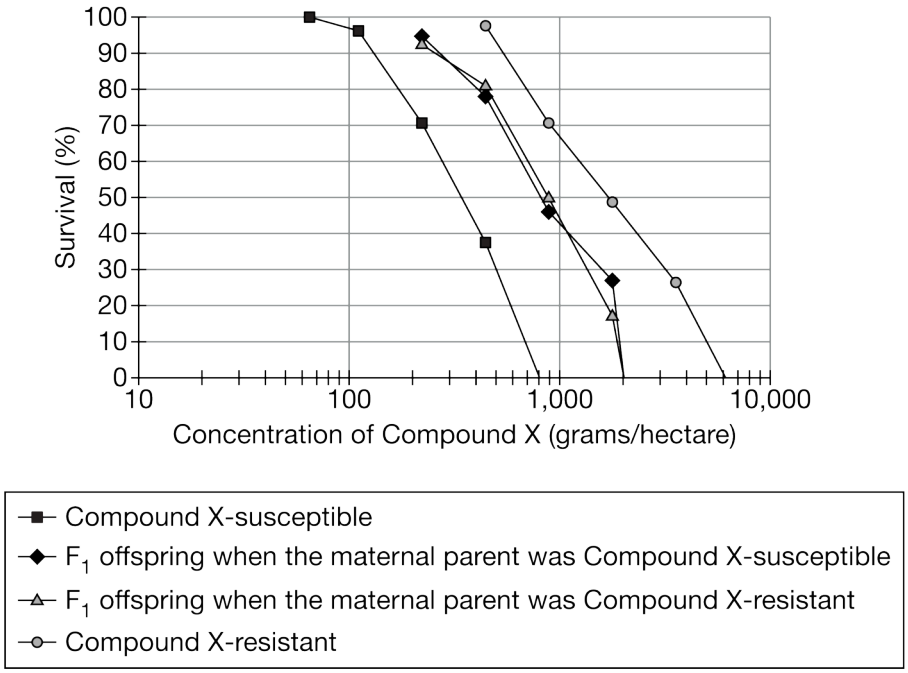


Figure 3. Percent survival of four populations of *L. rigidum* exposed to various concentrations of Compound X shown in a log scale. Data are an average from one experiment with four replicates conducted on seedlings growing in pots. $1\ \text{Hectare} = 10{,}000\ \text{m}^2$

(a) **Explain** how Compound X can affect binding of S3P and PEP to EPSP synthase.

(b) **Identify** one dependent variable in the experiment shown in Figure 2. **Identify** a control group in the experiment shown in Figure 3. Using the data provided in Figure 3, **identify** the concentration of Compound X, in grams/hectare, required to kill 50% of the Compound X-susceptible (S) population. Using the graph in Figure 2, **calculate** the mass, in pg, of EPSP that would be produced by this reaction in one minute at a PEP concentration of 50 µM and a Compound X concentration of 18 µM.

(c) Researchers claim that a single gene is responsible for the Compound X-resistance phenotype. In a follow-up experiment with *L. rigidum*, the researchers crossed an F_1 male with a female from the original resistant (R) population. The resulting offspring were treated with 900 grams/hectare of Compound X. Based on the data presented in Figure 3, **predict** ONE genotype of the resulting offspring and the percent survival of the offspring with this genotype after treatment with Compound X. Researchers claim that if the Compound X-resistance gene

FRQ

were carried on a chloroplast chromosome, the F_1 offspring would not have a phenotype intermediate to that of the two parents. Provide reasoning to **justify** the claim. Researchers discovered a variant of the Compound X -resistance gene in *L. rigidum* in which there was a single base pair insertion upstream of the promoter region of one copy of the gene. **Predict** the most likely effect on the phenotype of the plant.

(d) Researchers have engineered Compound X-resistant corn using prokaryotic genes. Other genetic engineering experiments have placed eukaryotic genes into prokaryotic cells for mass production of their gene products. **Explain** how the eukaryotic DNA would need to be altered in order for prokaryotic cells to express the eukaryotic gene. **Explain** why the *EPSP* synthase gene from a prokaryote can be properly expressed in a corn plant.

Part A

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1 ✓

The response includes an acceptable explanation. Acceptable explanations include the following.

· The binding of Compound X to EPSP synthase alters the shape and/or charge of the active site such that S3P and PEP are no longer compatible.

· Compound X can bind to the active site and block the binding of S3P and PEP.

Part B

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1	2	3	4 ✓

The response includes all of the following criteria.

- ☐ An identification that the dependent variable is the reaction rate
- ☐ An acceptable identification of a control group. Acceptable identifications include the following.
  · Compound X-susceptible
  · Compound X-resistant
- ☐ An identification that the concentration of Compound X, in grams / hectare, required to kill 50% of the Compound X-susceptible (S) population is 350 (grams / hectare); NOTE: acceptable student responses can be between 300 and 400
- ☐ The mass, in pg, of EPSP that would be produced by the reaction in one minute at a PEP concentration of $50\mu M$ and a Compound X concentration of $18\mu M$ is calculated to be 21 pg.

FRQ

Part C

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1	2	3 ✓

The response includes all of the following criteria.

- ☐ An acceptable prediction of the percent survival of one genotype of the resulting offspring. Acceptable predictions include the following.
  - 48% survival RS; NOTE: acceptable student responses can be between 45% and 50%.
  - 71% survival RR (resistant); NOTE: acceptable student responses can be between 70% and 72%.
- ☐ A justification that the offspring would inherit the Compound X resistance gene only from the maternal parent, and the phenotype would either be fully susceptible or fully resistant
- ☐ An acceptable prediction of the mostly likely effect on the phenotype of the plant. Acceptable predictions include the following.
  - No change (to the phenotype)
  - Compound X resistance

Part D

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1	2 ✓

The response includes both of the following criteria.

- ☐ An acceptable explanation. Acceptable explanations include the following.
  - Introns would need to be removed because prokaryotes do not perform RNA splicing.
  - The gene would need to be under the control of a prokaryotic promoter.
- ☐ An explanation that the genetic code is shared in all living organisms (and therefore the protein would have the same amino acids)

FRQ

5. Read each question carefully. Write your response in the space provided for each part of each question. Answers must be written out in paragraph form. Outlines, bulleted lists, or diagrams alone are not acceptable and will not be scored.

Solid tumors are clusters of cancer cells and often contain blood vessels. When molecule B binds to the wild-type Brec protein in the plasma membrane of certain solid tumor cancer cells (Figure 1), the cancer cells express the membrane protein A and sometimes stimulate increased growth of blood vessels into the tumors.

Cells with a particular mutation in the *Brec* gene (Brec-MUT cells) have much increased expression levels of A and stimulate greater growth of blood vessels than do cancer cells with the wild-type Brec (Brec-WT cells); the cells with the mutant Brec can trigger intracellular signaling in the absence of B.

Researchers proposed that the signaling pathway modeled in Figure 1 is triggered by activation of the wild-type Brec and is associated with phosphorylation and activation of kinase D, expression of A, and the ability of the cancer cells to stimulate blood vessel growth.

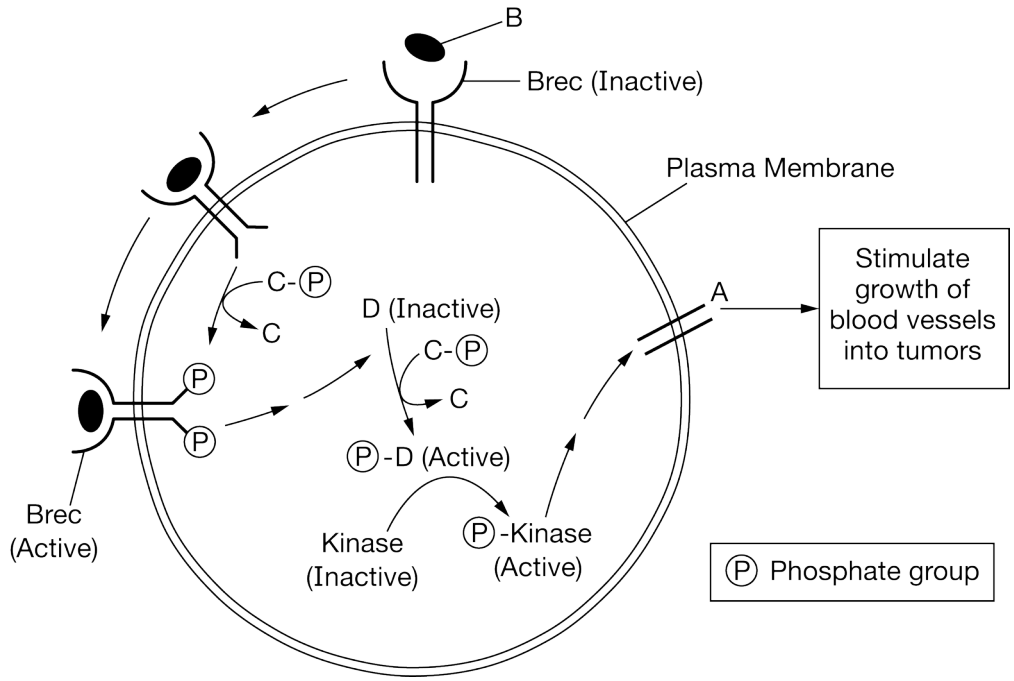


Figure 1. A simplified model of the normal signaling pathway hypothesized to play a role in certain cancer cells expressing A and stimulating blood vessel growth into solid tumors

(a) Based on the signaling model shown in Figure 1, **describe** the role of molecule C-P in the activation of D.

(b) **Explain** how a molecule such as kinase D can amplify an extracellular signal. **Explain** how the mutant Brec protein is able to activate kinase D even in the absence of its ligand.

(c) Based on the proposed signaling pathway, **predict** the relative activity of kinase D in the Brec-WT cells compared with kinase D in the Brec-MUT cells when both cell types are grown in nutrient broth lacking B.

FRQ

Justify your prediction.

(d) The researchers claim that treatment with a compound that inhibits dephosphorylation of the intracellular domain of the Brec protein will inhibit expression of A in Brec-WT cells but not in Brec-MUT cells. Based on the information provided, provide reasoning to **refute** their claim.

Part A

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1 ✓

The response includes a description that molecule C-P provides the energy and the phosphate (to activate D).

Part B

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1	2 ✓

The response includes both of the following criteria.

- ☐ An explanation that kinase D can act on multiple intracellular substrates/phosphorylate multiple kinases
- ☐ An explanation that the mutant Brec protein already has the appropriate shape to be phosphorylated/ activate kinase D/cause the release of a second messenger

Part C

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1	2 ✓

The response includes both of the following criteria.

- ☐ A prediction that there will be less kinase D activity in the Brec-WT cells than in the Brec-MUT cells
- ☐ A justification that the Brec-WT cells require B to stimulate intracellular signaling including activation of kinase D, but the Brec-MUT cells do not

FRQ

Part D

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1 ✓

The response includes a refutation of the claim by stating that if dephosphorylation of Brec (with WT or mutant) is blocked, then downstream signaling will continue to occur, and A will be expressed in both cell types.

FRQ

6. Read each question carefully. Write your response in the space provided for each part of each question. Answers must be written out in paragraph form. Outlines, bulleted lists, or diagrams alone are not acceptable and will not be scored.

Glycogen is a complex polymer of glucose molecules involved in the storage of energy in many organisms. Depending on the signals received by a cell, different enzymes are activated, which will lead to either the synthesis or breakdown of glycogen. The product of glycogen breakdown, glucose 1-phosphate, can be modified into glucose 6-phosphate, an intermediate of glycolysis (Figure 1).

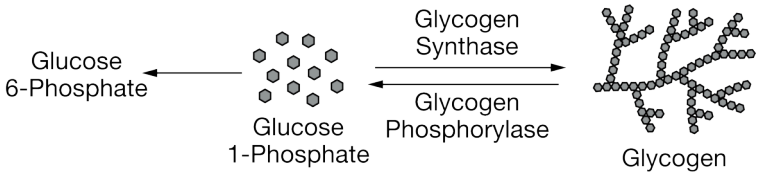
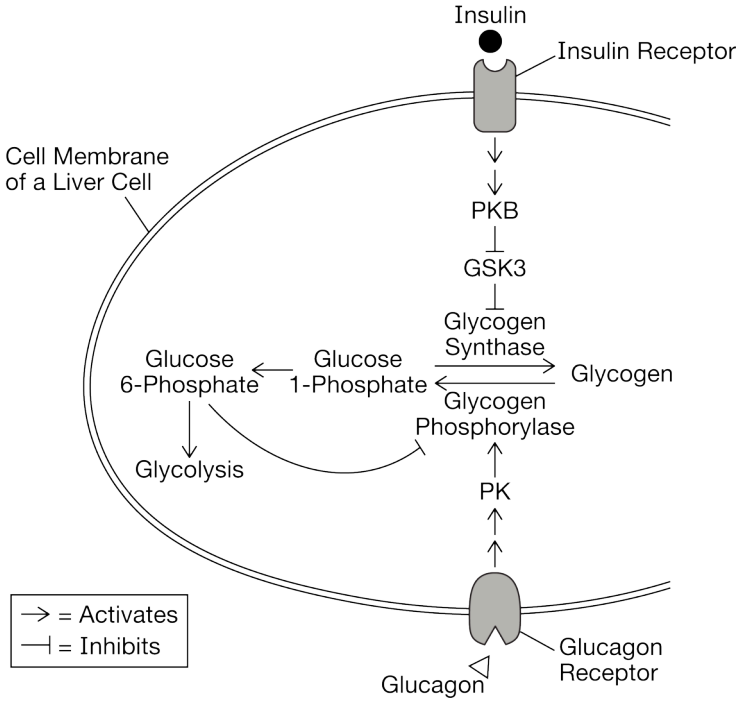


Figure 1. Glycogen synthesis and breakdown

In the liver, glycogen synthase and glycogen phosphorylase are regulated by kinases. These kinases transfer phosphate groups from ATP to amino acids in glycogen synthase and glycogen phosphorylase. Phosphorylase kinase B (PKB) is a kinase that regulates glycogen synthase kinase 3 (GSK3) activity. GSK3 regulates the activity of glycogen synthase. Phosphorylase kinase (PK) regulates the activity of glycogen phosphorylase. The activity of these kinases depends on the activation of signaling pathways in response to the binding of specific cell signals, including insulin and glucagon, to their receptors (Figure 2).



FRQ

Figure 2. A simplified model of signaling pathways in a liver cell that are activated by insulin and glucagon

(a) A cell where PKB is always active has a mutation that results in the permanent activation of GSK3. Based on the information in Figure 2, **predict** the effect of this mutation on the activity of glycogen synthase. **Justify** your prediction.

(b) **Describe** the process that results in the activation of multiple copies of PKB in response to the binding of a single molecule of insulin to its receptor. **Explain** why insulin can stimulate the activation PKB of but not the activation of PK.

(c) **Explain** why an increase in glycogen phosphorylase activity might result in an increase in O_2 consumption in the cell.

(d) A certain type of tumor results in the overproduction of glucagon. Researchers claim that treatment with insulin can counteract the effects of the excess glucagon on the pathway shown in Figure 2. Provide reasoning to **justify** the researchers' claim.

Part A

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1	2 ✓

The response includes both of the following criteria.

- ☐ A prediction that glycogen synthase will be inactive
- ☐ A justification that when active, the role of GSK3 is to phosphorylate glycogen synthase, resulting in its inhibition, even in the presence of active PKB, an upstream signaling molecule

Part B

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1	2 ✓

The response includes both of the following criteria.

- ☐ An acceptable description. Acceptable descriptions include the following.
  · In the signal transduction pathway activated by insulin binding, each enzyme in the pathway can catalyze multiple phosphorylation reactions.
  The insulin receptor activates many second messengers.
- ☐ An explanation that insulin can bind to the insulin receptor but not the glucagon receptor, because the

FRQ

shapes are not compatible

Part C

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1 ✓

The response explains that an increase in glycogen phosphorylase activity can lead to an increase in glycolysis (and the Krebs/citric acid cycle) and the electron transport chain, which consumes O_2.

Part D

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1 ✓

The response includes a justification that glucagon stimulates glycogen breakdown, but insulin stimulates the synthesis of glycogen.

FRQ

7. Read each question carefully. Write your response in the space provided for each part of each question. Answers must be written out in paragraph form. Outlines, bulleted lists, or diagrams alone are not acceptable and will not be scored.

The SIRT3 protein has been reported to increase the growth of certain types of cancer cells. Researchers studying the metabolism and development of stomach cancer cells investigated the potential role of the SIRT3 protein in these cells. In the first investigation, the researchers quantified the relative amount of SIRT3 in normal stomach cells and in each of four different lines of stomach cancer cells (Figure 1). Using data from a group of 100 patients with stomach cancer, the researchers also analyzed the relative amount of SIRT3 protein detected in samples of the cancer cells and in normal cells immediately adjacent to the cancer cells (Figure 2).

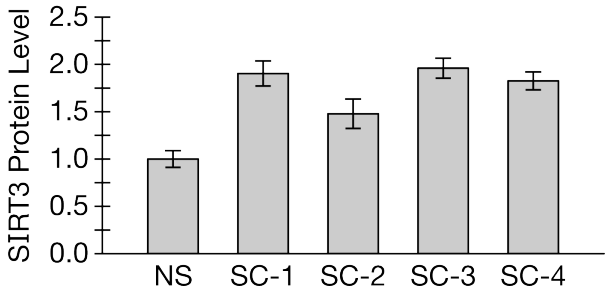


Figure 1. SIRT3 protein levels in normal stomach cells (NS) and stomach cancer cell lines from four different patients (SC-1, SC-2, SC-3, SC-4), shown relative to the level in normal stomach cells (NS). Error bars represent $\pm SE_{\bar{x}}$.

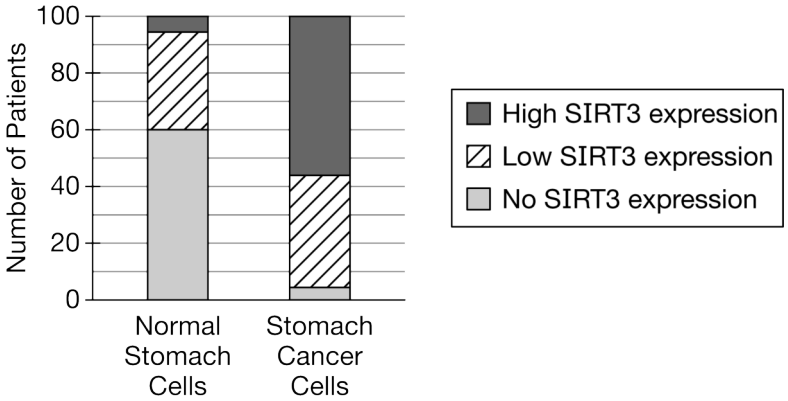
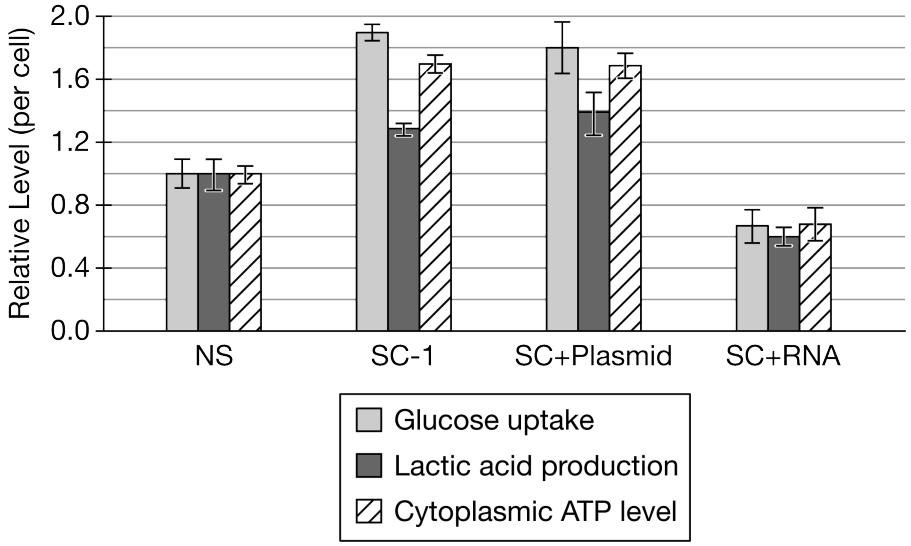


Figure 2. The number of patients, from 100 tested, whose stomach cancer cells or normal stomach cells adjacent to the cancer cells express high, low, or no SIRT3 protein

- In the second investigation, the researchers grew four groups of stomach cells in petri dishes containing nutrient broth:• NS, normal stomach cells• SC-1, a line of stomach cancer cells that the researchers had examined for SIRT3 expression• SC + RNA, the SC-1 line of stomach cancer cells to which a plasmid was added that encodes a small RNA complementary to a portion of the *SIRT3* mRNA. When these small RNAs bind to the complementary portion of *SIRT3* mRNA,

the *SIRT3* mRNA is broken down rather than translated.• SC + plasmid, the SC-1 line of stomach cancer cells to which a plasmid was added that does NOT encode the complementary small RNA

- After 24 hours of growth, the researchers measured several indicators of metabolism by an equal number of all four cell groups: glucose uptake, lactic acid production, and the level of cytoplasmic ATP production in the presence of a compound that blocks the electron transport chain (Figure 3). The values were calculated relative to the values obtained for normal stomach cells.



- Figure 3. Glucose uptake, lactic acid production, and cytoplasmic ATP levels in stomach cells. NS, normal stomach cells; SC-1, stomach cancer cells; SC+ plasmid, SC-1 cells with added plasmid only; SC + RNA, SC-1 cells with added plasmid encoding small RNA complementary to a portion of *SIRT3*. Error bars represent $\pm \mathrm{SE}_{\bar{x}}$.

- (a) A stomach cell line growing in nutrient broth has a mutation in the gene encoding a particular type of tRNA. Assuming the mutation is not lethal to the cells, **explain** why the amount of *SIRT3* mRNA in this cell line is identical to the amount produced by a normal stomach (NS) cell line but the amount of SIRT3 protein is less than in the NS cells.

(b) **Identify** a dependent variable in the researchers' experiment shown in Figure 3. **Justify** the analysis of normal stomach cells (NS) in the experiments shown in Figure 3. Based on the data in Figures 1 and 3, **describe** the relationship between SIRT3 expression and glucose uptake. **Calculate** the percent change in lactic acid production by SC-1 cells compared with SC+RNA cells.

(c) Researchers analyzing regulation of *SIRT3* gene expression engineer a *SIRT3* gene that is identical to the normal gene except that the normal *SIRT3* gene promoter is replaced by the promoter for a gene encoding a protein typically found only in human nerve cells. The researchers introduce this gene into normal human stomach cells to determine how much *SIRT3* mRNA AND protein will be expressed from the engineered *SIRT3* gene in the stomach cells. **Predict** the results of this experiment. The researchers claim that metabolic activity increases in cancerous cells. Provide reasoning based on the data provided to **justify** the claim. Based on the information provided, **predict** the cause of a decrease in CO_2 production by the three groups of cancer cells (SC-1, SC + plasmid, and SC + RNA) treated with the compound that blocks the electron transport chain compared

FRQ

with the same groups of cells not treated with the compound.

(d) Based on the data provided, **explain** how testing for $SIRT3$ gene expression in samples of stomach cells might be used by physicians to predict which individuals have an increased risk of developing stomach cancer. Human stomach cells have plasma membrane receptors for the binding of certain extracellular signaling molecules such as growth factors. **Explain** why the irreversible binding of chemicals that mimic the growth factors will most likely change the normal cells into cancer cells.

Part A

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1 ✓

The response includes an explanation that tRNAs are involved in translation not transcription, (so amount of $SIRT3$ mRNA should be the same in both cell lines). If a tRNA is mutated, it will be unable to carry the correct amino acid or unable to bind to the $SIRT3$ mRNA, so the cell line with the mutation will produce less $\mathrm{SIRT3}$ protein.

Part B

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1	2	3	4 ✓

The response includes all of the following criteria.

- ☐ An identification that the dependent variable is (relative level of) glucose uptake/lactic acid production/ cytoplasmic ATP
- ☐ A justification that normal stomach cells are used (as a control) for comparison to determine whether the metabolic indicators of the other groups of cells have changed
- ☐ An acceptable description. Acceptable descriptions include the following.
  · The greater the expression of $SIRT3$, the greater the amount of glucose uptake by the cells
  · There is a positive correlation.
- ☐ The percent change in lactic acid production by $\mathrm{SC\text{-}1}$ cells compared with $\mathrm{SC+RNA}$ cells is calculated to be 117%.

Part C

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

FRQ

0	1	2	3 ✓

The response includes all of the following criteria.

- ☐ A prediction that no mRNA or protein will be produced from the introduced gene
- ☐ A justification that the cancer cells/ $\mathrm{SC\text{-}1}$ cells/ $\mathrm{SC+plasmid}$ cells studied in Figure 3 demonstrate increased glucose uptake/lactic acid production/ ATP production, indicating that there has been an increase in glycolysis/fermentation/cellular respiration
- ☐ An acceptable prediction. Acceptable predictions include the following.
  · If the ETC is not active, the rate of NADH accumulation will increase, so the rate of the Krebs/citric acid cycle will decrease, and therefore less $\mathrm{CO_2}$ will be produced.
  · Cells switched to fermentation because Krebs/citric acid cycle stopped, and in the absence of the Krebs cycle, $\mathrm{CO_2}$ is not produced.
  · Pyruvate will be unable to enter the mitochondria because the hydrogen ion gradient is not being maintained and therefore pyruvate cannot enter the Krebs/citric acid cycle, which produces $\mathrm{CO_2}$.

Part D

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1	2 ✓

The response includes both of the following criteria.

- ☐ ð An explanation that because $SIRT3$ is expressed at higher levels in most stomach cancer cells than in normal stomach cells, determining the level of expression might help indicate if an individual is likely to develop stomach cancer
- ☐ An explanation that because irreversible attachment can continually stimulate a signaling cascade, this can lead to continued stimulation of the cell cycle/continued cell division

FRQ

8. Read each question carefully. Write your response in the space provided for each part of each question. Answers must be written out in paragraph form. Outlines, bulleted lists, or diagrams alone are not acceptable and will not be scored.

Friedreich's ataxia (FA) is an autosomal recessive genetic disorder that causes a lack of muscle control in patients. Heart defects are common in FA patients. FA is associated with a mutated *FXN* gene (*f* allele) that results in decreased levels of the mitochondrial protein frataxin, the *FXN* gene product. Muscle cells in individuals with FA produce decreased amounts of ATP . The normal *FXN* gene contains several copies of a triplet of nucleotides in an intron. Mutant alleles of the gene have many more copies of the nucleotide triplet, which causes the *FXN* gene to be abnormally condensed.

To investigate the effects of the *FXN* mutation, researchers studied heart cells from homozygous dominant individuals (*FF*) without FA (genotype A), heart cells from patients with FA (genotype B), and heart cells from individuals heterozygous for the *FXN* gene (genotype C). The researchers isolated two different proteins from an equal number of each of the three cell genotypes. For each cell genotype, they compared the amount of frataxin protein with the amount of glyceraldehyde 3-phosphate dehydrogenase (GAPDH), an enzyme that functions in glycolysis (Figure 1). They then quantified the relative amount of frataxin protein in each cell genotype (Figure 2).

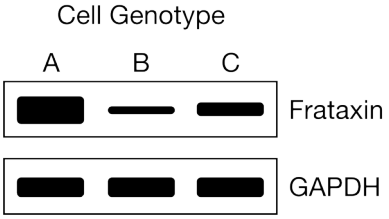


Figure 1. Amounts of frataxin and GAPDH proteins in the three different cell genotypes A, B, and C, as revealed by band width following protein gel electrophoresis

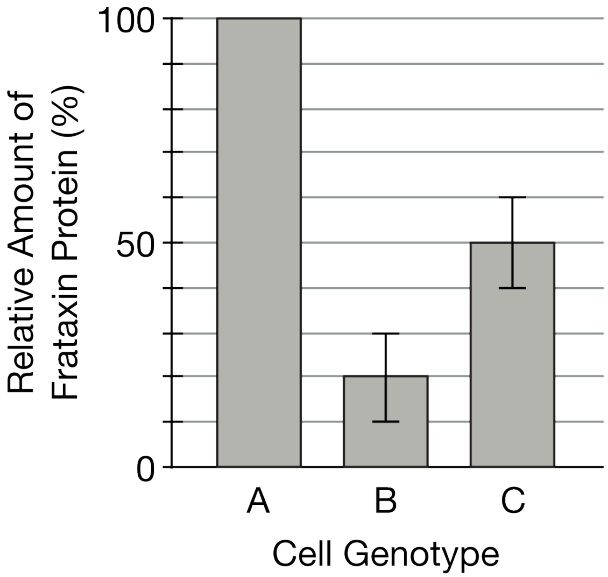


Figure 2. Quantified relative amount of frataxin protein in cell genotypes A, B, and C. Data represent the average from six trials, as shown in Figure 1. Error bars represent $\pm 2\mathrm{SE}_{\bar{x}}$.

FRQ

To investigate a potential role for frataxin in cellular respiration, another group of researchers introduced the human FXN gene into a mouse cell line and overexpressed the frataxin protein. Identical control cells were similarly treated but did not receive the human FXN gene. The researchers measured the activity of a Krebs cycle enzyme in an equal number of control cells and cells overexpressing the human FXN gene (Figure 3). In a related experiment, an equal number of both cell types were grown in medium with $^{14}\mathrm{C}$-labelled glucose, and the researchers measured the amount of $^{14}\mathrm{C}$-labelled $\mathrm{CO_2}$ produced by the cells (Figure 3). They also determined the concentration of lactic acid in both cell types (Figure 3).

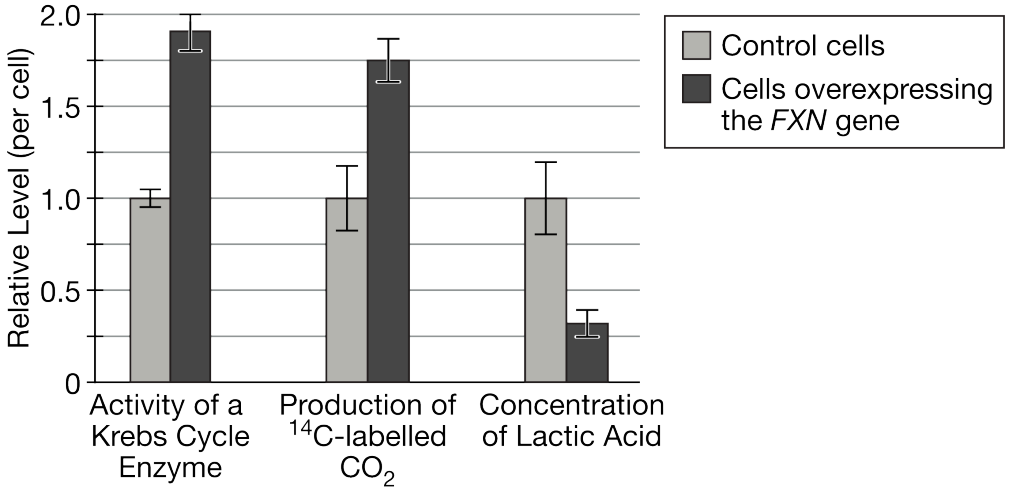


Figure 3. The activity of a Krebs cycle enzyme, the production of $^{14}\mathrm{C}$-labelled $\mathrm{CO_2}$ after growth in a medium with $^{14}\mathrm{C}$-labelled glucose, and the concentration of lactic acid in control cells and in cells overexpressing the FXN gene. Values are shown relative to the values in the control cells. Error bars represent $\pm 2\mathrm{SE}_{\bar{x}}$.

(a) **Describe** the possible genotypes of the heart cells of an individual if meiosis in this individual results in two sperm cells carrying the f allele and two sperm cells carrying the F allele.

(b) **Identify** a dependent variable for the experiment with data shown in Figure 2. **Justify** the measurement of $^{14}\mathrm{C}$-labelled $\mathrm{CO_2}$ in the experiment with data shown in Figure 3. **Describe** the difference between the mouse cells overexpressing the FXN gene and the control mouse cells with respect to the $^{14}\mathrm{C}$-labelled $\mathrm{CO_2}$ production shown in Figure 3. Based on the data in Figure 3, **calculate** how many times more lactic acid is produced on average in the control cells as compared with cells overexpressing FXN.

(c) Researchers produced mice that contained three copies of the FXN gene in each cell, producing genotype fff, rather than the typical two copies. Based on the data provided, **predict** the amount of frataxin protein most likely to be produced by the mice compared with the amount of frataxin in cells with genotypes A, B, and C. **Justify** your prediction.

(d) **Explain** why the phenotypes associated with the human FXN gene can be studied in a mouse cell. Unlike skeletal muscle cells, heart muscle cells cannot rely on fermentation for long. **Explain** the inability of heart muscle cells to function using only fermentation for metabolism. ATP acts as a negative feedback regulator of phosphofructokinase, one of the enzymes in glycolysis. **Predict** the eventual effect of ATP on pyruvate production in heart cells that express excess amounts of the normal frataxin.

Part A

FRQ

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1 ✓

The response includes an acceptable description. Acceptable descriptions include the following.

· All the heart cells are heterozygous/Ff.

· The only possible genotype is Ff.

Part B

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1	2	3	4 ✓

The response includes all of the following criteria.

- ☐ An identification that the dependent variable is the relative amount of frataxin protein
- ☐ A justification that the measurement of ^{14}C-labelled CO_2 allows researchers to analyze the rate at which the cells are performing respiration/the Krebs/citric acid cycle
- ☐ An acceptable description. Acceptable descriptions include the following.
  · The cells overexpressing the FXN gene produced about 1.75 times as much CO_2as the control cells.
  · Cells overexpressing FXN produced more (^{14}C -labelled) CO_2.
- ☐ The number of times more lactic acid produced on average in the control cells as compared with cells overexpressing FXN is calculated to be 3.33.

Part C

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1	2 ✓

The response includes both of the following criteria.

- ☐ An acceptable prediction. Acceptable predictions include the following.
  · The amount of frataxin protein should be about 30% of the amount produced by genotype A.
  · The amount of frataxin protein should be similar to/the same as the amount of frataxin in cells with genotype B.

FRQ

· The amount of frataxin protein should be slightly greater than the amount of frataxin in cells with genotype B but less than the amount of frataxin in cells with genotype C.

☐ An acceptable justification. Acceptable justifications include the following.

· Because the three mutant alleles are expressed at a low level, the total amount of frataxin will be similar to (or slightly more than) the amount in a cell that is homozygous recessive.

· Two f alleles produce about 20%, so three f alleles could be expected to produce 30% of the amount produced by an individual with FF.

Part D

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1	2	3 ✓

The response includes all of the following criteria.

☐ An acceptable explanation. Acceptable explanations include the following.

· Metabolic pathways are conserved across all domains/mammals, so similar genes should produce similar phenotypes in different mammals.

· The genetic code is shared by mice and humans as are the mechanisms by which genes are expressed.

☐ An acceptable explanation. Acceptable explanations include the following.

· Fermentation (and glycolysis) cannot generate sufficient ATP for the high energy demands of heart muscle cells.

· Fermentation cannot regenerate NAD^{+} fast enough to support glycolysis at a rate that would produce sufficient ATP.

☐ An acceptable prediction. Acceptable predictions include the following.

· Pyruvate production will decrease.

· Pyruvate production will stay the same.

FRQ

9. Read each question carefully. Write your response in the space provided for each part of each question. Answers must be written out in paragraph form. Outlines, bulleted lists, or diagrams alone are not acceptable and will not be scored.

The SIRT3 protein has been reported to increase the growth of certain types of cancer cells. Researchers studying the metabolism and development of stomach cancer cells investigated the potential role of the SIRT3 protein in these cells. In the first investigation, the researchers quantified the relative amount of SIRT3 in normal stomach cells and in each of four different lines of stomach cancer cells (Figure 1). Using data from a group of 100 patients with stomach cancer, the researchers also analyzed the relative amount of SIRT3 protein detected in samples of the cancer cells and in normal cells immediately adjacent to the cancer cells (Figure 2).

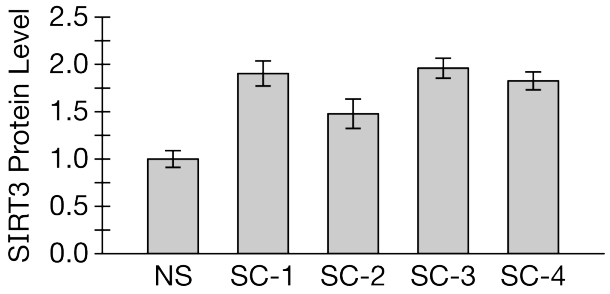


Figure 1. SIRT3 protein levels in normal stomach cells (NS) and stomach cancer cell lines from four different patients (SC-1, SC-2, SC-3, SC-4), shown relative to the level in normal stomach cells (NS). Error bars represent $\pm SE_{\bar{x}}$.

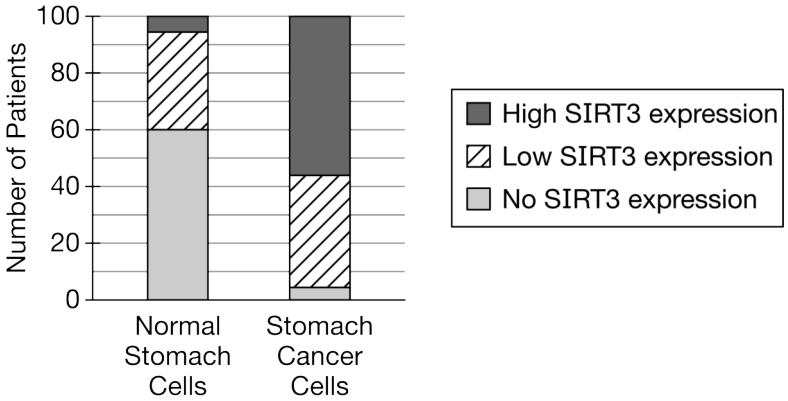
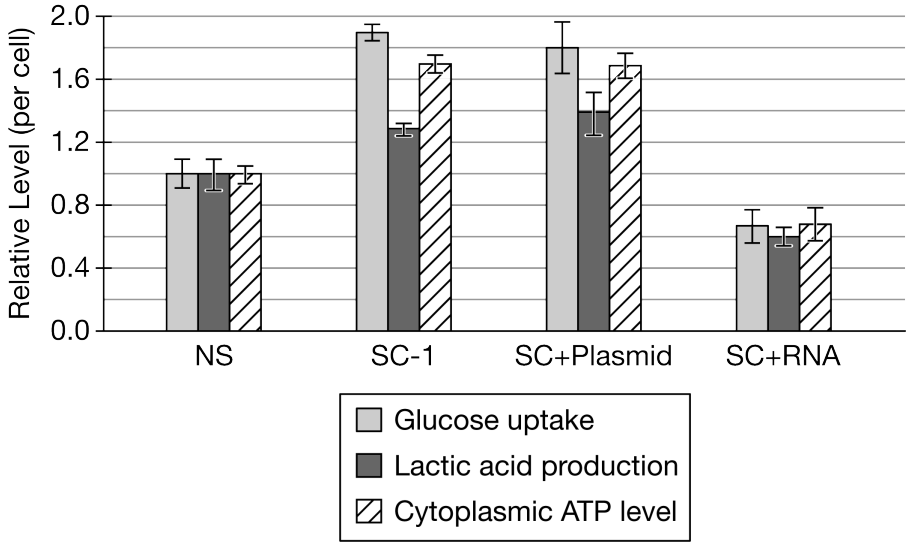


Figure 2. The number of patients, from 100 tested, whose stomach cancer cells or normal stomach cells adjacent to the cancer cells express high, low, or no SIRT3 protein

- In the second investigation, the researchers grew four groups of stomach cells in petri dishes containing nutrient broth:• NS, normal stomach cells• SC-1, a line of stomach cancer cells that the researchers had examined for SIRT3 expression• SC + RNA, the SC-1 line of stomach cancer cells to which a plasmid was added that encodes a small RNA complementary to a portion of the *SIRT3* mRNA. When these small RNAs bind to the complementary portion of *SIRT3* mRNA,

the *SIRT3* mRNA is broken down rather than translated.• SC + plasmid, the SC-1 line of stomach cancer cells to which a plasmid was added that does NOT encode the complementary small RNA

- After 24 hours of growth, the researchers measured several indicators of metabolism by an equal number of all four cell groups: glucose uptake, lactic acid production, and the level of cytoplasmic ATP production in the presence of a compound that blocks the electron transport chain (Figure 3). The values were calculated relative to the values obtained for normal stomach cells.



- Figure 3. Glucose uptake, lactic acid production, and cytoplasmic ATP levels in stomach cells. NS, normal stomach cells; SC-1, stomach cancer cells; SC+ plasmid, SC-1 cells with added plasmid only; SC + RNA, SC-1 cells with added plasmid encoding small RNA complementary to a portion of *SIRT3*. Error bars represent $\pm \mathrm{SE}_{\bar{x}}$.

- (a) **Explain** why an mRNA transcribed from the template strand of the *SIRT3* gene in normal stomach (NS) cells is translated to produce a SIRT3 protein of correct sequence, but if an mRNA were transcribed from the nontemplate strand of the *SIRT3* gene in these cells, it would not produce a SIRT3 protein of correct sequence.

(b) **Identify** a control group for the analysis shown in Figure 3. **Justify** analyzing SIRT3 protein level in four different cancer cell lines, as shown in Figure 1. Based on Figures 1 and 3, **describe** the relationship between SIRT3 expression and cytoplasmic ATP levels. **Calculate** the percent change in cytoplasmic ATP levels by SC + RNA cells compared with SC + plasmid cells.

(c) Researchers analyzing the *SIRT3* gene sequence engineer a *SIRT3* gene that is identical to the normal gene except that a single nucleotide has been deleted from the $5'$ end of the second exon of the gene; the deletion does not affect splicing. The researchers introduce this gene into normal stomach cells to determine how the primary AND tertiary structures of the protein expressed from the engineered *SIRT3* gene will compare with the protein expressed from the normal *SIRT3* gene in these cells. **Predict** the results of this experiment. The researchers claim that the data in Figure 3 provide evidence that the stomach cancer cells perform fermentation to a greater extent than do normal stomach cells. Provide reasoning to **justify** the claim. Based on the information provided, **predict** the effect on the ratio of $\mathrm{NADH}/\mathrm{NAD}^+$ in the cells treated with the compound that blocks the electron

FRQ

transport chain compared with similar cells not treated with the compound.

(d) Based on the data provided, **explain** how cellular processes must be altered in stomach cancer cells compared with normal stomach cells to result in the different levels of $\mathrm{SIRT3}$ expression observed. A mutation to the gene encoding a cyclin-dependent kinase (CDK) in human stomach cells results in a CDK that is continually active in the cells. **Explain** why the continually active CDK will most likely change the normal cells into cancer cells.

Part A

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1 ✓

The response includes an explanation that the nontemplate strand has a sequence that is complementary to the template strand, so transcription of this strand followed by translation will produce a protein of a different sequence from the normal $\mathrm{SIRT3}$ protein.

Part B

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1	2	3	4 ✓

The response includes all of the following criteria.

- ☐ An acceptable identification. Acceptable identifications include the following.
  - · Normal stomach cells (NS)
  - · $\mathrm{SC+plasmid}$ cells
- ☐ A justification that analyzing several different cell lines that all show an increase in $SIRT3$ expression allows researchers to attribute the increases in $\mathrm{SIRT3}$ protein to cancer rather than to variations among individuals
- ☐ An acceptable description. Acceptable descriptions include the following.
  - · The greater the expression of $SIRT3$, the greater the cytoplasmic ATP level.
  - · There is a positive correlation.
- ☐ The percent change in cytoplasmic ATP levels by $\mathrm{SC+RNA}$ cells compared with $\mathrm{SC+plasmid}$ cells is calculated to be -59%.

Part C

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

FRQ

0	1	2	3 ✓

The response includes all of the following criteria.

- ☐ An acceptable prediction. Acceptable predictions include the following.
  · The protein will be a different length if the nucleotide deletion changes the position of the stop codon. The polypeptide produced will fold differently/have a different 3D structure from the normal protein.
  · Every amino acid encoded following the nucleotide deletion is different from that encoded by the normal gene. The polypeptide produced will fold differently from the normal protein.
- ☐ A justification that the increase in lactic acid production in the cancer cells/ SC-1 and SC+plasmid means that there is an increase in fermentation by these cells
- ☐ A prediction that the ratio will increase/be comparatively greater in the treated cells

Part D

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1	2 ✓

The response includes both of the following criteria.

- ☐ An acceptable explanation. Acceptable explanations include the following.
  · An increase in cell signaling that leads to increased rates of transcription/translation results in higher levels of SIRT3 in cancer cells.
  · Increased rates of transcription and translation result in higher levels of SIRT3 in cancer cells.
- ☐ An explanation that CDK will push the cell cycle forward/stimulate cell division by moving the cell through checkpoints regardless of the levels of cyclin, because the CDK is active

FRQ

10. Read each question carefully. Write your response in the space provided for each part of each question. Answers must be written out in paragraph form. Outlines, bulleted lists, or diagrams alone are not acceptable and will not be scored.

Hemoglobin is an iron-containing protein in red blood cells that transports oxygen from the lungs to cells throughout the body. Hemoglobin is a protein complex that is composed of four polypeptide subunits. Typically, two of the polypeptide subunits are encoded by the *HBA1* gene and are called alpha-globin subunits, while the other two polypeptide subunits are encoded by the *HBB* gene and are called beta-globin subunits. Each globin subunit contains a single iron atom capable of binding oxygen.

Prior to birth and during early infancy, expression of the *HBB* gene is low and, instead, cells produce a third polypeptide called gamma-globin. During this time, the hemoglobin complex consists of two alpha-globin subunits and two gamma-globin subunits and is referred to as fetal hemoglobin. Fetal hemoglobin is able to bind oxygen more tightly than does the typical adult hemoglobin, which enables the fetus (unborn offspring) to remove oxygen from the hemoglobin in the mother's blood. A model of the expression levels of all three types of globin polypeptides during the weeks leading up to and following birth is shown in Figure 1.

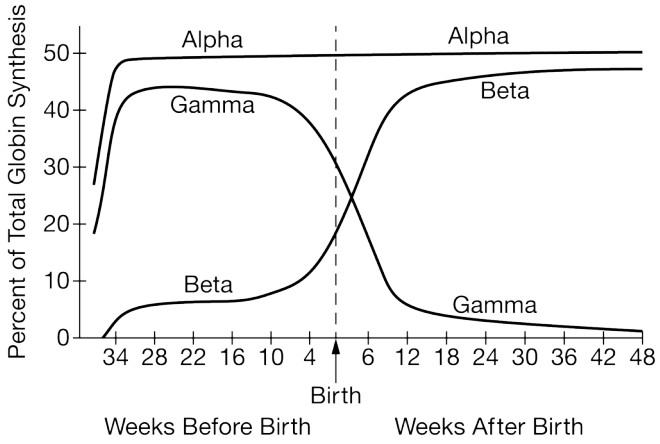
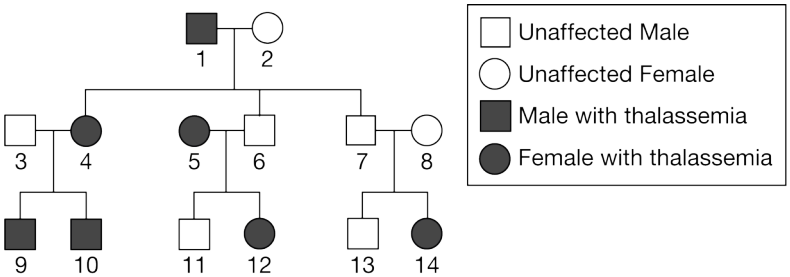


Figure 1. Model of globin expression in fetal and infant tissues

Mutations in the genes that encode globin subunits may affect the function of hemoglobin. Beta-thalassemia is an inherited disorder that results from a variety of mutations to the *HBB* gene that encodes the beta-globin subunit. The incidence of beta-thalassemia in one particular family is shown in Figure 2.



FRQ

Figure 2. Incidence of beta-thalassemia in three generations of a family

The inability to produce functional beta-globin results in an imbalance of alpha- and beta-globin subunits. The accumulation of free alpha-globin subunits causes red blood cells to lyse (burst) or undergo apoptosis (Figure 3). The extra alpha-globin subunits are unable to correctly fold into a hemoglobin protein complex and instead form structures called alpha-hemichromes. The alpha-hemichromes are unable to bind oxygen molecules and become insoluble in the cytoplasm of the red blood cells. The alpha-hemichromes are then deposited in solid form on the surface of the red blood cells, causing the red blood cells to lyse.

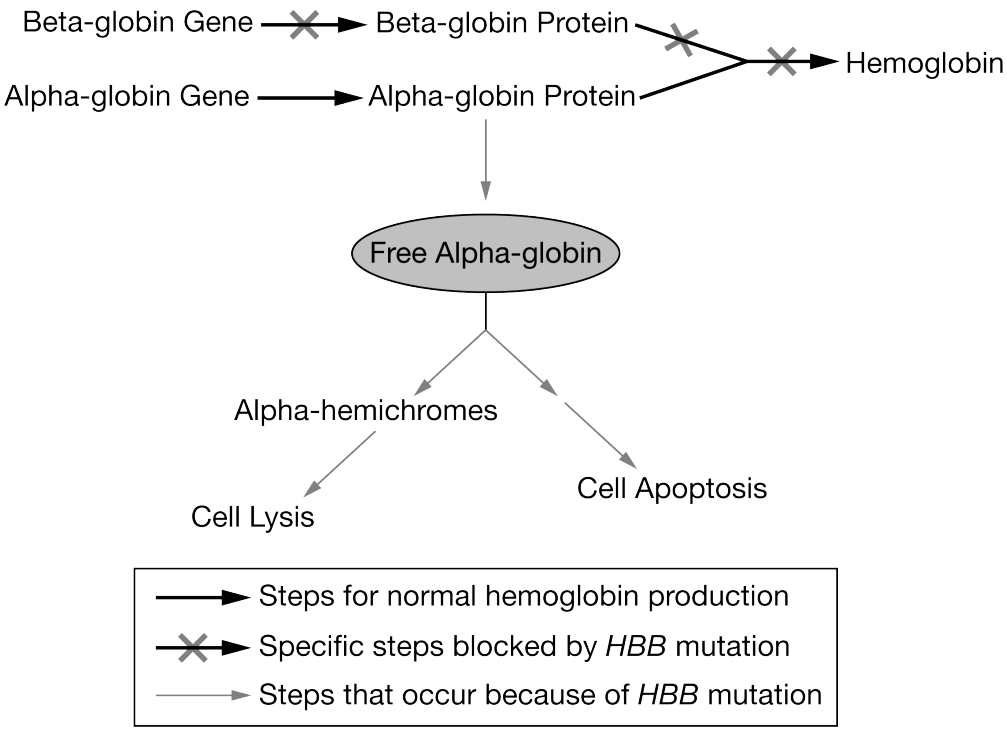


Figure 3. Model of globin synthesis and hemoglobin assembly and the effects of beta-thalassemia. Black arrows with gray Xs indicate steps in the normal hemoglobin assembly that are blocked by the HBB mutation. Gray arrows indicate steps that occur because of the HBB mutation.

(a) Based on the pedigree shown in Figure 2, **describe** the HBB alleles inherited by individual 6, including the parent from which each allele came.

(b) **Explain** why the symptoms associated with beta-thalassemia are often undetectable at time of birth. A study of the gene encoding the gamma-globin subunit showed that, shortly after birth, the gene becomes methylated (CH_3 groups are added to the DNA). **Explain** how DNA methylation could cause the change in the amount of gamma-globin protein that is represented in Figure 1.

(c) **Predict** the effect of a treatment that reduces the expression of $HBA1$ on the symptoms of individuals with beta-thalassemia. **Justify** your prediction.

(d) Researchers claim that the ability of fetal hemoglobin to bind oxygen more tightly than the hemoglobin of the mother increases the fitness of the fetus. Provide reasoning to **support** the claim.

FRQ

Part A

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1 ✓

The response includes a description that individual 6 inherited a mutant allele from his father and a normal allele from his mother.

Part B

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1	2 ✓

The response includes both of the following criteria.

- ☐ An acceptable explanation. Acceptable explanations include the following.
  · The symptoms are the result of nonfunctional beta-globin subunits, which are not part of hemoglobin until a few months after birth.
  · At birth, the baby would have enough gamma-globin subunits to carry oxygen, so the lack of beta globin would not be noticed.
- ☐ An explanation that methylation of DNA can lead to repression of gene expression, which could result in the decrease in the amount of gamma-globin present after birth

Part C

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1	2 ✓

The response includes both of the following criteria.

- ☐ An acceptable prediction of the effect of a treatment. Acceptable predictions include the following.
  · Their symptoms would be less severe.
  · There would be less cell lysis/apoptosis.
- ☐ A justification that the individuals will produce less alpha-globin, so there will be less free alpha-globin to form hemichromes and result in the cells lysing or undergoing apoptosis

FRQ

Part D

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1 ✓

The response includes the reasoning that the ability of fetal hemoglobin to bind oxygen more tightly allows the fetus to obtain the oxygen needed to perform cellular respiration/make the ATP/transform the energy, so they can survive.

FRQ

11. Read each question carefully. Write your response in the space provided for each part of each question. Answers must be written out in paragraph form. Outlines, bulleted lists, or diagrams alone are not acceptable and will not be scored.

Solid tumors are clusters of cancer cells and often contain blood vessels. When molecule B binds to the wild-type Brec protein in the plasma membrane of certain solid tumor cancer cells (Figure 1), the cancer cells express the membrane protein A and sometimes stimulate increased growth of blood vessels into the tumors.

Cells with a particular mutation in the *Brec* gene (Brec-MUT cells) have much increased expression levels of A and stimulate greater growth of blood vessels than do cancer cells with the wild-type Brec (Brec-WT cells); the cells with the mutant Brec can trigger intracellular signaling in the absence of B.

Researchers proposed that the signaling pathway modeled in Figure 1 is triggered by activation of the wild-type Brec and is associated with phosphorylation and activation of kinase D, expression of A, and the ability of the cancer cells to stimulate blood vessel growth.

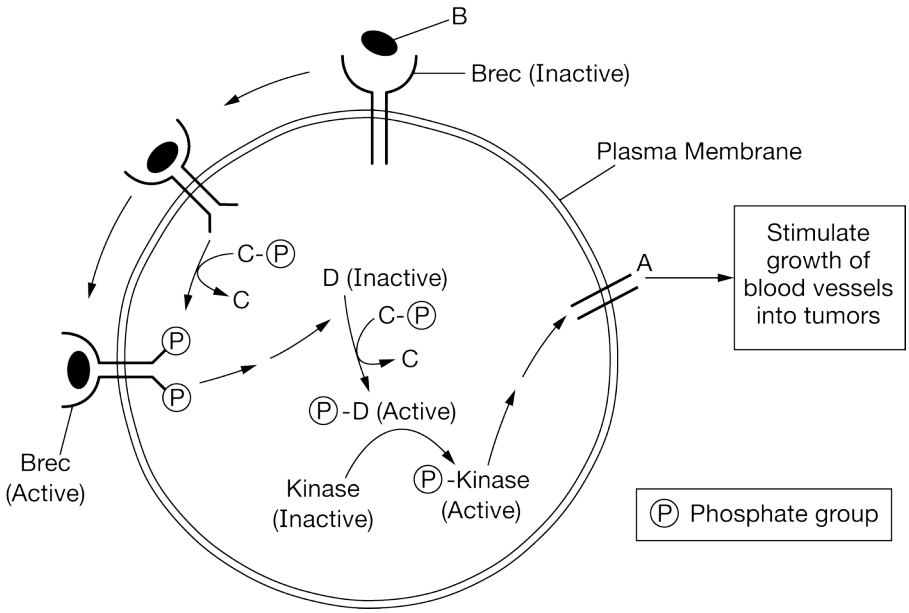


Figure 1. A simplified model of the normal signaling pathway hypothesized to play a role in certain cancer cells expressing A and stimulating blood vessel growth into solid tumors

(a) Based on the signaling model shown in Figure 1, **describe** the role of kinase D.

(b) A certain group of Brec-MUT cells expresses low levels of protein A, unlike the high expression levels of most Brec-MUT cells. Researchers determine that the cells have an excess of ATP and that there are no mutations in the gene encoding protein A. **Describe** a change in the Brec signaling pathway that would lead to the decreased expression of protein A even in the presence of Brec-MUT. **Explain** why B is able to bind to Brec but not to A.

(c) Based on the proposed signaling pathway, **predict** the relative amount of phosphorylated to unphosphorylated

FRQ

mutant Brec in the Brec-MUT cells when the cells are grown in nutrient broth lacking B. **Justify** your prediction.

(d) The researchers claim that treatment with a compound that inhibits phosphorylation of kinase D will affect expression of A in the Brec-WT cells but not in the Brec-MUT cells. Based on the information provided, provide reasoning to **refute** their claim.

Part A

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1 ✓

The response includes an acceptable description. Acceptable descriptions include the following.

· Kinase D phosphorylates and activates downstream kinases in the pathway.

· Kinase D activates downstream kinases, stimulating the production of protein A.

Part B

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1	2 ✓

The response includes both of the following criteria.

- ☐ An acceptable description. Acceptable descriptions include the following.
  · There is a mutation in the gene encoding any protein in the pathway between Brec and protein A.
  · Any protein in the pathway between Brec and protein A is inhibited.
- ☐ An explanation that B has a shape/charge that is complementary to the binding site of Brec but not to any domain of A

Part C

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1	2 ✓

FRQ

The response includes both of the following criteria.

- [ ] A prediction that there will be more of the phosphorylated Brec protein than of the unphosphorylated
- [ ] A justification that the mutant Brec protein does not require B for activation/phosphorylation

Part D

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1 ✓

The response includes a refutation for the claim by stating that because the signaling pathway is blocked after activation of the receptor protein/Brec, there will be no difference in expression of A between the two cell types.

FRQ

12. Read each question carefully. Write your response in the space provided for each part of each question. Answers must be written out in paragraph form. Outlines, bulleted lists, or diagrams alone are not acceptable and will not be scored.

Human hair grows in a cycle that includes a growth phase and a rest phase. A small fraction of the hairs on the scalp are in the rest phase, while the majority of the hairs on the scalp are in the growth phase, which allows the hair on the scalp to grow long. Conversely, the majority of hairs in areas such as the eyelashes are in the rest phase, which keeps those hairs short.

Familial trichomegaly (FT) is a genetic condition that results in excessively long eyelashes in affected individuals. In an investigation into the causes of FT, a team of researchers identified mutations in the gene that codes for fibroblast growth factor-5 (FGF5) that are associated with FT. FGF5 is a protein that is secreted by certain cells and functions as a signaling molecule that influences the ratio of growth phase to rest phase in hair. The structure of the *FGF5* gene includes 3 exons and 2 introns. Alternate splicing of the pre-mRNA transcribed from the gene creates a long version of the mRNA, *FGF5-L*, and a short version of the mRNA, *FGF5-S* (Figure 1).

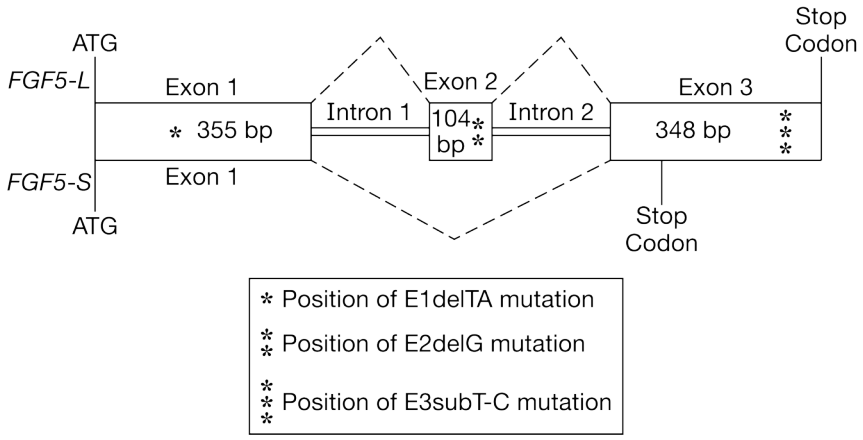


Figure 1. A map of the splicing patterns that lead to the production of both the FGF5-L protein and the FGF5-S protein. Labels below the diagram refer to gene components for *FGF5-S*, and labels above the diagram refer to gene components for *FGF5-L*. Dotted lines indicate the portion(s) of the sequence that will be removed from the transcribed pre-mRNA to form either *FGF5-L* or *FGF5-S* mRNA .

- Researchers identified three different mutations in the *FGF5* gene that have been associated with FT. These mutations, the positions of which are shown in Figure 1, include: • E1delTA: a mutation in which a thymine and an adenine are deleted from Exon 1 of the *FGF5* gene• E2delG: a mutation in which a guanine is deleted from Exon 2 of the *FGF5* gene• E3subT-C: a mutation in which a thymine is substituted for a cytosine in Exon 3 of the *FGF5* gene

The researchers collected data on a family with a history of FT and created a pedigree representing the instances of FT in the family as shown in Figure 2.

FRQ

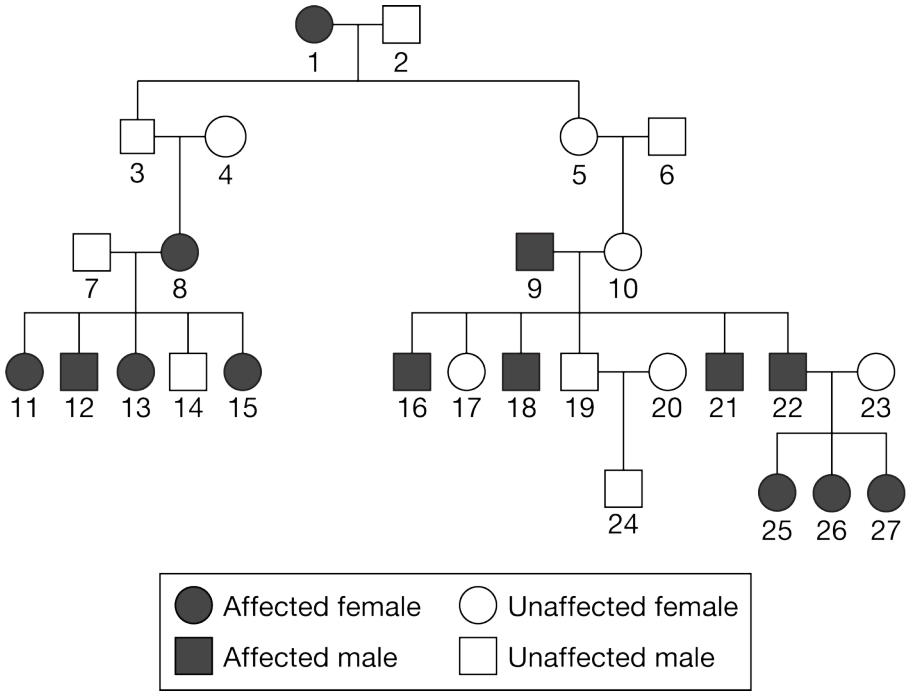


Figure 2. A pedigree representing five generations of a family with a history of FT

The researchers analyzed the length of hairs from individuals with FT and from unaffected individuals and plotted the length of each hair as shown in Figure 3A. They also determined the ratio of hairs in growth phase to hairs in rest phase as shown in Figure 3B.

FRQ

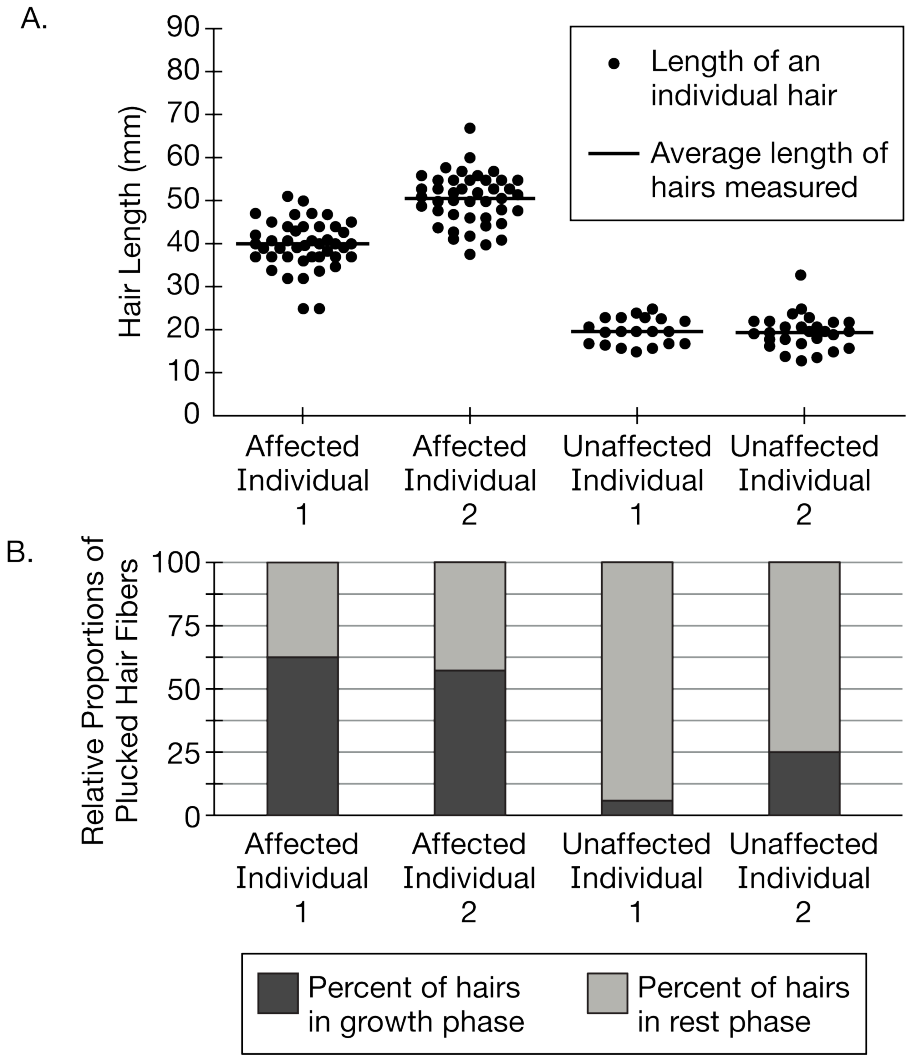


Figure 3A. Length of hairs taken from individuals with and without FT
Figure 3B. Proportion of hairs examined that were in growth phase compared with the proportion of examined hairs in rest phase

In a final experiment, the researchers grew human hair in a petri dish with and without adding FGF5 to the dish. The growth of the hair over an 8-day period was measured. The data from this experiment are shown in Figure 4.

FRQ

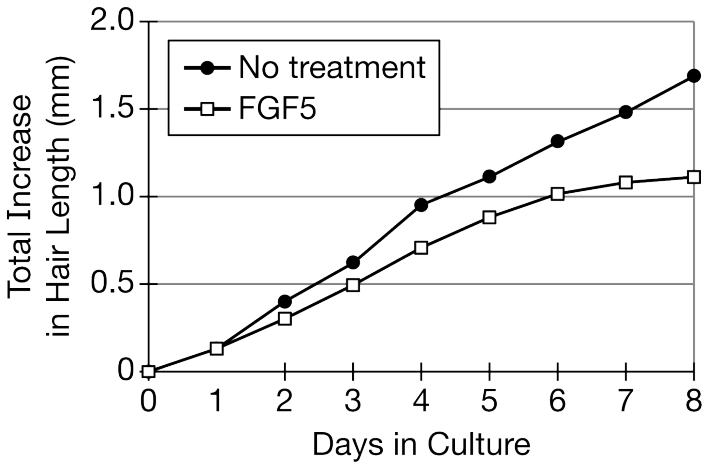


Figure 4. Growth of human hair in culture with and without FGF5

(a) **Describe** what happens to an *FGF5* mRNA to which a poly(A) tail has not been added.

(b) **Identify** a dependent variable in the experiment represented in Figure 4. **Justify** using the culture without FGF5 as the control in the experiment represented in Figure 4. Based on an analysis of the pedigree provided in Figure 2, **describe** the most likely mode of inheritance of FT in this family. Based on Figure 2, if individuals 3 and 4 had another child, **calculate** the probability that the child would have been an affected female.

(c) **Predict** the effect of the E3subT-C mutation on the function of FGF5-S. **Justify** your prediction. Researchers identify a new mutation that results in an FGF5 receptor that can bind the FGF5 protein but cannot stimulate production of second messengers in the cell. Based on the data provided in Figure 4, **predict** the effect of the new mutation on hair growth, regardless of hair type.

(d) Researchers introduced a single copy of the normal mouse *FGF5* allele into human cells homozygous for the *FGF5* mutation and found that, while the human cells could express the mouse *FGF5* gene, the mouse FGF5 protein was not able to restore normal FGF5 signaling in the human cells. **Explain** how the mouse FGF5 protein could be expressed but not activate the FGF5 pathway in the human cells. Bacterial cells can be used to produce large quantities of eukaryotic proteins. **Explain** why a functional FGF5 protein is not produced when bacteria are transformed with the full-length eukaryotic *FGF5* gene, even when the eukaryotic promoter is replaced with a bacterial promoter.

Part A

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1 ✓

The response includes an acceptable description. Acceptable descriptions include the following.

· The mRNA is degraded more rapidly (in the cytoplasm).

FRQ

· The mRNA will be unable to move from the nucleus to the cytoplasm.

Part B

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1	2	3	4 ✓

The response includes all of the following criteria.

- ☐ An acceptable identification of a dependent variable. Acceptable identifications include the following.
  · Hair growth (in the culture system)
  · Increase in hair length
- ☐ An acceptable justification. Acceptable justifications include the following.
  · Comparing the hair grown with $\mathrm{FGF5}$ to the hair without $\mathrm{FGF5}$ verifies whether $\mathrm{FGF5}$ has an effect on hair growth.
  · Growing hair in culture without $\mathrm{FGF5}$ verifies that hair will grow in the culture medium.
- ☐ A description that FT is most likely autosomal recessive
- ☐ The probability that the child would have been an affected female is calculated to be 0.125 OR $1/8$ OR 12.5% .

Part C

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1	2	3 ✓

The response includes all of the following criteria.

- ☐ A prediction that $\mathrm{FGF5\text{-}S}$ will be unaffected
- ☐ A justification that the mutation occurs after the stop codon in the $FGF5-S$ transcript and therefore does not affect the translation of the protein
- ☐ A prediction that hair growth should increase

Part D

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

FRQ

0	1	2 ✓

The response includes both of the following criteria.

- [ ] An explanation that the genetic code is conserved in all species, so human cells can transcribe and translate the mouse allele, but the shape/charge of the mouse protein must be different enough from the human $\mathrm{FGF5}$ such that it is unable to bind to and/or activate the receptor.
- [ ] An explanation that the full length eukaryotic $\mathrm{FGF5}$ gene contains introns, which are not removed in bacterial cells, therefore these introns will be translated, and the resulting protein will be different from the eukaryotic protein

FRQ

13. Read each question carefully. Write your response in the space provided for each part of each question. Answers must be written out in paragraph form. Outlines, bulleted lists, or diagrams alone are not acceptable and will not be scored.

Crossing-over between homologous chromosomes during meiosis is necessary for the proper segregation of the chromosomes. In the fly species *Drosophila melanogaster*, two conserved genes, $mei\text{-}9$ and $pch2$, are thought to play a role in crossing-over. The protein encoded by $mei\text{-}9$ is part of a large protein complex required for the pairing of homologous chromosomes, while the protein encoded by $pch2$ is associated with a delay in prophase I that is believed to serve as a cell cycle checkpoint.

To investigate the role of $mei\text{-}9$ and $pch2$ in crossing-over, researchers determined the frequency of recombination for each of four regions of chromosome 2 in wild-type flies (WT), flies expressing a mutant form of $mei\text{-}9$ ($\mathrm{mei\text{-}9\text{-}mut}$ flies), and flies expressing mutant forms of both $mei\text{-}9$ and $pch2$ ($\mathrm{mei\text{-}9\text{-}mut/pch2\text{-}mut}$ flies). The frequency of recombination of each region in both types of mutant flies is shown in Figure 1 as a percentage of the frequency of recombination observed in WT flies.

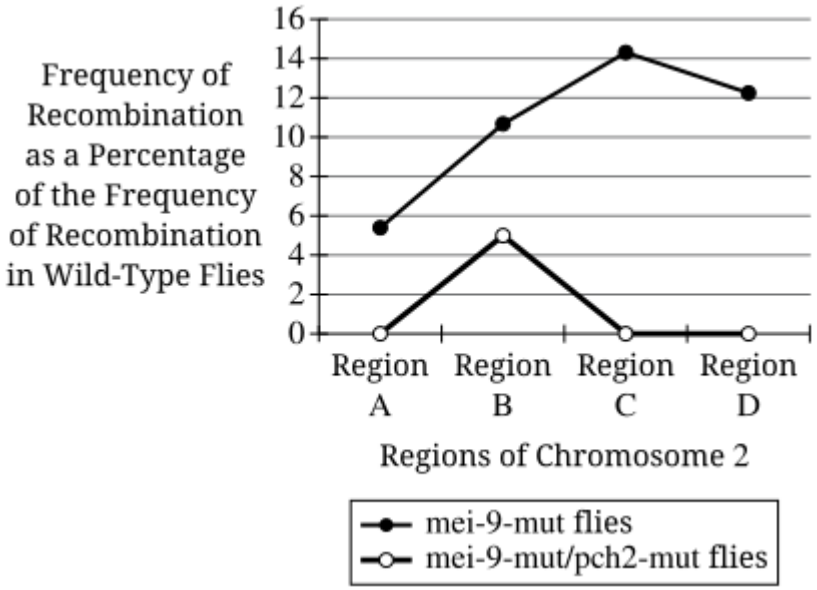


Figure 1. Frequency of recombination of four regions of chromosome 2 in $\mathrm{mei\text{-}9\text{-}mut}$ flies and $\mathrm{mei\text{-}9\text{-}mut/pch2\text{-}mut}$ flies as a percentage of the frequency of recombination in the same region in wild-type flies

(a) Based on the data in Figure 1, **identify** the region of chromosome 2 with the greatest frequency of recombination in $\mathrm{mei\text{-}9\text{-}mut/pch2\text{-}mut}$ flies.

(b) Based on the data in Figure 1, **describe** the frequency of recombination in $\mathrm{mei\text{-}9\text{-}mut}$ flies compared with the frequency of recombination in WT flies.

(c) Scientists hypothesize that the function of the protein encoded by wild-type $pch2$ allows some of the defects caused by the $mei\text{-}9$ mutation to be overcome, increasing the likelihood that crossing-over will occur. Use the

FRQ

data in Figure 1 to **evaluate** their hypothesis.

(d) Based on the information provided, **explain** why the fertility of mei-9-mut flies might be reduced compared with that of WT flies.

Part A

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1 ✓

The response includes an identification that region of chromosome 2 with the greatest frequency of recombination in mei-9-mut/pch2-mut flies is Region B.

Part B

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1 ✓

The response includes an acceptable description. Acceptable descriptions include the following.

- The frequency of recombination (in each region) is lower (in mei-9-mut flies than it is in WT flies).
- The frequency of recombination for mei-9-mut does not go above 15% of the WT.

Part C

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1 ✓

The response provides an acceptable evaluation of the scientists' hypothesis. Acceptable evaluations include the following.

- The scientists' hypothesis is supported because the frequency of recombination is higher in mei-9-mut flies (where the protein encoded by *pch2* is functional than it is in the mei-9-mut/pch2-mut flies where it is not).
- The scientists' hypothesis is supported because the frequency of recombination is lower in the mei-9-mut/pch2-mut flies (where the protein encoded by *pch2* is nonfunctional than it is in mei-9-mut

FRQ

flies where it is functional).

Part D

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1 ✓

The response includes an acceptable explanation. Acceptable explanations include the following.

- The reduced incidence of crossing-over/frequency of recombination observed in mei-9-mut flies could lead to (an increase in) nondisjunction/the formation of gametes with the incorrect numbers of chromosomes.
- Gametes with incompatible chromosome numbers are less likely to result in viable/fertile offspring.

FRQ

14. Read each question carefully. Write your response in the space provided for each part of each question. Answers must be written out in paragraph form. Outlines, bulleted lists, or diagrams alone are not acceptable and will not be scored.

Tuberculosis is an infectious disease caused by the bacterium *Mycobacterium tuberculosis.* Many of the mycobacteria are resistant to the drugs typically used for treatment. As a part of the process of developing new drugs, scientists identified two mycobacterial proteins, L10 and L12, that are part of the ribosome and that must bind to each other to allow ribosomal function. The scientists aimed to identify a compound that blocks the interaction between the two proteins, thereby preventing the survival and reproduction of *Mycobacterium tuberculosis*.

As a first step, the scientists performed an *in vitro* (in a test tube) experiment to test the ability of different compounds to block the binding between the L10 and L12 proteins. The scientists measured the amount of binding in the presence of increasing concentrations of three compounds: I, II, and III. Results are shown in Figure 1.

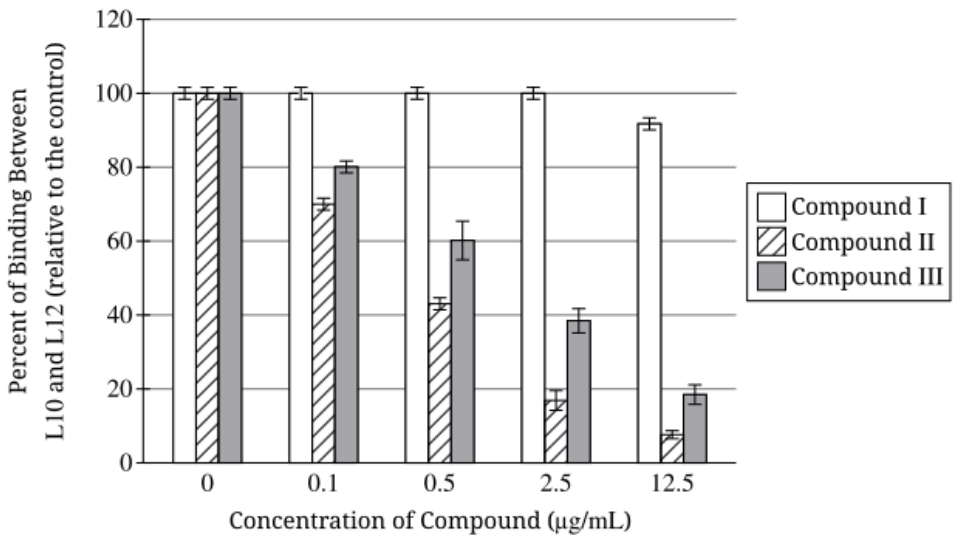


Figure 1. Average percent of binding between L10 and L12 in the presence of increasing concentrations of compounds I, II, and III, relative to the amount of binding in the absence of any compound ($0\ \mu g/mL$, the control). Error bars represent $\pm 2SE_{\bar{x}}$.

In a second experiment, the scientists performed an *in vitro* experiment to test the amount of inhibition of translation in the presence of compounds I, II, and III. They added the three compounds to test tubes containing molecules needed for translation, including a specific mRNA, and ribosomes from one of three organisms: *Mycobacterium tuberculosis*, the bacterium *Escherichia coli*, or the eukaryote *Oryctolagus cuniculus* (rabbit). They measured the concentration of each compound required to reduce translation by 50% (Table 1).

TABLE 1. CONCENTRATION ($\mu g/mL$) REQUIRED TO REDUCE TRANSLATION BY 50% FOR COMPOUNDS I, II, AND III

FRQ

Compound	*M. tuberculosis* ribosomes	*E. coli* ribosomes	*O. cuniculus* ribosomes
I	162.7	143.1	30.2
II	16.9	117.7	118.1
III	25.9	34.2	60.1

(a) Based on the data in Figure 1, **identify** the concentration in $\mu g/mL$ of compound III that reduces the amount of binding between L10 and L12 to 60% of the control.

(b) Based on the data in Figure 1, **describe** the specific decrease in L10 and L12 binding in relation to each specific increase in compound III concentration from $0.1\,\mu g/mL$ to $12.5\,\mu g/mL$.

(c) Based on the data in Table 1, the scientists predict that compound II is the best of the three compounds to treat tuberculosis infections. **Evaluate** the data to support or refute their prediction.

(d) One of the compounds becomes routinely used as a drug to treat tuberculosis. Based on the experiments, **explain** how a mutation in *Mycobacterium tuberculosis* would most likely cause resistance to the drug.

Part A

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1 ✓

The response includes an identification that the concentration in $\mu g/mL$ of compound III that reduces the amount of binding between L10 and L12 to 60% of the control is 0.5.

Part B

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1 ✓

The response includes a description that each five-fold (five times) increase in concentration of compound III should decrease binding by an additional 20% (relative to the control).

Part C

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

FRQ

0	1 ✓

The response includes an acceptable evaluation of the data to support or refute the scientists' prediction. Acceptable evaluations include the following.

· The prediction is supported. Almost 7 times more/a much greater amount of/a much higher concentration of compound II is required to block translation by 50% in rabbits than to block translation in *M. tuberculosis*, and rabbits are a mammal model for humans.

· The prediction is supported. Compound II works best against *M. tuberculosis* and is least effective against eukaryotes (the rabbit), and therefore is unlikely to be toxic to humans.

· The prediction is supported. Compound II works best against *M. tuberculosis* and is much less effective against *E. coli*, and therefore is unlikely to be toxic to those *E. coli* that are beneficial to humans.

Part D

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1 ✓

The response includes an explanation that a mutation to the DNA encoding ribosomal proteins/ L10 / L12 can change the structure of the proteins/ L10 / L12 in such a way that the compound can no longer block the binding between the two proteins (so the proteins still interact and allow the ribosomes to function).

FRQ

15. Read each question carefully. Write your response in the space provided for each part of each question. Answers must be written out in paragraph form. Outlines, bulleted lists, or diagrams alone are not acceptable and will not be scored.

Several studies indicate that a particular forest biome in the Northern Hemisphere will become significantly warmer over the next century. Researchers studying a population of trees in the forest hypothesize that warming will negatively impact the evolutionary fitness of many of the trees. The researchers also hypothesize that warming will shift the range of the population further north to cooler habitats.

(a) **Describe** the way the evolutionary fitness of an individual tree is measured.

(b) The range shift of the tree population may result in the fragmentation of the population into several smaller, isolated populations. **Explain** how the fragmentation may result in speciation.

(c) Recent research suggests that the range in which trees can grow is due in large part to their symbiotic relationships with soil microbes that are sensitive to temperature. **Predict** the effects on a population of microbes if the average temperature in an area increases above the temperature at which most of the microbes are able to successfully reproduce.

(d) **Justify** your prediction in part (c) using evidence from biological principles.

Part A

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1 ✓

The response includes an acceptable description of the way the evolutionary fitness of an individual tree is measured. Acceptable descriptions include the following.

· It is measured by counting the number of offspring/seedlings/alleles in the next generation that came from this tree.

· It is measured by determining the ability of a tree to produce offspring.

Part B

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1 ✓

The response includes the explanation that the separate populations will be reproductively isolated/have no gene flow

FRQ

between them, and genetic differences between the populations may accumulate over time due to natural selection/genetic drift/mutations.

Part C

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1 ✓

The response includes an acceptable prediction. Acceptable predictions include the following.

· Option 1: Future generations of the microbes would be composed of microbes able to reproduce at higher temperatures (shifting the gene frequency toward those alleles that enable reproduction at higher temperatures).

· Option 2: The population would go extinct.

· Option 3: The population would decrease in size.

Part D

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1 ✓

The response includes an acceptable justification for the prediction provided in part (c). Acceptable justifications include the following.

· Justification for option 1: The alleles that allow the microbes to survive would have been passed down to their offspring, and the alleles would become more prevalent in the population because of natural selection.

· Justification for option 1: Microbes with different heat tolerances have different degrees of reproductive success.

· Justification for option 1: More-heat-tolerant individuals reproduce more than do less-heat-tolerant individuals.

· Justification for option 2: The (effective) population size will be too small/will have insufficient genetic variation to allow continued survival of the population.

· Justification for option 3: Most of the microbes cannot reproduce (therefore there will be fewer offspring).

FRQ

16. Read each question carefully. Write your response in the space provided for each part of each question. Answers must be written out in paragraph form. Outlines, bulleted lists, or diagrams alone are not acceptable and will not be scored.

During the Late Devonian Period, there was a massive loss of marine biodiversity. An analysis of fossil marine invertebrates that existed during the Late Devonian has revealed that the rate of extinction was relatively constant, as compared with the rates of extinction before and after the Late Devonian. The fossil evidence also suggests that frequent increases in sea level during the Late Devonian effectively removed geographical barriers and allowed generalist species (species that can inhabit a wide range of habitats) to migrate into new areas.

(a) **Describe** how the geographical separation of two populations can lead to speciation.

(b) **Explain** how a decrease in biodiversity during the Late Devonian could occur even though the extinction rate was relatively constant.

(c) Scientists claim that the modern rate of biodiversity loss may be even greater than that observed during the Late Devonian. In recent decades, the number of invasive species has increased. **Predict** the most likely effect of the current increase in invasive species on speciation rates of native species.

(d) Provide reasoning to **justify** your prediction in part (c).

Part A

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1 ✓

The response includes an acceptable description. Acceptable descriptions include the following.

- The populations (on either side of the geographical barrier) are no longer able to exchange genetic material.
- There is no gene flow between the populations (on either side of the geographical barrier).
- The populations (on either side of the geographical barrier) are no longer able to reproduce with each other.
- The populations on either side of the geographical barrier have different selective pressures/have different mutations/experience genetic drift in different directions.

Part B

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

FRQ

0	1 ✓

The response includes an acceptable explanation. Acceptable explanations include the following.

- The speciation rate could have been lower than the extinction rate.
- A decrease in the speciation rate could reduce biodiversity (even if extinction rates do not change).
- Genetic biodiversity could be reduced (if species undergo selection), without an increase in extinction rate.

Part C

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1 ✓

The response includes the prediction that there will be/has been a decrease (in speciation rates, compared with previous rates).

Part D

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1 ✓

The response includes an acceptable justification. Acceptable justifications include the following.

- Invasive species can outcompete native species/specialists for resources, leading to the extinction of many native species/limiting the ability of native species to spread to new habitats.
- An invasive species can prey on and drive (native prey) species to extinction (which would prevent these species from diverging).

FRQ

17. Read each question carefully. Write your response in the space provided for each part of each question. Answers must be written out in paragraph form. Outlines, bulleted lists, or diagrams alone are not acceptable and will not be scored.

A population of invasive feral hogs in a wilderness area is approaching carrying capacity. The feral hogs in each age-group of this population are counted once a year. Figure 1 shows the age pyramid of this population in one particular year. Feral hogs reproduce between 1 and 5 years of age. Mothers care for their young until the young are three months old.

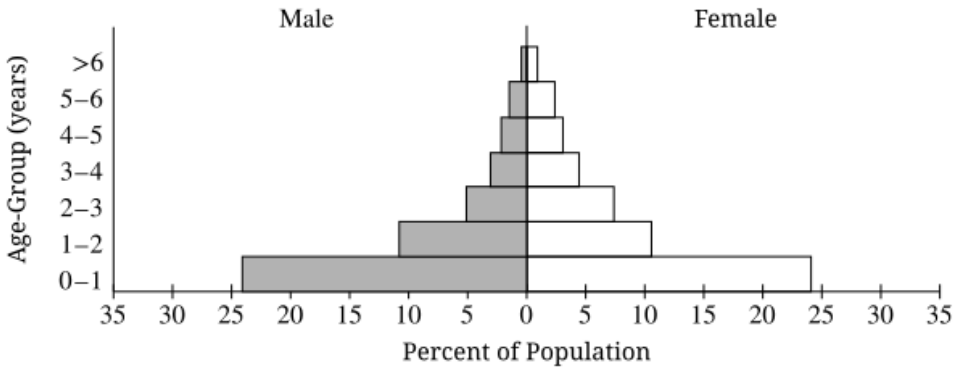


Figure 1. Age pyramid of a feral hog population in a wilderness area

Ten years later a researcher again counts the number of individual feral hogs in each age-group in the same population and produces the age pyramid shown in Figure 2.

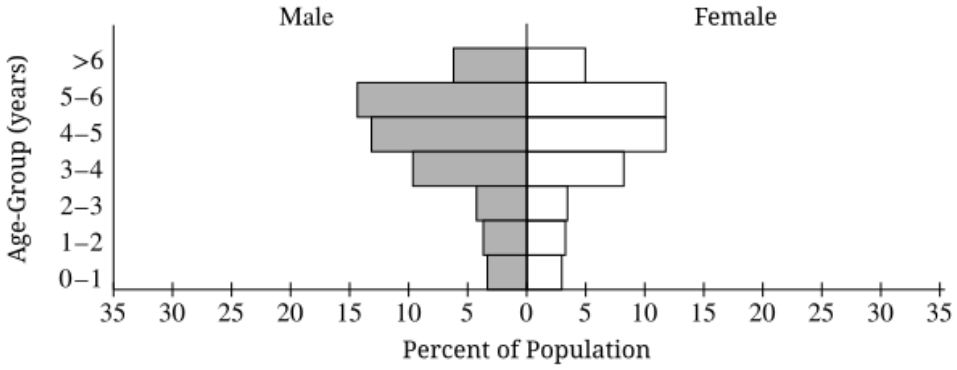


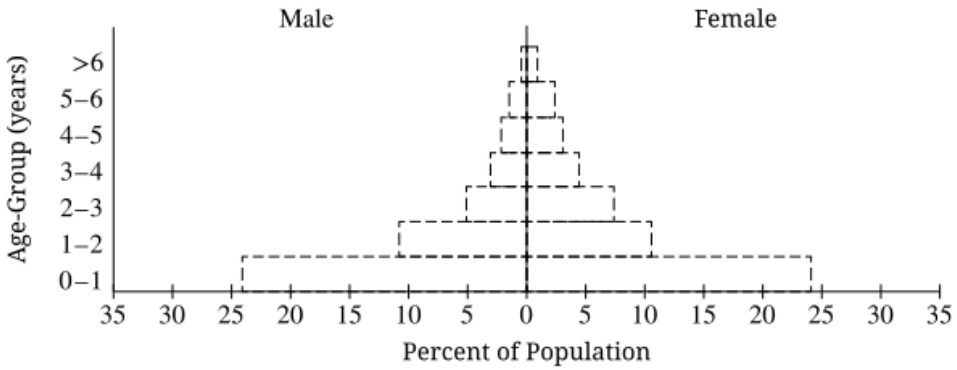
Figure 2. Age pyramid of the feral hog population ten years later

(a) **Describe** a characteristic of an ecosystem that allows an introduced species to become an invasive species.

FRQ

(b) A population age structure, as shown in Figure 1, is the result of multiple factors affecting birth and death rates of animals of various ages. **Explain** how ecological factors lead to the difference between the numbers of hogs in the $0-1$ year age-group and the number of hogs in the $1-2$ year age-group shown in Figure 1.

(c) An outbreak of a viral infection causes approximately half of the male hogs in each age-group to die. Using the template in the space provided for your response, **draw** new bars for each age-group to represent the approximate distribution of the entire hog population immediately after the death of the males. The dashed lines of the template represent the original age pyramid (Figure 1) before the viral outbreak.



(d) Using biological principles, **explain** what is likely to have caused the shift from the age structure in Figure 1 to the age structure shown in Figure 2.

Part A

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1 ✓

The response includes an acceptable description. Acceptable descriptions include the following.

· There is an available niche.

· There are sufficient resources.

· There are few/no competitors.

· There are few/no predators.

· There is a lack of parasites.

Part B

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

FRQ

0	1 ✓

The response includes an explanation that the 0-1 year age-group is more susceptible to predation/diseases/competition for food/abiotic fluctuations, and this is likely to contribute to higher death rates of individuals in this age group.

Part C

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1 ✓

The response includes a representation showing a decrease (approx 50%) in males on the left side and an increase in females on the right side.

Part D

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1 ✓

The response includes an acceptable explantion. Acceptable explanations include the following.

· The population exceeded the carrying capacity of the environment, and younger hogs were unable to compete for resources.

· The carrying capacity of the environment was reduced, and younger hogs became unable to compete for resources.

· There was a decrease in the reproductive rate, so there are fewer young hogs.

· There is a (new) predator/disease that causes lower survival rates among young hogs (compared with older hogs).

FRQ

18. Read each question carefully. Write your response in the space provided for each part of each question. Answers must be written out in paragraph form. Outlines, bulleted lists, or diagrams alone are not acceptable and will not be scored.

Vernal ponds, ponds that fill with water each spring and then evaporate in late summer, pose particular challenges to the organisms that live in them. Copepods, a type of crustacean, produce eggs that survive in dry sediment (the soil and other solid material at the bottom of a pond) for long periods of time. Once the water returns, the copepod larvae emerge from the eggs and begin to eat, consuming primarily algae. However, algae often emerge from the sediment later than the copepod larvae do, and the algal populations grow slowly in cool spring temperatures.

To investigate whether the copepods that live in vernal ponds have other adaptations to allow them to supplement their diet of algae, scientists examined related species that they categorized as large-bodied or small-bodied copepods. After examining the mouthparts of each species, the scientists determined a mouthpart carnivory score for each. A high score indicates a tendency toward a more carnivorous diet. The scientists also recorded the gut contents of each species in both early spring and summer. They hypothesized that large-bodied copepods have mouth adaptations that enable them to eat small animals that emerge early from sediments, while the small-bodied copepods do not have such adaptations.

(a) **Describe** the effect of periodic changes in resource availability on population density.

(b) **Justify** the researchers' performing their measurements and analyzing gut contents in both early spring and summer.

(c) **State** a null hypothesis for this investigation.

(d) Climate change is causing water in vernal ponds to evaporate earlier in the summer. Scientists claim that under these conditions, copepod species with a lower mouthpart carnivory score are more likely to go extinct in these vernal ponds than are species with a greater mouthpart carnivory score. Provide reasoning to **support** the scientists' claim.

Part A

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1 ✓

The response includes an acceptable description. Acceptable descriptions include the following.

- The population density will fluctuate.
- The population will fluctuate around the carrying capacity.

Part B

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

FRQ

0	1 ✓

The response justifies that the researchers' performing their measurements in both early spring and summer allowed the scientists to see whether/how the diets of the copepods changed (with changing conditions).

Part C

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1 ✓

The response states an acceptable null hypothesis. Acceptable null hypotheses include the following.

- There is no (statistical) difference in diet among copepods of all sizes.
- There is no difference in (mouthpart) adaptations between large-bodied and small-bodied copepods.
- The (mouth) adaptations of a copepod do not correlate with the copepod's diet.
- The mouthpart carnivory score of a copepod does not correlate with the copepod's diet.
- The time of year does not correlate with diet.

Part D

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1 ✓

The response includes the support that (if the ponds dry up) the algae will die, so the copepods (that have a lower mouthpart carnivory score) will die before they have a chance to lay eggs.

2023

AP® Biology

Scoring Guidelines

Question 1: Interpreting and Evaluating Experimental Results with Experimental Design — 9 points

In eukaryotic microorganisms, the PHO signaling pathway regulates the expression of certain genes. These genes, *Pho* target genes, encode proteins involved in regulating phosphate homeostasis. When the level of extracellular inorganic phosphate (Pi) is high, a transcriptional activator Pho4 is phosphorylated by a complex of two proteins, Pho80–Pho85. As a result, the *Pho* target genes are not expressed. When the level of extracellular Pi is low, the activity of the Pho80–Pho85 complex is inhibited by another protein, Pho81, enabling Pho4 to induce the expression of these target genes. A simplified model of this pathway is shown in Figure 1.

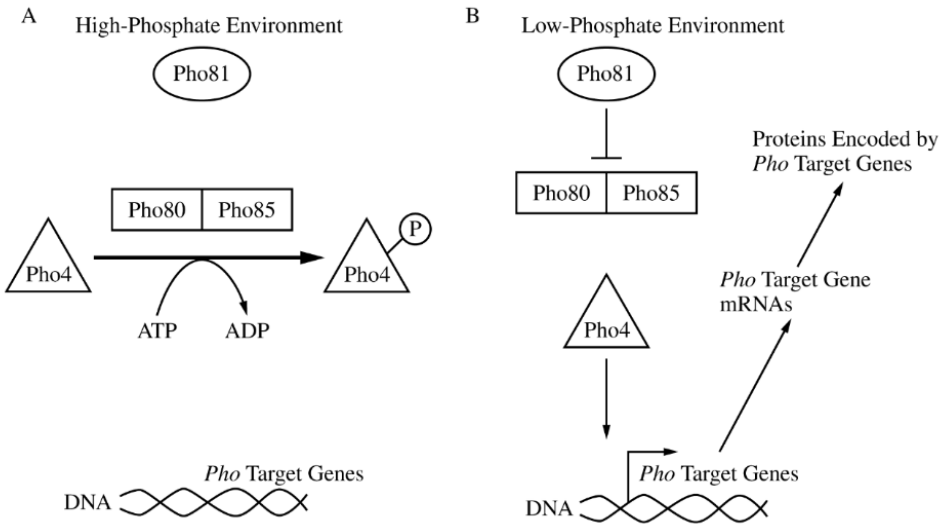


Figure 1. A simplified model of the regulation of expression of *Pho* target genes in (A) a high-phosphate (high-Pi) environment and (B) a low-phosphate (low-Pi) environment

To study the role of the different proteins in the PHO pathway, researchers used a wild-type strain of yeast to create a strain with a mutant form of Pho81 (*pho81mt*) and a strain with a mutated form of Pho4 (*pho4mt*). In each of these mutant strains, researchers measured the activity of a particular enzyme, APase, which removes phosphates from its substrates and is encoded by *PHO1*, a *Pho* target gene (Table 1). They then determined the level of *PHO1* mRNA relative to that of the wild-type yeast strain, which was set to 10.

TABLE 1. APase ACTIVITY AND RELATIVE AMOUNTS OF *PHO1* mRNA IN WILD-TYPE AND MUTANT STRAINS OF YEAST IN HIGH- AND LOW-PHOSPHATE ENVIRONMENTS

Yeast Strain	Mutation	APase Activity in High-P_i Environment (mU/mL/OD_{600}) $\pm 2SE_{\bar{x}}$	APase Activity in Low-P_i Environment (mU/mL/OD_{600}) $\pm 2SE_{\bar{x}}$	Relative Amounts of *PHO1* mRNA in High-P_i Environment $\pm 2SE_{\bar{x}}$	Relative Amounts of *PHO1* mRNA in Low-P_i Environment $\pm 2SE_{\bar{x}}$
Wild-type	None	0.5 ± 0.1	17.3 ± 0.9	0.1 ± 0.0	10 ± 2.0
pho81mt	Nonfunctional Pho81	0.4 ± 0.1	0.6 ± 0.1	0.7 ± 0.2	0.9 ± 0.8
pho4mt	Nonfunctional Pho4	0.5 ± 0.0	0.8 ± 0.2	0.6 ± 0.4	0.3 ± 0.1

(a) **Describe** the effect the addition of a charged phosphate group can have on a protein that would cause the protein to become inactive. **1 point**

- It changes the structure/shape of the protein.

Explain how a signal can be amplified during signal transduction in a pathway such as the PHO signaling pathway. **1 point**

- Each enzyme (in a signal transduction pathway) can act on many copies of a protein.

Total for part (a) **2 points**

(b) Based on Table 1, **identify** a dependent variable in the researchers' experiment. **1 point**

Accept one of the following:

- APase activity
- (Relative) amount of *PHO1* (mRNA)

Justify the researchers' using the wild-type strain for the creation of the mutant strains. **1 point**

Accept one of the following:

- It ensures that any observed differences (in experimental results) between the strains are due to the introduced mutations (and not to other genetic differences between the yeast strains).
- It ensures that the strains are genetically identical except for the introduced mutations.

Justify the researchers' using mutant strains in which only a single component of the pathway was mutated in each strain. **1 point**

Accept one of the following:

- It allows them to test the effect of each mutation separately.
- It allows them to (better) determine which component is responsible for any observed differences.

Total for part (b) **3 points**

(c)	Based on the data in Table 1, **identify** the yeast strain and growth conditions that lead to the highest relative amount of *PHO1* mRNA. • Wild-type yeast in a low-Pi environment	**1 point**
	Calculate the percent change in APase activity in wild-type yeast cells in a high-Pi environment compared with that of wild-type cells in a low-Pi environment. Accept one of the following: • 3,360% [(17.3–0.5)/0.5 × 100%] • -97% [(0.5–17.3)/17.3 × 100%]	**1 point**
	Total for part (c)	**2 points**
(d)	In a follow-up experiment, researchers created a strain of yeast with a mutation that resulted in a nonfunctional Pho85 protein. Based on Figure 1, **predict** the effects of this mutation on *PHO1* expression in the mutant strain in a high-Pi environment. • It/*PHO1*/Target genes will be expressed.	**1 point**
	Provide reasoning to **justify** your prediction. • (In a high-Pi environment) a nonfunctional Pho85 will be unable to phosphorylate/inhibit Pho4.	**1 point**
	Total for part (d)	**2 points**
	Total for question 1	**9 points**

Question 2: Interpreting and Evaluating Experimental Results with Graphing — 9 points

Elevated levels of CO_2 increase the rate of photosynthesis and growth in plants. Scientists studying the mechanisms involved in these increases examined a variety of species and found that when plants are exposed to elevated levels of CO_2, there is an increase in the number of chloroplasts per cell. To investigate whether the elevated levels of CO_2 have a similar effect on the number of mitochondria in plant cells, the scientists then selected six of these species to quantify the number of mitochondria per cell when the plants were exposed to both normal and elevated levels of CO_2 (Table 1).

TABLE 1. AVERAGE NUMBER OF MITOCHONDRIA IN PLANTS EXPOSED TO NORMAL AND ELEVATED LEVELS OF CO_2

Species	Mitochondria at Normal CO_2 (per 100 µm^2 of cell area) $\pm 2SE_{\bar{x}}$	Mitochondria at Elevated CO_2 (per 100 µm^2 of cell area) $\pm 2SE_{\bar{x}}$
1	1.0 ± 0.10	1.6 ± 0.10
2	0.4 ± 0.05	0.9 ± 0.08
3	0.5 ± 0.07	0.9 ± 0.10
4	0.3 ± 0.03	0.6 ± 0.06
5	0.7 ± 0.06	1.5 ± 0.22
6	1.3 ± 0.15	2.4 ± 0.22

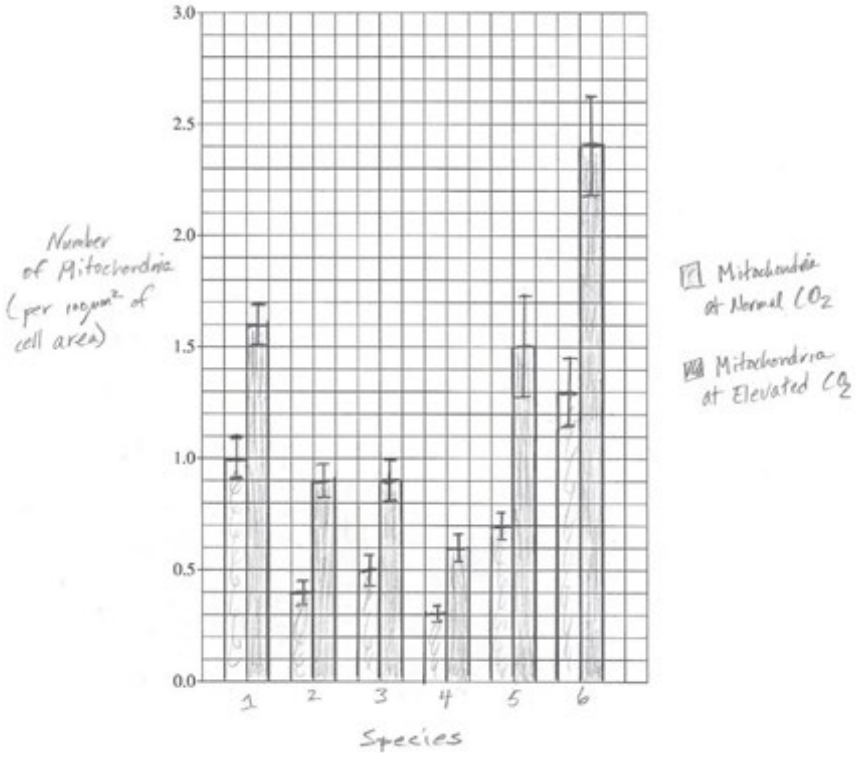
(a) **Describe** the role of the inner mitochondrial membrane in cellular respiration. **1 point**

Accept one of the following:

- It provides the location for the components of the electron transport chain/ATP synthase/oxidative phosphorylation.
- It separates (reactions in) the intermembrane space from (reactions in) the matrix.
- It allows the establishment of a proton gradient.

(b) Using the template in the space provided for your response, **construct** an appropriately labeled graph that represents the data in Table 1. **1 point**

Sample response:



- Appropriate labelling

Using the template in the space provided for your response, **construct** an appropriately labeled graph that represents the data in Table 1. **1 point**

- Data are represented in a bar/modified bar graph.

Using the template in the space provided for your response, **construct** an appropriately labeled graph that represents the data in Table 1. **1 point**

- Data points and error bars are correctly plotted.

Determine which species show(s) a difference in the number of mitochondria between normal and elevated levels of CO_2. **1 point**

- All of the species

Total for part (b) **4 points**

(c) Based on the data in Table 1, **describe** the relationship between the level of CO_2 and the average number of mitochondria per unit area of a cell. **1 point**

Accept one of the following:

- The number of mitochondria is greater under conditions of elevated CO_2 (than under normal CO_2).
- It is a positive relationship/correlation.

(d)	The leaves of a particular plant species are typically green, but scientists notice a plant in which the leaves have white stripes. They determine that the stripes result from a mutation in mitochondrial DNA that interferes with the development of chloroplasts. The scientists crossed plants using pollen from the plant with white-striped leaves and ovules from a plant with green leaves. **Predict** the phenotype(s) of the leaves of offspring produced from this cross. • The leaves will be (all) green/not have white stripes.	**1 point**
	Provide reasoning to **justify** your prediction. • (All offspring will have the same leaf phenotype as the ovule-producing plant because) mitochondria are maternally inherited/transferred by the ovule.	**1 point**
	Explain why plants with the same genotype are able to differ in the structure and/or number of certain organelles in response to changes in atmospheric levels of CO_2. • (Plants have different phenotypes because) changes in CO_2 levels/the environment affect the expression of certain genes.	**1 point**
	Total for part (d)	**3 points**
	Total for question 2	**9 points**

Question 3: Scientific Investigation — 4 points

Sand lances of the genus *Ammodytes* are small fish that function as keystone organisms in several coastal ecosystems. These sand lances are prey fish that support organisms at higher trophic levels. Scientists performed experiments to examine how sand lance populations are likely to be affected by the rising temperatures and CO_2 levels associated with climate change.

Sand lance embryos typically develop and mature into adult fish at low temperatures (approximately 5°C) and stable, low CO_2 levels (approximately 400 µatm). Over the course of two years, the scientists measured the survival rate of sand lance embryos allowed to develop and mature in a laboratory at three different temperatures, 5°C, 7°C, and 10°C, with the level of CO_2 maintained at 400 µatm, 1,000 µatm, and 2,100 µatm for each temperature.

(a) **Describe** the effect of increased biodiversity on the resilience of an ecosystem in a changing environment. — **1 point**

- (Ecosystem) resilience/it will be greater (with increased biodiversity).

(b) **Justify** the scientists' selecting 5°C as the lowest temperature and 400 µatm as the lowest CO_2 level in their study of sand lance embryo survival. — **1 point**

Accept one of the following:

- These are the normal/current conditions at which the embryos develop and were used as a basis for comparison.
- These (current) conditions were used as a basis to compare the effects of changes in environmental conditions/increases in temperature and CO_2.

(c) **State** a null hypothesis for the experiment. — **1 point**

Accept one of the following:

- Climate change will have no effect on sand lance (embryo) survival/sand lance development/the size of sand lance populations.
- (Increases in) temperature/ CO_2 levels will have no effect on sand lance (embryo) survival/sand lance development /the size of sand lance populations.
- There will be no difference in the sand lance (embryo) survival rates/sand lance development/the size of sand lance populations measured at all/different temperatures and CO_2 levels.

(d) The scientists claim that a reduction in the population size of the *Ammodytes* sand lances will affect the stability of the entire coastal ecosystem. Provide reasoning to **support** the scientists' claim. — **1 point**

Accept one of the following:

- There will be a negative effect on other trophic levels because the sand lance provides food for many other species.
- There will be a negative effect on other trophic levels because there will be reduced energy to transfer (to higher trophic levels).

Total for question 3 — 4 points

Question 4: Conceptual Analysis — 4 points

Noncyclic electron flow and cyclic electron flow are two major pathways of the light-dependent reactions of photosynthesis. In noncyclic electron flow, electrons pass through photosystem II, then components of a chloroplast electron transport chain, and then photosystem I before finally reducing $NADP^+$ to NADPH. In cyclic electron flow, electrons cycle through photosystem I and some components of the electron transport chain (Figure 1).

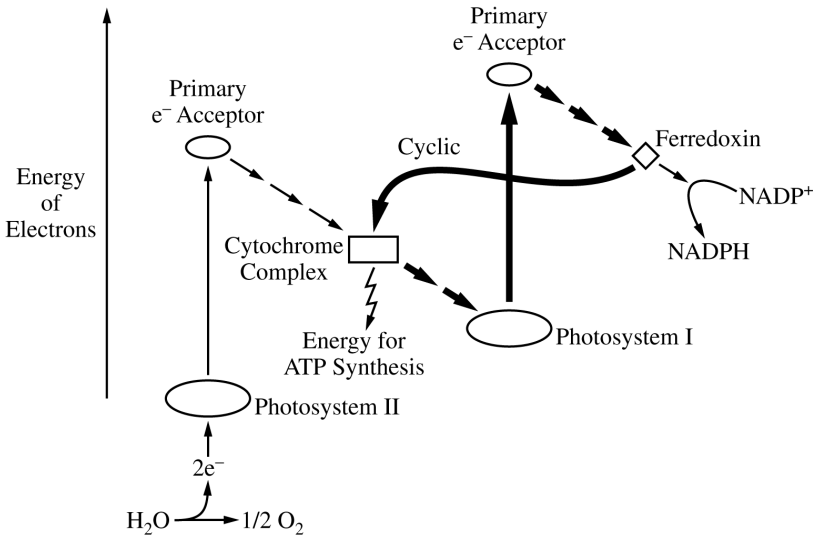


Figure 1. The pathways of noncyclic and cyclic (heavy arrows) electron flow. The cytochrome complex is a component of the electron transport chain between the two photosystems.

(a)	**Describe** the role of chlorophyll in the photosystems of plant cells. Accept one of the following: • Chlorophyll captures/absorbs light (energy). • Chlorophyll receives electrons (from water)/receives electrons (from an electron transport chain)/transfers electrons (to an electron transport chain).	**1 point**
(b)	Based on Figure 1, **explain** why an increase in the ratio of NADPH to $NADP^+$ will cause an increase in the flow of electrons through the cyclic pathway. • There is less/no $NADP^+$ to accept the electrons, so the electrons pass (instead) to the cyclic pathway/from ferredoxin to the cytochrome complex.	**1 point**
(c)	Using rice plants, scientists examined the effect of a mutation that results in the loss of the protein CRR6. CRR6 is a part of the photosystem I complex, and its absence reduces the activity of photosystem I. **Predict** the effect of the mutation on the rate of biomass (dry weight) accumulation. • The rate (of biomass accumulation)/Biomass/It will be lower (in comparison with plants without the mutation).	**1 point**
(d)	**Justify** your prediction in part (c). • There will be insufficient ATP/NADPH produced for the synthesis of carbohydrates/the Calvin cycle.	**1 point**
		Total for question 4 — 4 points

Question 5: Analyze Model or Visual Representation of a Biological Concept or Process — 4 points

Ruminants are hoofed animals, including cattle and sheep, that have a unique four-chambered stomach specialized to digest tough, fiber-filled grasses. Researchers studying ruminants are investigating the morphological and molecular characteristics of different ruminant families in order to determine the evolutionary relationships among the families. Cladograms of several ruminant families were constructed based on morphological data (Figure 1A) and molecular data (Figure 1B). Table 1 shows a sample of the morphological characteristics present in each family used to construct the cladogram in Figure 1A.

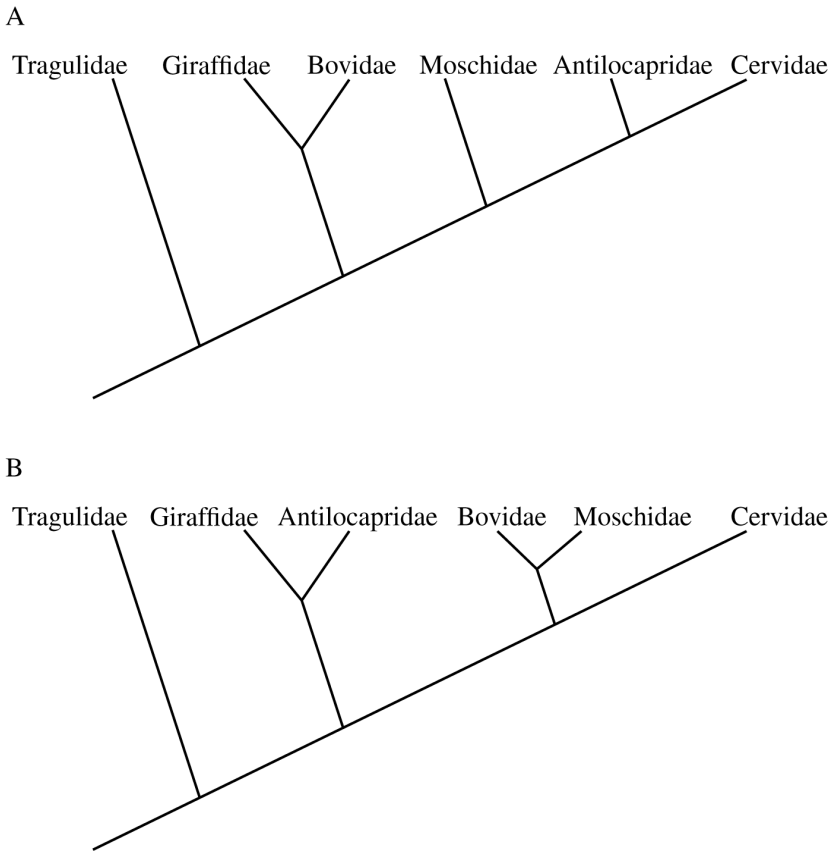
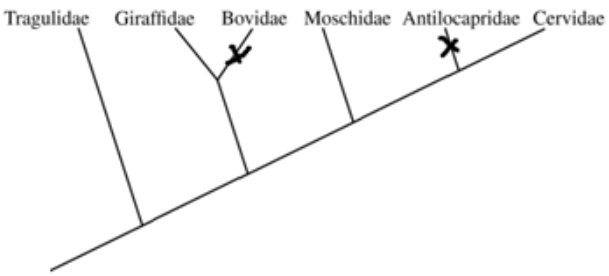


Figure 1. Cladogram of six ruminant families based on (A) morphological data and (B) molecular data

TABLE 1. MORPHOLOGICAL CHARACTERISTICS FOUND IN EACH RUMINANT FAMILY

Characteristic Number	Morphological Characteristic	Tragulidae	Giraffidae	Bovidae	Moschidae	Antilocapridae	Cervidae
1	Extra tooth material			X		X	
2	Third stomach		X	X	X	X	X
3	Double opening for tear ducts					X	X

(a)	**Describe** how a scientist would use a comparison of the DNA sequences of different organisms to suggest the most likely evolutionary relationship among the organisms. • The (DNA) sequences of organisms that are more closely related are more similar (than those of organisms that are less closely related).	**1 point**
(b)	Based on Figure 1, **explain** why Bovidae is likely to be more closely related to Moschidae than it is to Giraffidae. Accept one of the following: • (The molecular data/Figure 1B support(s) this relationship, and) molecular data are more reliable (than are morphological data). • (The molecular data/Figure 1B support(s) this relationship.) Morphological similarities may not reliably indicate evolutionary relatedness.	**1 point**
(c)	Using the template in the space provided for your response, **represent** the point(s) at which characteristic 1, listed in Table 1, evolved by marking "X" on the line(s) of the cladogram in the correct location(s). Sample Response:  • Response must show an X placed on the line leading to Bovidae and an X placed on the line leading to Antilocapridae.	**1 point**
(d)	Based on Figure 1A, **explain** why a characteristic found only in the Cervidae and Bovidae families is more likely evidence of convergent evolution than it is of common ancestry. Accept one of the following: • There are other families that have the same common ancestor as the Bovidae and Cervidae families but do not have the characteristic. • It is more likely that the characteristic arose independently in Cervidae and Bovidae than it arose in their common ancestor and was lost in Giraffidae, Moschidae, and Antilocapridae.	**1 point**
		Total for question 5 **4 points**

Question 6: Analyze Data — 4 points

Housekeeping genes encode proteins involved in universally important processes such as transcription, translation, and glycolysis. Because these genes appear to be expressed in all cells at constant levels, the expression of housekeeping genes is often used as a control when comparing how the expression of other genes varies under different conditions.

Researchers studying the effect of pesticides on declining bee populations wanted to determine whether the expression of four housekeeping genes (*GAPDH*, *RPL32*, *RPS5*, and *TBP-AF*) was in fact constant in bees across different variables. The researchers collected samples of mRNA for each of the four genes and compared how their expression varied across the developmental stage of the bee, the sex of the bee, and the cell type from which the sample was taken. The mRNA from the samples was reverse transcribed to produce DNA copies of each gene. PCR was then used to amplify the DNA, and the Cq value was determined. The Cq value is the number of PCR cycles needed to produce a specified number of DNA copies (Figure 1). A high Cq value for a sample indicates the gene was expressed at a low level.

To analyze whether any of the examined variables affected expression of the housekeeping genes, researchers examined the range of Cq values for each gene in response to each variable. Genes with a wide range of Cq values were determined to be affected by the variable, while genes with a narrow range of Cq values were determined to be unaffected by the variable.

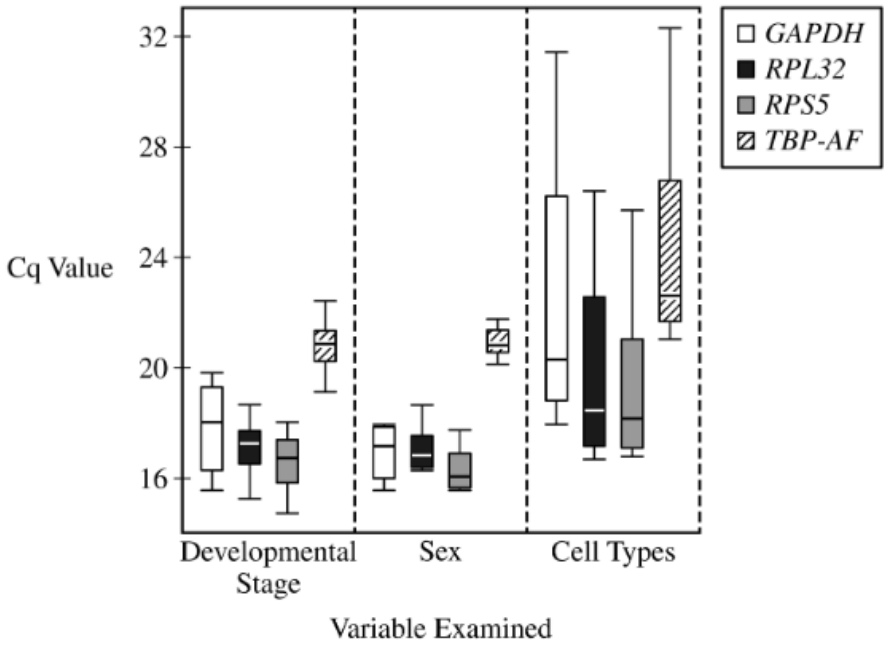


Figure 1. The effect of developmental stage, sex, and cell type on the Cq value of four housekeeping genes

(a)	Based on the data in Figure 1, **identify** the gene that had the lowest median Cq value when bees of different developmental stages were compared. • *RPS5*	**1 point**
(b)	The Cq value is inversely proportional to the amount of mRNA from that gene in the starting sample. Based on the data in Figure 1, **identify** the gene that has the lowest level of gene expression regardless of variable. • *TBP-AF*	**1 point**
(c)	The scientists investigated the effect of pesticides on the expression of other genes in one cell type of a group of bees containing males and females of the same developmental stage. They hypothesized that *TBP-AF* would serve as the best control gene for this experiment. Use the data to **evaluate** their hypothesis. • Their hypothesis is supported because *TBP-AF* has the smallest Cq range/most constant expression (when comparing sexes).	**1 point**
(d)	**Explain** how expression of a gene such as *GAPDH* can vary from one cell type to another within the same bee. • Different cell types contain different levels of/different transcription factors, and therefore regulate the expression of genes in different ways.	**1 point**
	Total for question 6	**4 points**

AP® Biology

Scoring Guidelines

Question 1: Interpreting and Evaluating Experimental Results with Experimental Design — 9 points

Crossing over in meiosis I is required for homologous chromosomes to properly align during metaphase and segregate during the first cell division.

Some regions of a chromosome called hotspots display a higher frequency of crossing over than other regions do. Crossing over is suppressed in chromosomal regions near the centromeres. The centromere region of a duplicated chromosome includes a collection of proteins that form a structure called the kinetochore. Scientists hypothesized that one or more of these kinetochore proteins are responsible for suppressing crossing over around the centromere.

To investigate their hypothesis, scientists modified chromosome 8 in yeast such that, in each cell, one chromosome from the pair of homologous chromosome 8s contained the gene encoding red fluorescent protein (RFP), while the other chromosome from the pair contained the gene encoding green fluorescent protein (GFP). Cells expressing RFP emit (give off) red light, and cells expressing GFP emit green light. Models of the modified chromosome 8 both before and after crossing over are shown in Figure 1.

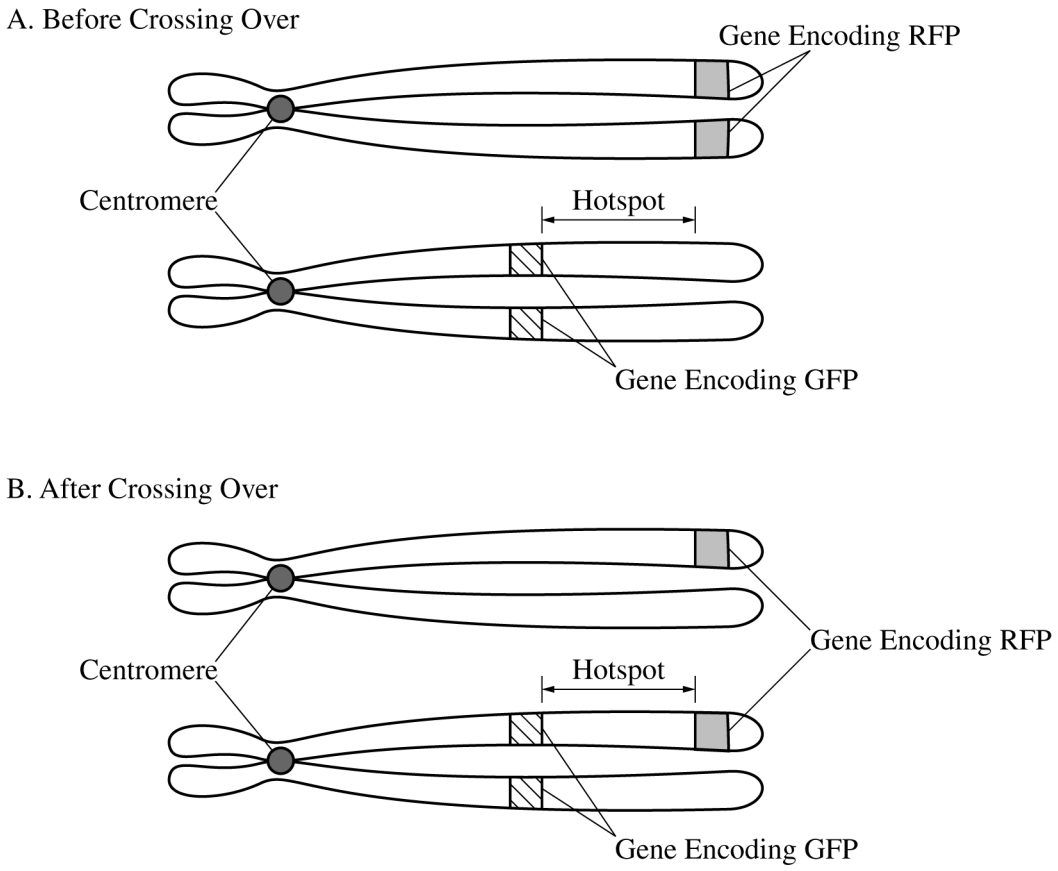


Figure 1. Models of modified chromosome 8 used in the experiment (A) before and (B) after crossing over occurs at the hotspot

The scientists then investigated whether attaching individual kinetochore proteins to a specific DNA sequence present in a known crossing-over hotspot on chromosome 8 affected the frequency of crossing

over at this location. In their first experiment, they examined three groups of yeast cells containing the modified chromosome 8. Group 1 contained no kinetochore proteins attached to the hotspot, group 2 contained the kinetochore protein CTF attached to the hotspot, and group 3 contained the kinetochore protein IML attached to the hotspot. For each group, the scientists determined the frequency of crossing over between the RFP and GFP genes. To determine the frequency, the scientists added the number of cells emitting both red and green light to the number of cells that emitted no light and divided by the total number of cells (Figure 2).

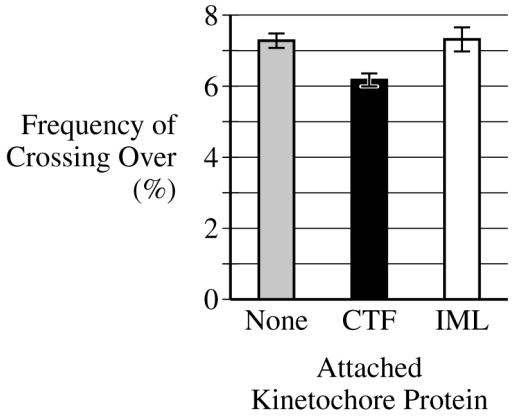


Figure 2. The frequency of crossing over in a hotspot on yeast chromosome 8 for cell groups treated with different kinetochore proteins. Error bars represent $\pm 2\mathrm{SE}_{\bar{X}}$.

(a)	**Describe** the function of S phase of interphase. Accept one of the following: • (The function of S phase is to) replicate/duplicate/synthesize the DNA/chromosomes. • (The function of S phase is to) double the amount of DNA.	**1 point**
	Explain why some haploid cells formed after meiosis in this experiment will have only one fluorescent marker. Accept one of the following: • Some cells will receive a chromosome/(sister) chromatid that did not undergo crossing over/recombination (in the hotspot). • Two (or a multiple of two) crossing-over events occurred (in the hotspot). • Each daughter cell will receive one of the four sister chromatids. Two of the sister chromatids shown in Figure 1 have only one fluorescent protein gene. • The two chromosomes that were not involved in crossing over will have only the RFP gene or the GFP gene. After meiosis II, a gamete with one of these chromosomes will have only the RFP or the GFP gene.	**1 point**
		Total for part (a) 2 points
(b)	**Identify** the control group for the scientists' first experiment, shown in Figure 2. Accept one of the following: • Group 1 • (The group called) None • Cells with no (kinetochore) protein (attached to the chromosome)	**1 point**

	In a follow-up experiment, the scientists created a modified version of CTF in which the DNA-binding portion had been removed. They compared the frequency of crossing over in yeast cells in the presence and absence of unmodified CTF with that in yeast cells in the presence and absence of the modified CTF protein (data not shown). In the follow-up experiment, **justify** why the scientists used a modified CTF protein that is unable to bind to DNA as a control. Accept one of the following: • (Using a modified CTF enabled the scientists) to determine whether DNA binding of the CTF/kinetochore protein affects/inhibits crossing over/recombination. • (Using a modified CTF enabled the scientists) to determine whether just the presence of the CTF/kinetochore protein is enough to affect/inhibit the crossing over/recombination frequency.	**1 point**
	Identify the independent variable in the follow-up experiment. Accept one of the following: • The type of CTF used • The presence or absence of (modified/unmodified) CTF	**1 point**
		Total for part (b) 3 points
(c)	Based on Figure 2, **describe** the effect on the frequency of crossing over when CTF is attached to the chromosome 8 hotspot compared with the effect when IML is attached to the hotspot. Accept one of the following: • CTF attachment results in a decreased/lower frequency of crossing over/recombination, (whereas IML had no effect). • IML attachment results in an increased/a higher frequency of crossing over/recombination (compared with CTF attachment).	**1 point**
(d)	**Predict** the effect on the number of copies of chromosome 8 likely to be present in the resulting daughter cells when CTF is attached to the hotspot. Accept one of the following: • There will be zero/two (copies). • There will be one less/one extra (copy).	**1 point**
	Provide reasoning to **justify** your prediction. • Cells (with attached CTF molecules) undergo crossing over/recombination at a lower frequency, so it is more likely that nondisjunction would occur/chromosomes would not separate properly.	**1 point**
	Explain how the presence of hotspots (Figure 1) could increase the likelihood that a population will survive in the presence of selective pressures. • Hotspots would increase genetic diversity; therefore, it would be more likely that some individuals would survive and reproduce.	**1 point**
		Total for part (d) 3 points
		Total for question 1 9 points

Question 2: Interpreting and Evaluating Experimental Results with Graphing — 9 points

To investigate how increases in environmental temperatures affect the metabolism of certain organisms, researchers incubated liver cells from toads at different temperatures and measured two markers of metabolic activity (Table 1): the rate of oxygen consumption and the rate of ATP synthesis.

TABLE 1. RATE OF OXYGEN CONSUMPTION AND ATP SYNTHESIS AT DIFFERENT TEMPERATURES

Metabolic Marker	20°C	25°C	30°C
Rate of Oxygen Consumption (nmol/min/mg of mitochondrial protein $\pm 2\mathrm{SE}_{\bar{X}}$)	12.8 ± 2.2	16.5 ± 2.0	22.1 ± 0.7
Rate of ATP Synthesis (nmol/min/mg of mitochondrial protein $\pm 2\mathrm{SE}_{\bar{X}}$)	12.6 ± 1.6	16.8 ± 2.0	21.07 ± 0.8

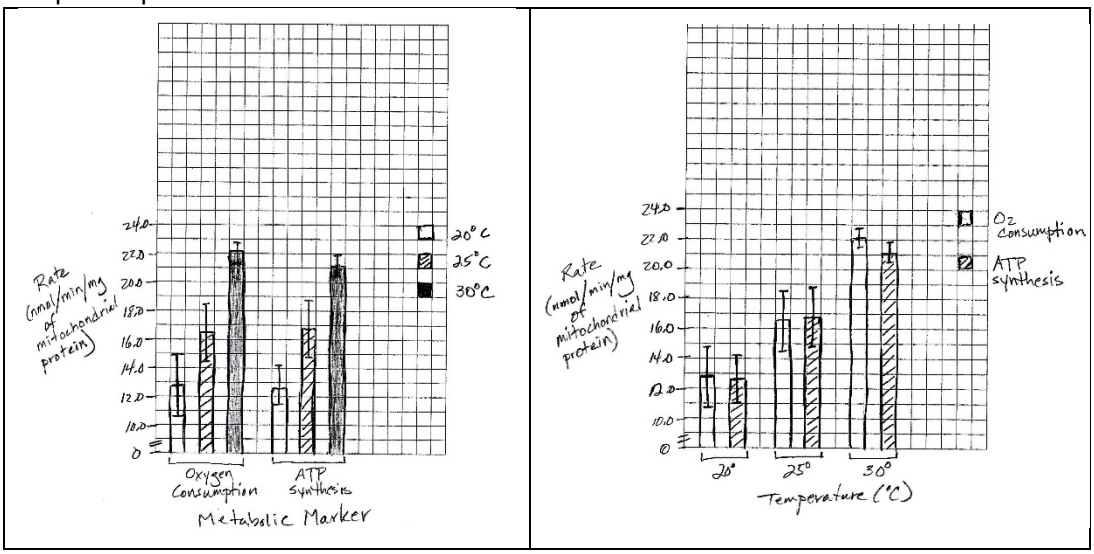
(a) **Describe** the role of water in the hydrolysis of ATP. **1 point**

Accept one of the following:

- Water is added in the process of cleaving/splitting (a phosphate from) ATP.
- Water breaks down/splits ATP.

(b) Using the template in the space provided for your response, **construct** a bar graph that represents the data shown in Table 1. Your graph should be appropriately plotted and labeled. **1 point**

Sample Responses:



- Data are represented in a bar graph.

Using the template in the space provided for your response, **construct** a bar graph that represents the data shown in Table 1. Your graph should be appropriately plotted and labeled. **1 point**

- Graph is appropriately labeled.

	Using the template in the space provided for your response, **construct** a bar graph that represents the data shown in Table 1. Your graph should be appropriately plotted and labeled. • Data points and error bars are correctly plotted.	**1 point**
	Based on the data provided, **determine** the temperature in $°C$ at which the rate of oxygen consumption is different from the rate of oxygen consumption at $25°C$. • 30	**1 point**
		Total for part (b) 4 points
(c)	Based on the data in Table 1, **describe** the effect of temperature on the rate of ATP synthesis in liver cells from toads. Accept one of the following: • As the temperature increases, the rate of ATP synthesis also increases. • There is a positive relationship (between temperature and ATP synthesis). • Temperature and ATP synthesis are directly correlated.	**1 point**
	Based on the data in Table 1**, calculate** the average amount of oxygen consumed, in $nmol$, for 10 mg of mitochondrial protein after 10 minutes at $25°C$. 1,650 [$16.5\ nmol/min/mg \times 10\ mg \times 10\ min$]	**1 point**
		Total for part (c) 2 points
(d)	Oligomycin is a compound that can block the channel protein function of ATP synthase. **Predict** the effects of using oligomycin on the proton gradient across the inner mitochondrial membrane. Accept one of the following: • (The proton gradient) will increase/become steeper (and may eventually plateau). • The difference in the concentration of protons/pH (across the inner mitochondrial membrane) will increase. • There will be an increase in the concentration of protons/a decrease in pH in the intermembrane space relative to that found within the mitochondrial matrix.	**1 point**
	Justify your prediction. Accept one of the following: • (Without protons being able to flow back into the matrix through ATP synthase), more protons will accumulate in the intermembrane space/between the two mitochondrial membranes. • (Without protons being able to flow back into the matrix through ATP synthase), there will be a lower pH in the intermembrane space/between the two mitochondrial membranes. • Protons will not be able to flow across the membrane (through ATP synthase), but the electron transport chain will still pump protons into the intermembrane space.	**1 point**
		Total for part (d) 2 points
		Total for question 2 9 points

Question 3: Scientific Investigation — 4 points

To investigate whether red blood cells of animals lose the ability to take in glucose from their environment as they age, scientists collected red blood cells from guinea pigs that ranged in age from one day old to seven months old. Scientists incubated an equal number of red blood cells in separate culture dishes that contained a $300\ \text{nM}$ solution of radioactively labeled glucose. The amount of radioactively labeled glucose present inside the red blood cells of each group was measured over time.

(a) **Describe** a difference between passive transport and active transport. **1 point**

Accept one of the following:

- Active transport requires energy/ATP (while passive transport does not).
- Passive transport does not require energy/ATP (while active transport does).
- In passive transport, substances move from a high concentration to a low concentration, (while in active transport substances move from a low concentration to a high concentration).
- In active transport, substances move from a low concentration to a high concentration (while in passive transport substances move from a high concentration to a low concentration).

(b) **Justify** why the scientists used an equal number of red blood cells in each culture dish as a control. **1 point**

Accept one of the following:

- (Scientists used an equal number of cells) to attribute differences in results/glucose transport to guinea pig age, (rather than to the number of cells used in the experiment).
- (Scientists used an equal number of cells) to compare results from the different dishes (containing cells from guinea pigs of different ages).
- (Scientists used an equal number of cells) to eliminate the number of cells as a variable (that might affect the amount of glucose in each group).
- (Scientists used an equal number of cells) because the number of cells used might affect the results/amount of glucose (present inside the red blood cells).

(c) Glucose transporters are required for the facilitated diffusion of glucose into red blood cells. The scientists claim that the expression of the gene encoding these transporters decreases as guinea pigs age. If the scientists' claim is supported by experimental data, **predict** the effect of increased age on the amount of radioactively labeled glucose present inside the cells of each group. **1 point**

- (As guinea pig age increases) the amount of glucose (inside the cells) decreases.

(d) **Justify** your prediction in part (c). **1 point**

- With fewer transporters, fewer glucose molecules will be moved into the cells.

Total for question 3 **4 points**

Question 4: Conceptual Analysis — 4 points

The common wild oat is native to regions of Europe and Asia but is an invasive species in central California grasslands. In California, the common wild oat has almost completely replaced some species of native bunchgrass. Researchers found that aphids, a type of small insect that often carries plant viruses, have a much higher reproductive rate in grasslands that include the common wild oat than in grasslands composed of only native bunchgrass species. Additionally, the viruses carried by the aphids appear to affect only the native bunchgrasses and not the common wild oat. Native bunchgrasses infected by the virus have much higher death rates than do native bunchgrasses that are not infected.

(a)	**Describe** the change in the resilience of an ecosystem when there is a decrease in the number of species. • (The resilience of the ecosystem) will decrease.	**1 point**
(b)	**Explain** how the addition of the common wild oat affects the number of native bunchgrass plants that can be supported by the California grasslands ecosystem. Accept one of the following: • (The addition of the wild oat) limits the resources available (to the native plants), resulting in a decrease (in the population size). • (The addition of the wild oat) enables the aphid population to increase/increases the exposure to viruses, resulting in a decrease (in the population size of the native bunchgrass).	**1 point**
(c)	Researchers suggest adding ladybugs, predators of aphids, to the California grasslands. **Predict** the effect of adding ladybugs on the abundance of the native bunchgrass population. • (The native bunchgrass population) will increase (in abundance).	**1 point**
(d)	**Justify** your prediction in part (c). • (Adding ladybugs) will decrease the number of aphids, which will cause a decrease in the transmission of/infection by plant viruses.	**1 point**
	Total for question 4	**4 points**

Question 5: Analyze Model or Visual Representation of a Biological Concept or Process — 4 points

Researchers study mechanisms that enable or prevent speciation.

New genes can evolve from noncoding regions of DNA. It is not until certain regulatory elements are present in the DNA that a noncoding region becomes a new, functional gene that encodes a protein. These regulatory elements include a promoter, a 5′ untranslated region (UTR) followed by a start codon, and a 3′ UTR following a stop codon (Figure 1).

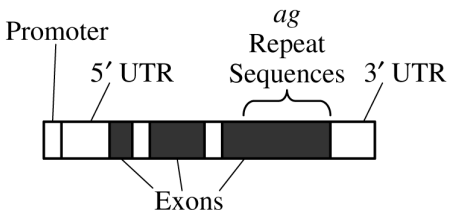
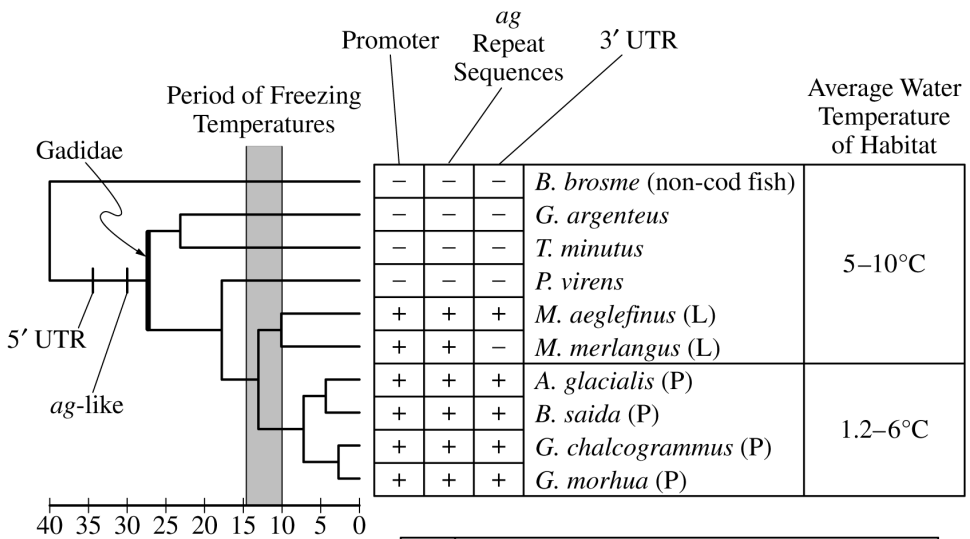


Figure 1. Basic structure of a functional *ag* gene

Researchers studied the evolution of the family of antifreeze-glycoprotein (AG) encoding genes in Gadidae, a family of marine fish known as cods. When present in the fish, these glycoproteins reduce the freezing temperature of the fish. The researchers compared genomic sequences in nine cod species and one non-cod fish species, *B. brosme*. They recorded the presence or absence of the elements of functional *ag* genes as well as *ag*-like sequences that are similar to a functional gene but have undergone mutation and do not contain all the elements required to enable protein production (Figure 2).



–	–	–	*B. brosme* (non-cod fish)	5–10°C
–	–	–	*G. argenteus*	
–	–	–	*T. minutus*	
–	–	–	*P. virens*	
+	+	+	*M. aeglefinus* (L)	
+	+	–	*M. merlangus* (L)	
+	+	+	*A. glacialis* (P)	1.2–6°C
+	+	+	*B. saida* (P)	
+	+	+	*G. chalcogrammus* (P)	
+	+	+	*G. morhua* (P)	

–	Element is absent.
+	Element is present.
L	Functional *ag* genes are lost.
P	Functional *ag*-encoded proteins are present.

Figure 2. Phylogenetic tree showing the evolution of *ag* genes

(a) **Describe** a post-zygotic mechanism that prevents gene flow and thus enables speciation. **1 point**

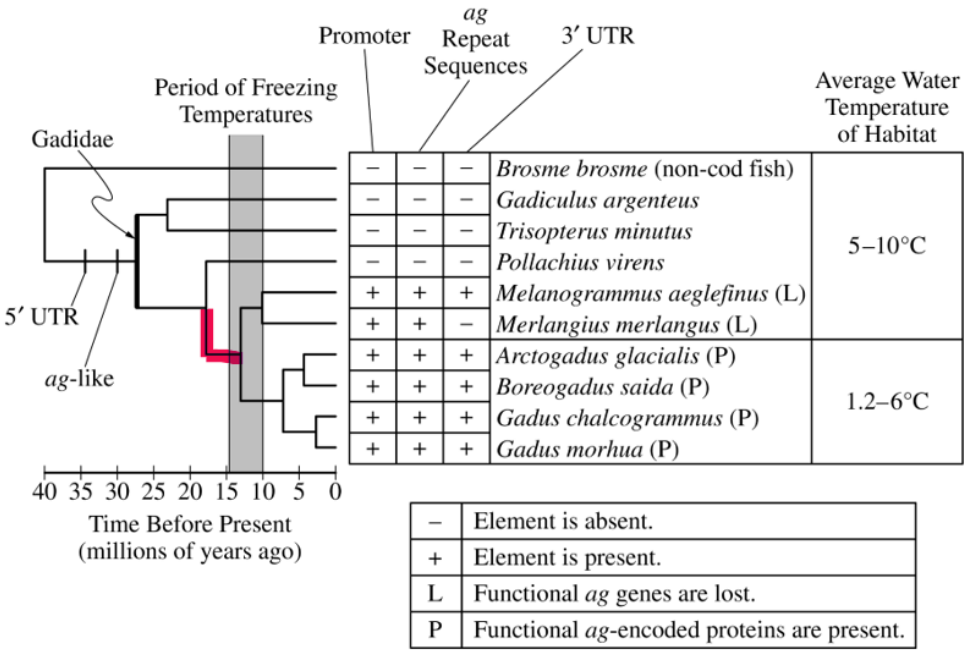
- The offspring do not survive (long enough to reproduce)/cannot (successfully) reproduce.

(b) Based on the data in Figure 2, **explain** how changes to the genome enabled cods to survive and reproduce after a period of freezing temperatures between 10 and 15 million years ago. **1 point**

- (Over time) the addition of the promoter/*ag* repeat sequences/3′ UTR/regulatory elements led to emergence of new genes/*ag* genes/functional gene products (that prevent freezing).

(c) Using the template in the space provided for your response, place an "X" on the phylogenetic tree to **represent** the origin of the functional *ag* gene. **1 point**

- An X is placed anywhere along the colored L-shaped line in the figure.



–	–	–	*Brosme brosme* (non-cod fish)	5–10°C
–	–	–	*Gadiculus argenteus*	
–	–	–	*Trisopterus minutus*	
–	–	–	*Pollachius virens*	
+	+	+	*Melanogrammus aeglefinus* (L)	
+	+	–	*Merlangius merlangus* (L)	
+	+	+	*Arctogadus glacialis* (P)	1.2–6°C
+	+	+	*Boreogadus saida* (P)	
+	+	+	*Gadus chalcogrammus* (P)	
+	+	+	*Gadus morhua* (P)	

–	Element is absent.
+	Element is present.
L	Functional *ag* genes are lost.
P	Functional *ag*-encoded proteins are present.

(d) Based on Figure 2, **explain** how genetic differences among the species in the Gadidae family determine the habitats in which they can survive. **1 point**

- Species with the functional *ag* gene/antifreeze glycoprotein are able to live in colder water/lower temperatures (than are species without the functional gene).

Total for question 5 **4 points**

Question 6: Analyze Data — 4 points

Scientists can quantify the rate of translation as ribosomes move along an mRNA from one codon to the next. Using a procedure called ribosome profiling, the scientists measured how long a ribosome remains stationary at each codon of each mRNA. They determined the average translation rate across all codons is 5.2 amino acids per second but that the average translation rate for specific codons in different mRNA sequences can vary widely. These variations in translation rates are thought to facilitate correct folding of the protein being produced. The rate at which three different codons were translated was measured in 100 different mRNAs. The scientists determined the distribution of rate (number of times each rate was recorded) for each of the three codons: GAC (Figure 1A), AUU (Figure 1B), and UGG (Figure 1C).

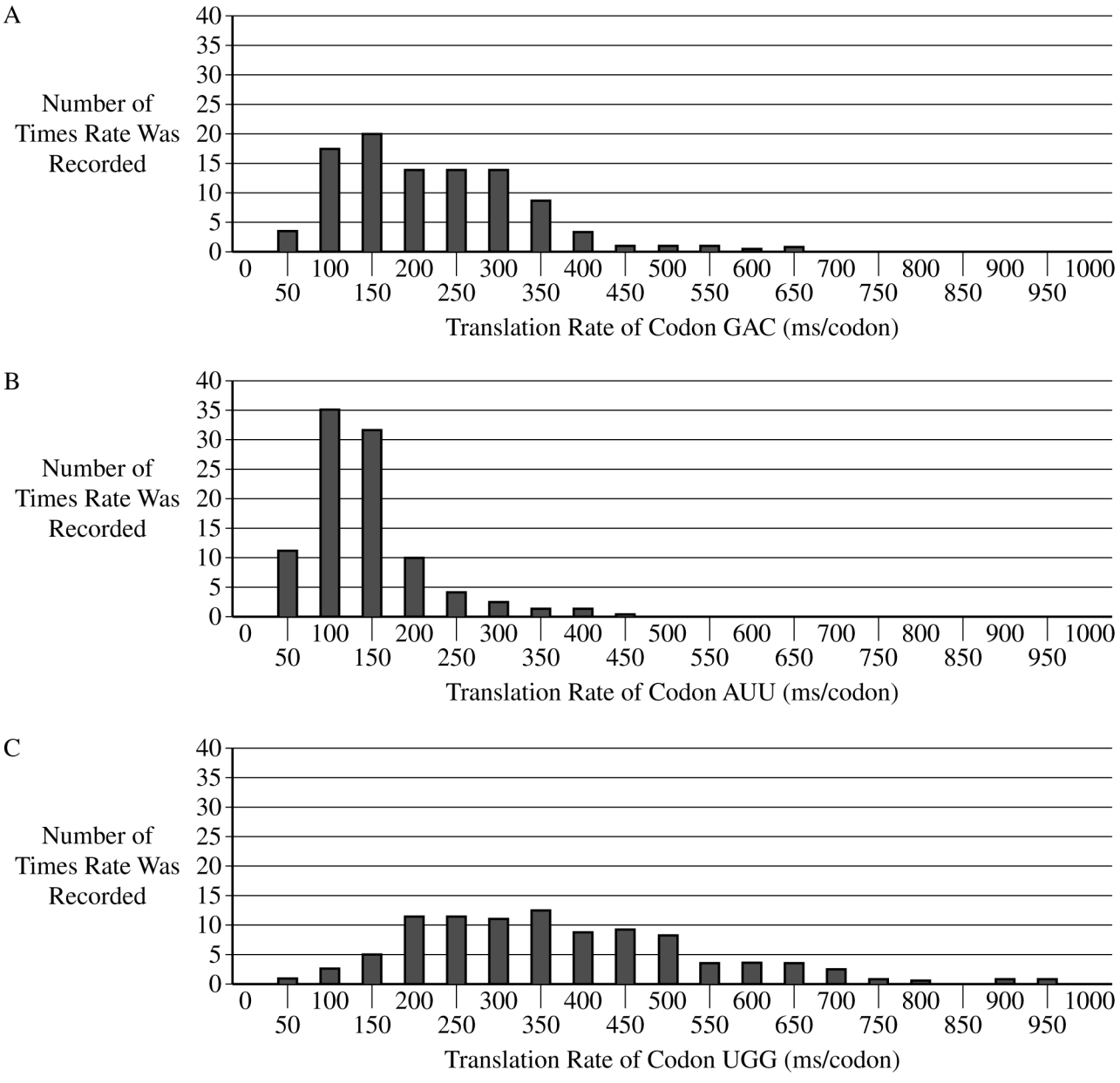


Figure 1. The distribution of translation rates for three different codons (A) GAC, (B) AUU, and (C) UGG

(a)	Using the data in Figure 1, graph A, **identify** the rate (in ms/codon) that was recorded the greatest number of times for the GAC codon. • 150	**1 point**
(b)	Using the data in Figure 1, graphs B and C, **describe** the variation in translation rate of the AUU codon compared with that of the UGG codon. Accept one of the following: • There is greater variation in (the translation rate of) UGG codons (than in the translation rate of AUU codons). • (The translation rate of) UGG ranges from 50 (ms/codon) to 950 (ms/codon), while (the translation rate of) AUU ranges from 50 (ms/codon) to 450 (ms/codon).	**1 point**
(c)	Scientists hypothesize that tRNA molecules that bind to UGG codons are available in lower abundance than are tRNAs that bind to AUU codons. **Support** the scientists' hypothesis using the data in Figure 1. Accept one of the following: • The (average) translation rate of UGG is slower (than that of AUU). • Translation of UGG takes longer (per codon than does translation of AUU). • More of the UGG codons were translated at slower rates (than AUU codons were).	**1 point**
(d)	Amino acids can be encoded by multiple codons. In many organisms, certain codons for the same amino acid occur more frequently in an mRNA than do other codons. Based on the data provided, **explain** why the use of one codon over another for the same amino acid might result in increased levels of protein production from a particular mRNA. • Certain codons are translated at faster rates than are others (and result in increased levels of protein production from a particular mRNA).	**1 point**
	Total for question 6	**4 points**

AP® Biology

Scoring Guidelines

Question 1: Interpreting and Evaluating Experimental Results 9 points

Most proteins that are secreted from a cell must be transported to the endoplasmic reticulum (ER) either during translation or after translation.

For proteins transported during translation, this process begins in the cytosol and pauses when a specific sequence of amino acids is translated. The translation complex is then transported to the surface of the ER where translation continues. Proteins that are transported after translation are translated entirely in the cytosol and then transported to the ER. In both instances, the translated proteins enter the ER through a protein channel in the membrane of the ER.

Researchers studying the two types of protein transport identified that the ER membrane protein SR is necessary for transport during translation, while the ER membrane protein Sec62 is necessary for transport after translation. To investigate which transport mechanism is used for different proteins, researchers first created small interfering RNAs (siRNAs) that reduce expression of either SR or Sec 62. They then treated groups of cells with either the SR siRNA or the Sec62 siRNA and determined the relative amount of SR and Sec 62 protein in each group of cells compared with cells treated with a control siRNA. (Figure 1).

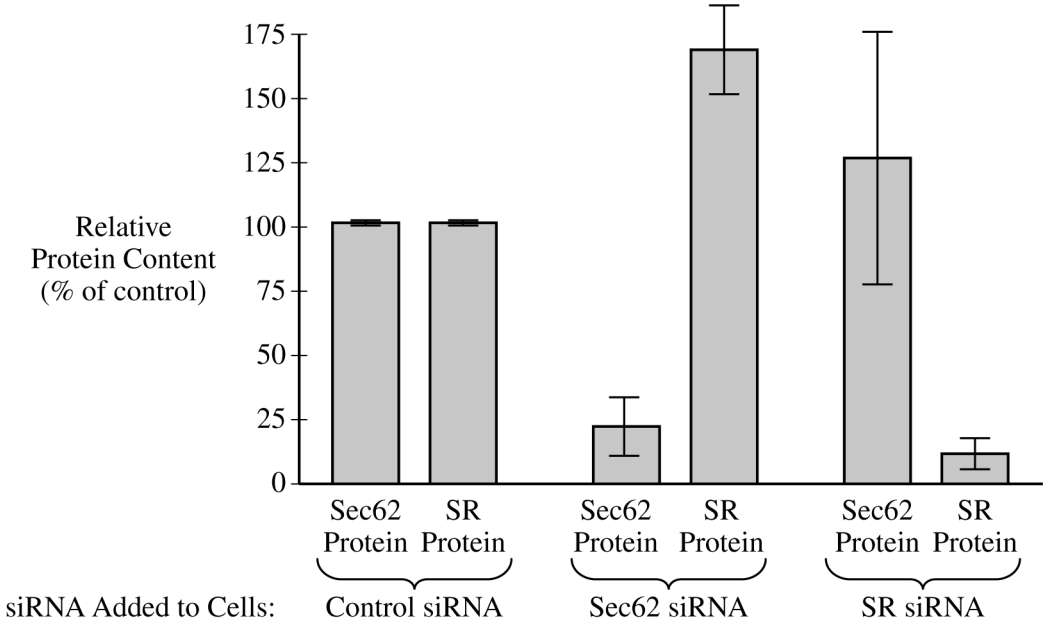


Figure 1. Average relative amounts of Sec62 and SR proteins in cells treated with control siRNA, Sec62 siRNA or SR siRNA. Error bars represent $\pm SE_{\bar{X}}$.

The researchers then measured the amount of each of three different proteins that was transported to the ER in cells treated with Sec62 siRNA or SR siRNA. The researchers calculated the percent transported relative to the cells treated with control siRNA (Figure 2).

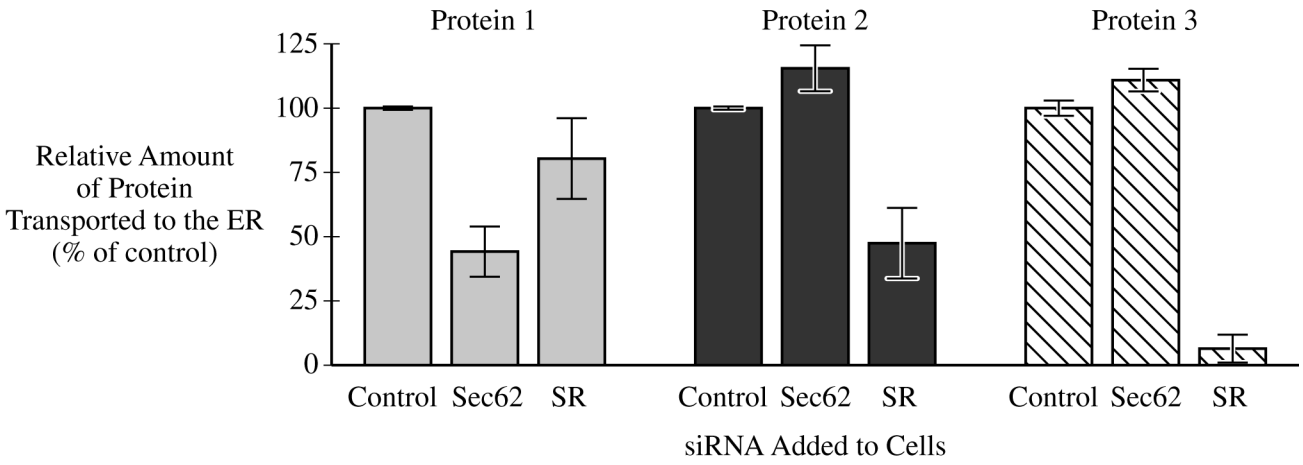


Figure 2. Average relative amounts of three proteins that were transported to the ER when treated with control siRNA, Sec62 siRNA, or SR siRNA. Error bars represent $\pm SE_{\bar{X}}$.

A	**Describe** the function of ribosomes. Examples of acceptable responses may include the following: • Ribosomes synthesize polypeptides/proteins. • Ribosomes perform translation. • Ribosomes are sites of polypeptide/protein synthesis.	**Point A1**
B	(i) **Identify** the dependent variable in the experiments shown in Figure 1. • The (relative) amount of protein/protein content	**Point B1**
	(ii) **Justify** why the researchers included the control of measuring the relative amounts of both Sec62 and SR proteins in cells that were treated with Sec62 siRNA only (data shown in Figure 1). Examples of acceptable responses may include the following: • (The control allowed the researchers) to determine whether the (Sec62) siRNA reduced/affected the content of both proteins (relative to protein content in the presence of the control siRNA). • (The control allowed the researchers) to determine whether the (Sec62) siRNA reduced/affected Sec62 (protein content) only.	**Point B2**
	(iii) Based on Figure 1, **describe** the effect on the production of SR protein when cells are treated with Sec62 siRNA. Examples of acceptable responses may include the following: • (SR protein production) increased. • (SR protein production) increased by 65% (accept 50–80%).	**Point B3**

C	(i) **Identify** the independent variable in the researchers' second experiment (data shown in Figure 2). Examples of acceptable responses may include the following: • (The type of) siRNA (added to the cells) • The type of protein whose amount was measured	**Point C1**
	(ii) Based on Figure 2, **identify** the protein(s) that when treated with Sec62 siRNA showed an increase in percent transport to the ER compared with the control. Examples of acceptable responses may include the following: • Proteins 2 and 3 • Protein 2 • Protein 3	**Point C2**
	(iii) Protein 1 is encoded by 234 nucleotides, while protein 2 is encoded by 495 nucleotides. Assuming all nucleotides for both proteins encode amino acids, **calculate** the difference in the number of amino acids between the two proteins. • 87 (amino acids) [(495/3)−(234/3)]	**Point C3**
D	(i) Researchers claim that protein 1 is the only tested protein that is transported to the ER following its complete translation in the cytosol. Using data from Figure 2, **support** the researchers' claim. • Only protein 1 showed a reduced amount/percentage of transport when cells were treated with Sec62 siRNA.	**Point D1**
	(ii) For any protein that enters the ER, researchers claim that amino acids close to the protein's amino terminus determine how likely the protein is to pass through the protein channel within the ER membrane. **Justify** the researchers' claim based on your understanding of factors that affect the transport of proteins across membranes. • Amino acids (at the amino terminus) that have similar polarity to/opposite charge to (the R groups of amino acids lining) the protein channel are more likely to pass through the channel (than are proteins where the amino terminus contains amino acids with dissimilar polarity or similar charge).	**Point D2**

Question 2: Interpreting and Evaluating Experimental Results with Graphing — 9 points

Many insects rely on pheromones (chemical signals) that are released by the females to find mating partners. Scientists hypothesize that, in a certain type of moth, the behavior of male moths in response to pheromones is regulated by the extracellular signaling molecule 20E.

To investigate whether the binding of 20E to its receptor, DopEcR, affects behavior in moths, scientists injected male moths with saline (control solution) or with small interfering RNA molecules (siRNAs) that inhibit the expression of the gene encoding DopEcR. The scientists then exposed the moths to the pheromone and determined the percent of total time observed that the moths engaged in general activity, defined as movement in any direction. The scientists also determined the percent of the general activity time that the moths spent in oriented activity, defined as movement toward an area of high pheromone concentration (Table 1).

Table 1. Average General and Oriented Activity in Male Moths Injected With Saline or siRNA Molecules

Treatment	General Activity (percent of total time observed, average $\pm 2SE_{\bar{x}}$)	Oriented Activity (percent of general activity, average $\pm 2SE_{\bar{x}}$)
Male moths injected with saline (control solution)	95 ± 5	60 ± 4
Male moths injected with siRNAs that inhibit expression of the gene encoding DopEcR	90 ± 8	25 ± 6

DopEcR is a G protein-coupled receptor. When 20E binds to DopEcR, GTP displaces the GDP bound to the G protein, and a signaling pathway is activated. The scientists hypothesize that this leads to the transcription of genes associated with the oriented activity observed in the male moths (Figure 1).

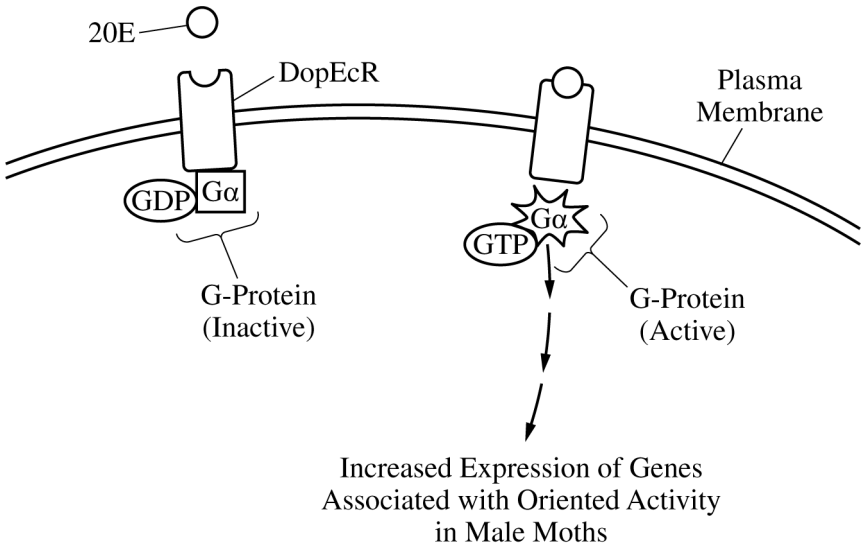
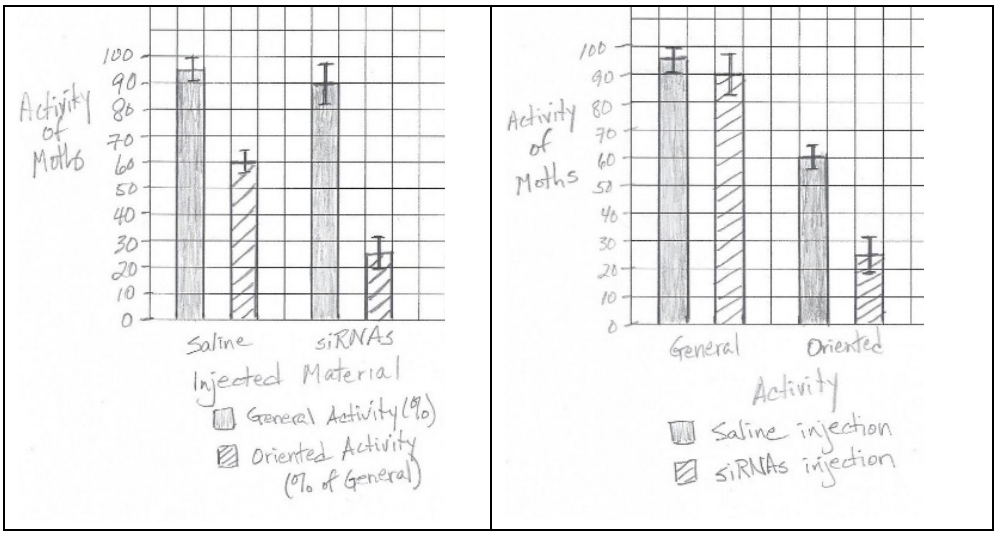


Figure 1. A simplified model of a signaling pathway activated by the binding of 20E to its receptor, DopEcR

A	Many receptors are embedded in the plasma membrane. **Describe** the polarity of the portion of the receptor that is inside the membrane. Examples of acceptable responses may include the following: • (The portion of the receptor inside the membrane) is nonpolar. • (The portion of the receptor inside the membrane) is hydrophobic.	**Point A1**
B	(i) Using the template in the space provided for your response, **construct** an appropriate type of graph that represents the data in Table 1. Your graph should be appropriately plotted and labeled. • Data are represented in a bar/modified bar graph. • Examples of acceptable responses may include the following: 	**Point B1**
	(i) Using the template in the space provided for your response, **construct** an appropriate type of graph that represents the data in Table 1. Your graph should be appropriately plotted and labeled. • Data and error bars are accurately plotted.	**Point B2**
	(i) Using the template in the space provided for your response, **construct** an appropriate type of graph that represents the data in Table 1. Your graph should be appropriately plotted and labeled. • Graph is appropriately labeled.	**Point B3**
	(ii) Based on the data in Table 1, **determine** the type of activity that was affected by inhibiting the expression of the DopEcR receptor. • Oriented activity (was affected).	**Point B4**

C	(i) Based on Table 1, identify the treatment group in which the oriented activity was greater than 50% of the general activity. Examples of acceptable responses may include the following: • The control/saline-treated group • Male moths injected with saline/control solution	**Point C1**
	(ii) The scientists studied some moths with a mutation in the gene encoding the G protein. The mutation prevents GTP from displacing the GDP bound to the G protein. Based on Figure 1, **predict** the effect of this mutation on the oriented activity in male moths exposed to the pheromone. • (The moths will show) decreased (oriented) activity.	**Point C2**
D	(i) Expression of the gene encoding DopEcR is low in the male moths during their first few days as adults, when they are sexually immature. Gene expression rapidly increases as the moths reach sexual maturity. The scientists claim that this increase in gene expression increases the likelihood of males finding females with whom to mate. Use evidence from the information provided to **support** the scientists' claim. Examples of acceptable responses may include the following: • (Increased expression of DopEcR) will increase oriented activity (and the moths' ability to find a mate). • (An increase in DopEcR expression) will enable the binding of 20E to more receptors. • (An increase in DopEcR expression) will make the males more sensitive to pheromones from the females. • (An increase in DopEcR expression) will enable the binding of 20E to more receptors and will make the males more sensitive to pheromones from the females.	**Point D1**
	(ii) Based on Figure 1, **explain** how an inhibitor of the DopEcR pathway might serve as an effective chemical to protect crops from moth damage. • (An inhibitor of the pathway) will reduce oriented activity/expression of genes associated with oriented activity and therefore decrease mating/population growth.	**Point D2**

Question 3: Scientific Investigation — 4 points

Buffelgrass, an invasive grass species in southwestern desert ecosystems, is threatening the saguaro cactus, a keystone species in these ecosystems. Buffelgrass is drought-tolerant and can survive wildfires. However, the dry buffelgrass also acts as fuel for wildfires, causing the fires to be more severe. Older saguaro cacti can survive wildfires; however, many of the young cacti cannot.

Scientists conducted an experiment to determine whether they could control the abundance of the buffelgrass population. The scientists identified several native grass species that, when grown with buffelgrass, might reduce the abundance of buffelgrass. They grew buffelgrass in the presence of several different native grass species in greenhouses, in either nondrought (watered every 3 days) or drought (watered every 9 days) conditions. After twelve weeks, they measured the height and dry weight of the buffelgrass in each treatment group.

A	**Describe** the effect that removing a keystone species will have on an ecosystem. Examples of acceptable responses may include the following: • (Removal of a keystone species) reduces biodiversity/diversity/resilience (of the ecosystem). • (Removal of a keystone species) will cause the ecosystem to collapse.	**1 point**
B	**Identify** a control group the scientists should include in their experiment. Examples of acceptable responses may include the following: • (A treatment group with) only buffelgrass planted • (A treatment group with) no native grass species	**1 point**
C	**State** the null hypothesis of the experiment in which buffelgrass is grown in the presence of native grass species. Examples of acceptable responses may include the following: • There will be no difference in/effect on the abundance/dry weight/height/size of buffelgrass (when grown alone as compared with) when grown with native plants. • There will be no difference in/effect on the abundance/dry weight/height/size of buffelgrass grown in drought and nondrought conditions.	**1 point**
D	Scientists have found that the population growth rates of native grasses are much slower than the population growth rate of buffelgrass following a wildfire. The scientists claim that wildfires will therefore increase the abundance of buffelgrass plants in the ecosystem. Based on the information given, **justify** the scientists' claim. Examples of acceptable responses may include the following: • (With fewer native plant species after a wildfire) buffelgrass will have less competition/no competition (for resources). • (With fewer native plant species after a wildfire) buffelgrass will have more resources (for growth, development, and reproduction).	**1 point**

Question 4: Conceptual Analysis — 4 points

Twenty million years ago the Caribbean Sea and Pacific Ocean were connected, and water flowed freely between the two bodies of water. Many of the same marine species were found in both areas. Over millions of years, the land referred to as the Isthmus of Panama formed, eventually closing off the connection between the Caribbean Sea and Pacific Ocean and creating two separate bodies of water. The ecology of these two marine habitats was dramatically altered by this land formation. The warmer Caribbean water could no longer flow west, so the Pacific water cooled and became more nutrient-rich, while the Caribbean water became warmer.

A	**Describe** the genetic evidence that evolution is occurring in a population. Examples of acceptable responses may include the following: • There are changes in allele/gene frequencies. • There are heritable changes in phenotypes.	**1 point**
B	**Explain** how the isolation of marine species by the formation of a land barrier can lead to divergent evolution of those species. Examples of acceptable responses may include the following: • Selective pressures could result in different allele/gene frequencies. • (The land barrier) results in reproductive isolation/lack of gene flow/allopatric speciation. • Different environmental conditions/pressures select for different alleles/genotypes/phenotypes.	**1 point**
C	The formation of the Isthmus of Panama connected two continents, North America and South America. Many North American land animal species migrated to South America after the formation of the isthmus and occupied similar niches as South American species. **Predict** the effect the formation of the isthmus had on resource availability for South American species. • (Resource availability) would have decreased.	**1 point**
D	**Justify** your prediction in part C. • More species would now be competing for the same resources, (resulting in fewer resources for each individual).	**1 point**

Question 5: Analyze Model or Visual Representation of a Biological Concept or Process — 4 points

Figure 1 shows the reactions of the metabolic pathway used to synthesize amino acid B from amino acid A in cells.

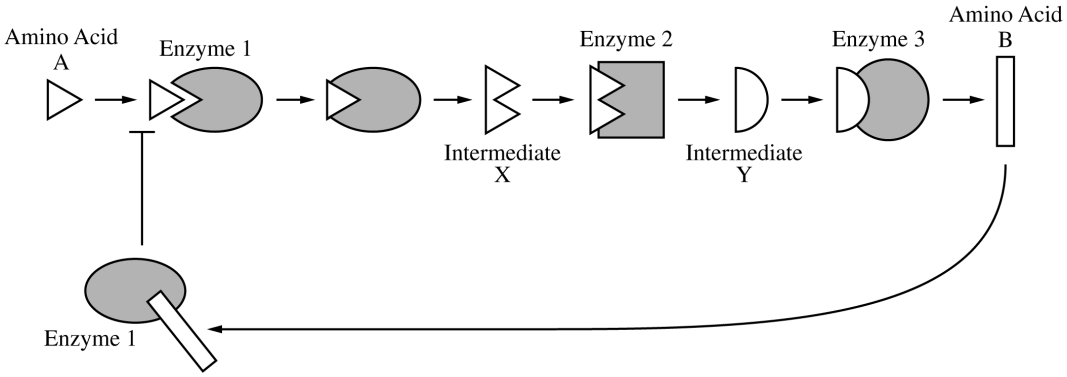


Figure 1. Synthesis of amino acid B from amino acid A

A	**Describe** a characteristic of an enzyme's active site that allows it to catalyze a specific chemical reaction. Examples of acceptable responses may include the following: • (The active site) is able to bind to a (specific) substrate/amino acid. • The shape/charge of the active site of an enzyme is compatible with a (specific) substrate/amino acid. • (The active site) positions the substrate/amino acid in a way that makes the reaction more likely. • Induced fit stresses bonds so that a reaction is more likely to occur.	**1 point**
B	Based on Figure 1, **explain** how the binding of amino acid A to enzyme 1 is regulated by amino acid B. Examples of acceptable responses may include the following: • Amino acid B inhibits the binding of amino acid A to enzyme 1. • Amino acid B binds to an allosteric site of enzyme 1 and prevents amino acid A from binding to enzyme 1. • The binding of amino acid B causes the enzyme to change shape and prevents amino acid A from binding to enzyme 1.	**1 point**
C	Using the information in Figure 1, **identify** the product of the reaction catalyzed by enzyme 2: intermediate X, intermediate Y, or amino acid B. • (Intermediate) Y	**1 point**
D	Based on Figure 1, **explain** how a change in pH could affect enzyme 3 in such a way that amino acid B cannot be produced. Examples of acceptable responses may include the following: • (A change in pH) could change the structure/active site/enzyme so that intermediate Y/the substrate cannot bind. • (A change in pH) could denature (enzyme 3) so that intermediate Y/the substrate cannot bind.	**1 point**

Question 6: Analyze Data — 4 points

The *ald* gene of fruit flies encodes the ALD protein, which is associated with both the centromeres of chromosomes and protein filaments produced during meiosis. In the absence of functional ALD proteins, gamete-producing cells enter anaphase I before homologous chromosomes are correctly aligned. As a result, the gametes produced do not contain the correct numbers of chromosomes.

Scientists generated four mutations in the *ald* gene: *ald1*, *ald3*, *ald23*, and *del*, which was a deletion of the gene. To study the role of the ALD protein in meiosis, scientists used gamete-forming metaphase cells from groups of flies with different *ald* genotypes. Some of the flies were homozygous for the wild-type allele of *ald: WT*/*WT*. Other flies were heterozygous for different *ald* alleles: *WT*/*del*; *ald1*/*del*; *ald3*/*ald23*; *ald23*/*del*. The scientists measured the percent of metaphase cells that contained ALD-associated filaments (Figure 1A) and the amount of ALD protein produced by each of the cell types (Figure 1B).

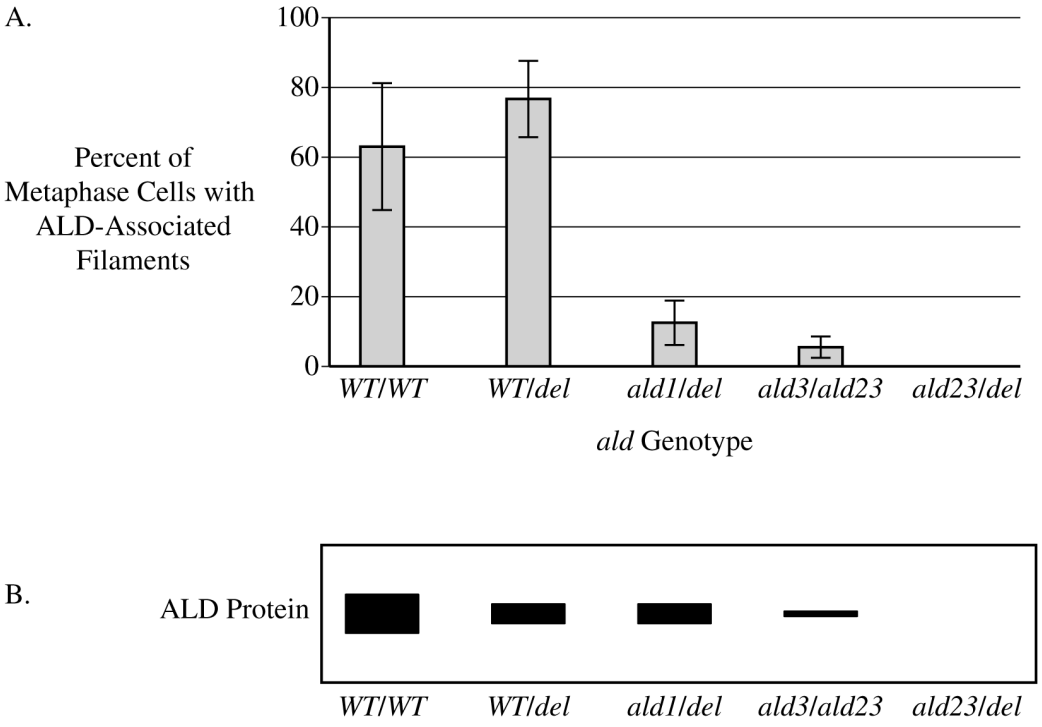


Figure 1. (A) The average percent of gamete-forming metaphase cells that contained filaments associated with ALD and (B) the amount of ALD protein produced by each cell type. A thicker band indicates a greater amount of ALD protein.

A	Based on Figure 1A, **identify** the fly genotype in which the average percent of metaphase cells with ALD-associated filaments is close to 12%. • *ald1/del*	**1 point**
B	Based on Figure 1B, **describe** the difference in ALD protein production between gamete-forming metaphase cells of flies with the genotype *ald3*/*ald23* and flies with the genotype *ald23*/*del*. Examples of acceptable responses may include the following: • More ALD protein is produced by *ald3*/*ald23* cells (than by *ald23*/*del* cells). • Less ALD protein is produced by *ald23*/*del* cells (than by *ald3*/*ald23* cells). • No ALD protein is produced by *ald23*/*del* cells (whereas it is produced by *ald3*/*ald23* cells).	**1 point**
C	Scientists hypothesize that gamete-forming metaphase cells can produce a normal amount of ALD-associated filaments even when they produce about half as much ALD protein as the wild-type cells produce. Use the data in Figures 1A and 1B to **support** the scientists' hypothesis. Examples of acceptable responses may include the following: • (With half as much protein) the *WT/del* cells show no difference in percent of (gamete forming metaphase cells with) ALD-associated filaments. • The data for the *WT/del* relative to/in comparison with the *WT/WT* cells support the scientists' hypothesis.	**1 point**
D	For gamete-forming metaphase cells of the *WT*/*del* and *ald1*/*del* flies, **explain** why the phenotypes observed in Figure 1A differ even though the amount of ALD protein produced (Figure 1B) does not. Examples of acceptable responses may include the following: • (The phenotypes) differ because only the *WT*/*del* flies produce enough functional (ALD) protein to generate a wild-type phenotype. • When one allele encodes functional ALD protein (in *WT/del* flies), the flies can generate a wild-type phenotype/produce ALD-associated filaments in a similar amount as *WT/WT* flies. • Both genotypes produced ALD protein, but the *ald1* mutation resulted in a protein with reduced function (compared with *WT*, resulting in a different phenotype).	**1 point**